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Chapter 1

Introduction

The Zeno effect has its name from a series of apparent paradoxes thought of in ancient Greece by the philosopher Zeno. Prominent examples include that of Archilles and the turtoise and the Arrow paradox. While these paradoxes were solved through the invention of infinitesimal calculus, a similar quantum effect may be more applicable in physics: the Quantum Zeno effect. Roughly speaking, the effect describes the phenomenon that a watched quantum state does not change. More precisely, the 'watching' part is done by a measurement on the quantum system. The effect relies on the fact that measurement may change the state of a quantum system. A prominent example for the Quantum Zeno effect is that of an atom decaying into the ground state. If the atom is always observed to be in the excited state $|e\rangle$, then it can not leave this state since it gets projected back onto the state with each measurement. In reality, no system can be always measured or no measurement can be performed with infinite frequency. Mathematically, however, this frequency limit poses an interesting problem. More recently, the existence of this limit was proven to exist for a wider class of quantum operations and time evolutions of the system. The resulting dynamics are called Zeno dynamics. The goal of this thesis is to investigate what effect such Zeno dynamics can have on the system. As the title of this thesis suggests, the broad idea is that if the dynamics of the system are restricted to a smaller subsystem due to the Zeno measurements, then the dynamics in that subsystem mabybe somewhat faster without the measurements. Since in arbitrary quantum systems there is no general notion of velocity, this can be investigated in many different aspects.

Chapter 2

Prerequisites

2.1 Mathematics

Here, there will be a few important mathematical preliminaries recalled, which are important for the study of quantum mechanics, mostly basics of linear algebra and functional analysis. Let it be mentioned here once that the listings below do not claim to be the most general results, e.g. a lot of the statements are valid not only for Hilbert spaces, but for vector or Banach spaces in general.

A Hilbert space $(\mathcal{H}, \langle\cdot,\cdot\rangle)$ is a vector space $\mathcal{H}$ over the complex or real numbers equipped with an inner product $\langle\cdot,\cdot\rangle$ such that the space is complete (every Cauchy sequence converges) with respect to the norm $||\cdot|| = \sqrt{\langle\cdot,\cdot\rangle}$ induced by the inner product. In the case of a complex Hilbert space the inner product is chosen to be conjugate linear in the first component, so standard physics notation. Two elements $v, w \in \mathcal{H}$ space are called orthogonal w.r.t. to this inner product if[1] $\langle v, w\rangle = 0$. A basis for this vector space is a set of vectors $\{e_i\}_{i=1}^{d} \subset \mathcal{H}$ [2] that is a linear independent family, i.e. if $\sum_{i=1}^{d} c_i e_i = 0$ for any coefficients c_i in the field, then it follows that $c_i = 0$ for all indices i and secondly a spanning system for $\mathcal{H}$ such that every element of the vector space can be written as a linear combination of the basis elements. $d =: \dim(\mathcal{H}) \in \mathbb{R}\cup\{\infty\}$ is then called the dimension of $\mathcal{H}$. This work restricts to finite dimensional spaces where $\dim(\mathcal{H}) < \infty$ since this simplifies the mathematics. Two of the most fundamental results of the study of linear algebra are that every finite dimensional vector space has a basis and that each basis has the same length, thus making the dimension of the Hilbert space well-defined. Naturally, every finite-dimensional Hilbert space is also separable (countable basis). A basis is called orthogonal if all its elements are mutually orthogonal and orthonormal if additionally every element of the basis is a unit element, i.e. has norm one. From now on Hilbert spaces will be over the field $\mathbb{C}$ unless stated otherwise and the inner product will be the standard inner product: For two vectors $v = (v_i)_{i=1}^{\dim(\mathcal{H})}$ and $w = (w_i)_{i=1}^{\dim(\mathcal{H})}$ it is

$$\langle v, w\rangle = \sum_{i=1}^{\dim(\mathcal{H})} \overline{v_i} w_i, \tag{2.1.1}$$

where the v_i and w_i are the coefficients of the vectors v and w with respect to a fixed chosen orthonormal basis. One then often only writes just $\mathcal{H}$ for the Hilbert space. The coefficients of a vector $v \in \mathcal{H}$ may be computed via $v_i = \langle e_i, v\rangle$ since for $v = \sum_i c_i e_i$ it is $\langle e_j, v\rangle = c_j$. From $v = \sum_i \langle e_i, v\rangle e_i$ follows the useful idenitity $\sum_i \langle e_i, \cdot\rangle e_i = \mathbb{1}_{\mathcal{H}}$

Of particular importance are linear operators. If the following map

$$(\forall O : \mathcal{H} \to \mathcal{H}) : ||\cdot||_{\text{op}} = \sup_{v\in\mathcal{H}, v\neq 0} \frac{||Ov||}{||v||} = \sup_{v\in\mathcal{H}, ||v||=1} ||Ov|| \tag{2.1.2}$$

is finite, it fulfills the property of a norm. Hence, the space of linear operators for which the operator norm is finite becomes a normed space itself, that space is denoted by $\mathcal{B}(\mathcal{H}) =$

[1] As a general rule: When defining properties, it is always the 'if and only if' relation meant, but only 'if' is written. So in this example there are no elements that are orthogonal but do not fulfil this inner product equality.

[2] $\subset$ here may also mean equality and it is written $\subsetneq$ if the equality case is excluded.

$\{O \text{ is linear and } ||O||_{\text{op}} < \infty\}$. This also works for operators between different Hilbert spaces (or in general normed spaces). $\mathcal{B}(\mathcal{H})$ is also again complete w.r.t. the operator norm since $\mathcal{H}$ is complete. For the operator norm the index op is often omitted entirely or substituted by the index ∞, coming from the fact the the operator norm is the limit of the p-norms for operators (which are not introduced below) when p goes to infinity. From the definition of the operator norm it imidieatly follows that for all $v \in \mathcal{H}$ the inequality $||Tv|| \leq ||T|| \cdot ||v||$ holds true. Indeed, the operator norm is the infimum of all numbers that fulfill this inequality for all vectors. The operator norm is submultiplicative since for all operators $A, B : \mathcal{H} \to \mathcal{H}$ and vectors $v \in \mathcal{H}$ it is

$$||(A \circ B)v|| = ||A(Bv)|| \leq ||A|| \cdot ||Bv|| \leq ||A|| \cdot ||B|| \cdot ||v||. \tag{2.1.3}$$

Another important statement is the following: A linear operator $O \in \mathcal{L}(\mathcal{H})$ is continuous if and only if it is bounded meaning its operator norm is finite. Additionally, for linear operators continuity is equivalent to continuity at the point $0 \in \mathcal{H}$ and boundedness is equivalent to the operator being $||O||$-Lipschitz, making all these properties equivalent in turn.

In the space $\mathcal{B}(\mathcal{H})$ there are again classes of elements that are of particular importance: Hermitian and positive semidefinite operators. To define Hermitian operators the (Hermitian) adjoint has to be introduced first for which the dual space is needed. The dual space of a Hilbert space $\mathcal{H}$ is $\mathcal{H}^* := \mathcal{B}(\mathcal{H}, \mathbb{C})$. Elements of the dual space are often called functionals. Any functional can be represented by an element from the original space, this is formalized in the Riesz representation theorem:

Proposition 2.1.1 (Riesz representation)**.** *The map*

$$J : \mathcal{H} \to \mathcal{H}^*, v \mapsto \langle v, \cdot \rangle$$

is isometric, i.e. $||J(v)|| = ||v||$ *for all* $v \in \mathcal{H}$*, and bijective.*

The Riesz representation justifies the physics notation for functionals since now every functional can be written as follows: Let $v \in \mathcal{H}$ then $\langle v|$ is an element of $\mathcal{H}^*$ via $w \mapsto \langle v|w\rangle := \langle v, w\rangle$. In this notation elements of the original Hilbert space $\mathcal{H}$ are written as $|v\rangle \in \mathcal{H}$, called kets. The $\langle v|$ are called bras. Obviously every $\langle v|$ is an element of the dual space but the Riesz representation theorem guarantees that every functional is of such form. The Riesz representation also proves the existence of the Hilbert adjoint[3]: Let $T \in \mathcal{B}(\mathcal{H}, \mathcal{H}')$ and $\mathcal{H}$ and $\mathcal{H}'$ two Hilbert spaces. Then setting $\langle Tw, v\rangle = \langle w, T^*v\rangle$ for all $v \in \mathcal{H}'$ and $w \in \mathcal{H}$ defines a linear continuous map $T^* \in L(\mathcal{H}', \mathcal{H})$ that depends on T. This can be proven by considering the map $(w \mapsto \langle Tw, v\rangle) \in \mathcal{H}^*$ for a fixed $v \in \mathcal{H}'$. There exists $z \in \mathcal{H}$ such that $\langle Tw, v\rangle = \langle z, w\rangle$ for each $v \in \mathcal{H}'$ and $w \in \mathcal{H}$. This determines the map $T^* : v \mapsto z$. One can show that the norms of T and T^* coincide which makes T^* also bounded and thus continuous and that $(T^*)^* = T$. Finally, the following definitions are introduced:

Definition 2.1.1. An operator $T \in \mathcal{B}(\mathcal{H})$ is called

(i) Hermitian or self-adjoint if $T = T^*$,

(ii) positive semidefinite, written $T \geq 0$, if for all $v \in \mathcal{H}$ it is $\langle v, Tv\rangle \geq 0$,

(iii) a projection or idempotent if $T^2 = T$

(iv) and an orthogonal projection if it is Hermitian and a projection.

Note that these definitions now imply the operators to be linear since $T \in \mathcal{B}(\mathcal{H})$. If an operator $T \in \mathcal{B}(\mathcal{H})$ is Hermitian, then all its eigenvalues are real. To see this let $Tv = \lambda v$ for $v \neq 0$ be an eigenvector equation and consider

$$\overline{\lambda}\langle v, v\rangle = \langle Tv, v\rangle = \langle v, Tv\rangle = \lambda\langle v, v\rangle, \tag{2.1.4}$$

implying $\lambda = \overline{\lambda}$. The converse is only true if T and T^* commute, so there are operators with only real eigenvalues that are not Hermitian, for example

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \tag{2.1.5}$$

[3]Admittetly, in the case of finite dimensional spaces one does not need the Riesz representation to prove the existence since one can just compute the conjugate transpose of the matrix representation and check that it fulfills the properties of the adjoint.

Eigenvectors corresponding to different eigenvalues of an Hermitian operator are orthogonal: Let μ and λ be different eigenvalues of $T = T^*$ with eigenvectors v and w, respectively. Then

$$\mu\langle v, w\rangle = \langle Tv, w\rangle = \langle v, Tw\rangle = \lambda\langle v, w\rangle. \tag{2.1.6}$$

This can only be true if $\langle v, w\rangle = 0$ since $\lambda \neq \mu$ by construction. $T \in \mathcal{B}(\mathcal{H})$ is positive semidefinite if and only if all eigenvalues are non-negative. A positive semidefinite operator is necessarily Hermitian. A bijective function U between Hilbert spaces (of the same dimension) that preserves inner products is called a unitary, i.e. for all $x, y \in \mathcal{H}$ it is $\langle Ux, Uy\rangle = \langle x, y\rangle$. This is equivalent to $UU^* = U^*U = \mathbb{1}$, so the adjoint is the inverse of the operator. One can also extend the notion of unitaries to operators between Hilbert spaces of different dimension where only $U^*U = \mathbb{1}$ is required.

One of the advantages of dealing with finite dimensional spaces is that linear operators between such spaces can be represented by matrices. Consider the linear operator $T : \mathcal{H} \to \mathcal{H}'$ between two finite dimensional Hilbert spaces $\mathcal{H}$ and $\mathcal{H}'$ with orthonormal bases $\{e_i\}_{i=1}^{d}$ and $\{e'_i\}_{i=1}^{d'}$. Then each linear operator induces a complex valued $d' \times d$ matrix $\hat{T}$ with coefficients $\langle e'_i, Te_j\rangle =: \hat{T}_{ij}$ from which the operator may be recovered as

$$Tv = \sum_{i=1}^{d'} \left(\sum_{j=1}^{d} \hat{T}_{ij} v_j \right) e'_i \tag{2.1.7}$$

It turns out that adjoint of operators corresponds to the conjugate transpose of matrices. This also allows to very easily take over the definition of the trace of a quadratic matrix to the trace of an operator $T \in \mathcal{B}(\mathcal{H})$, $\mathrm{tr}[T] := \mathrm{tr}[\hat{T}] = \sum_i \hat{T}_{ii} = \sum_i \langle e_i, Te_i\rangle$, such that the know properties as for example cyclicity are preserved. Additionally the trace of an operator is independent of the basis it is evaluated in which can be proven using the the identity for $\mathbb{1}_{\mathcal{H}}$ mentioned above.

Another useful notion is that of convex subsets K of a vector space V: A convex combination of elements $x, y \in V$ in the vector space is a combination $px + (1-p)y$ where $p \in [0, 1]$ are scalars. The convex combination is said to be non-trivial if $p \notin \{0, 1\}$ and x is not proportional to y. Convex sets are exactly the sets that are closed under convex combinations of their elements. An element $x \in K$ of a convex set K is called extremal if it can not be written as a non-trivial convex combination of other elements in K. Extremal points play an important role in applications since by definition every other point can be built from extremal ones via convex combinations. However, not every convex set has extremal points. For example the set $\mathbb{C}^n$ is convex but every point $x = (x + v)/2 + (x - v)/2 \in \mathbb{C}^n$ for a $x \neq v \in \mathbb{C}^n$ is not extremal. The question of existence of extremal points is answered by the (much stronger) Krein-Milman theorem (see e.g. 7.4 in [Con90]), which implies that if $K \subset V$ is non-empty convex subset of a normed vector space V and the set of extremal points of K is denoted as $\mathrm{ext}(K)$, then $\mathrm{ext}(K) \neq \emptyset$.

To describe measurements in quantum mechanics the notion of a projection onto a subspace is needed, i.e. to each element in a vector space one wants to assign an element in the subspace that is closest to it, which is then called the projection onto that subspace. Necessary for the existence of such a projection is that the subspace is closed. More specifically there is the following proposition (without proof in the text):

Proposition 2.1.2 (Projection, see e.g. 2.5 in [Con90])**.** *Let $\mathcal{H}$ be a Hilbert space and $U \subset \mathcal{H}$ a non-empty, convex and closed subset. Then for every $x \in \mathcal{H}$, there exists a $y \in U$ so that*

$$||x - y|| = \inf_{z \in U} ||x - z||.$$

This defines the map $P_U : \mathcal{H} \to U, x \mapsto y$ which is called the projection onto U

Naturally, if an element $x \in \mathcal{H}$ is already in U, then $P_U(x) = x$, making $P_U = P_U^2$ indeed a projection operator. If additionally $U \subset \mathcal{H}$ is a closed subspace (so closed under any linear combination), then P_U is a linear Hermitian map fulfilling $\mathrm{Ker}(P_U) = U^\perp$ and $\mathrm{Ran}(P_U) = U$ (see e.g. 2.7. and 3.3. in [Con90]), where $U^\perp := \{v \in \mathcal{H} | \langle v, u\rangle = 0 \ \forall \ u \in U\}$ is the orthogonal complement of U. The projection onto the orthogonal complement is $P_{U^\perp} = \mathbb{1} - P_U$[4]. It follows for any non-empty subspace $U \neq \{0\}$ and $0 \neq x \in \mathcal{H}$, $x \notin U^\perp$ that

$$0 = \langle P_{U^\perp} x, P_U x\rangle = \langle x, P_U x\rangle - \langle P_U x, P_U x\rangle \leq ||x|| \cdot ||P_U x|| - ||P_U x||^2 \tag{2.1.8}$$

[4]citation needed

This implies $||P_U x|| \leq ||x||$. Here, it was possible to divide by $||P_U x||^2$ since $P_U x \neq 0$ because of $x \notin U^\perp$. If $x \in U^\perp$, then $P_U x = 0$, so (2.1.8) implies that $||P_U|| \leq 1$ and realizing that for $x \in U$ it is $P_U x = x$ yields $||P_U|| = 1$. Now, for every subspace U and for every $x \in \mathcal{H}$ it holds $x = x - P_U x + P_U x = P_{U^\perp} + P_U x$, so that $\mathcal{H} = U \bigoplus U^\perp$. On the other hand, one could ask if every (orthogonal) projection corresponds to a projection onto a subspace: In general, if $P = P^* = P^2$ is an orthogonal projection then $\mathrm{Ker}(P) = \mathrm{Im}(P)^\perp$, see e.g. 3.3 in [Con90], and $\mathrm{Im}(P)$ is closed. To see the latter, note that $\mathrm{Im}(P) = \mathrm{Ker}(\mathbb{1} - P)$ and $\mathbb{1} - P \in \mathcal{B}(\mathcal{H})$ so that its kernel is closed. Consequently, $P_{\mathrm{Im}(P)}$ exists and is indeed equal to P[5].

In the case of a finite dimensional Hilbert space there is no need to worry about closedness since every subspace of a finite-dimensional vector space is closed: Let $W \subset V$ be a linear subspace of a $\mathbb{K}$-vector space V. Fix a basis of W and extend it to a basis of V. Consider the map $L : V \to \mathbb{K}^{\dim(V)-\dim(W)}$ that maps a vector v to the coefficients in front of the basis vectors that are not in W. Then $W = \ker(L) = L^{-1}(\{0\})$. Since every map between finite-dimensional normed spaces is continuous[6] and $\{0\}$ is closed, so is W.

The eigenvalues and eigenvectors already give a lot of information about the operator, in the case of Hermitian operators they determine the operator fully via the spectral decomposition:

Theorem 2.1.1 (Spectral decomposition)**.** *Let $A \in \mathcal{H} \mapsto \mathcal{H}$ be normal and $\mathcal{H} \simeq \mathbb{C}^n$, then there exists a unitary U on $\mathcal{H}$ and a diagonal operator[7] D such that*

$$A = UDU^* = \sum_{i=1}^{n} \lambda_i u_i \langle u_i, \cdot \rangle$$

where the λ_i are the eigenvalues of A and u_i is an eigenvector corresponding to λ_i. It is $\hat{D} = diag(\lambda_i)_i$ and $U = (u_1, \ldots, u_n)$

Note that a single eigenvalue may appear multiple times in the spectral decomposition. The operator $u_i \langle u_i, \cdot \rangle$ is exactly the projector onto the subspace spanned by u_i. If every eigenvalue only appears once, the spectral decomposition may also be written as $A = \sum_{\lambda \in \sigma(A)} \lambda P_\lambda$, where P_λ is the eigenspace spanned by all eigenvectors corresponding to λ. Note that in the finite-dimensional case the spectrum consists exactly of the eigenvalues. The spectral decomposition in turn is a special case of the Schur decomposition for matrices and will not be proven here. There is an even more general decomposition that works for every bounded operator, the singular value decomposition:

Theorem 2.1.2 (Singular value decomposition)**.** *Let $A \in \mathcal{B}(\mathcal{H}, \mathcal{H}')$, then there exist unitaries $U \in \mathcal{B}(\mathcal{H})$ and $V \in \mathcal{B}(\mathcal{H}')$ and a diagonal operator $\Sigma \in \mathcal{B}(\mathcal{H}, \mathcal{H}')$ such that*

$$A = U \Sigma V = \sum_{i=1}^{r} \sigma_i u_i \langle v_i, \cdot \rangle$$

The σ_i are called the singular values of A and if A admits a spectral decompostion then they are the absolute values of the eigenvalues.

The singular value decomposition can be proven via the spectral decompositions of A^*A and AA^* which are both Hermitian and thus normal.

To use operators in applications it is often useful to apply functions to operators just as one is used to do with numbers, this is called functional calculus. For normal operators a sensible way is via the spectral decomposition: Let $f : D \subset C \to \mathbb{C}$ be a function and A be a normal operator such that $\sigma(A) \subset D$ with spectral decomposition as in theorem 2.1.1. Then $f(A)$ is defined as

$$f(A) := \sum_i f(\lambda_i) u_i \langle u_i, \cdot \rangle, \tag{2.1.9}$$

where the zero eigenvalues need to be included in the spectral decomposition. This also implies the definition of functions $f : D \subset \mathbb{R} \to \mathbb{R}$ by viewing $\mathbb{R}$ as a subset of $\mathbb{C}$. For example, the

[5] citation acctually not true oder doch ??

[6] citation

[7] meaning the corresponding matrix has diagonal form

square root $\sqrt{\cdot} : \mathbb{R}_{\geq 0} \to \mathbb{R}$ is now only defined for normal operators whose eigenvalues are all non-negative, so for positive-semidefinite operators.

There are many norms on $\mathcal{B}(\mathcal{H})$ other than the operator norm. An interesting class of norms is that of the p-norms $||\cdot||_p := \mathrm{tr}[|\cdot|^p]^{1/p}$ for $p \in [1, \infty[$. Here, $|A| = \sqrt{A^*A}$ is the absolute value of an operator A. Note that A^*A is indeed normal and positive semidefinite since for any $v \in \mathcal{H}$ it holds $\langle v, A^*Av\rangle = \langle Av, Av\rangle = ||Av||^2 \geq 0$ and hence the square root is well-defined. When A has the SVD $A = \sum_i \sigma_i |u_i\rangle\langle v_i|$ then

$$A^*A = \sum_{ij} \sigma_i^* \sigma_j |v_i\rangle\langle u_i|u_j\rangle\langle v_j| = \sum_i |\sigma_i|^2 |v_i\rangle\langle v_i|, \tag{2.1.10}$$

so that $|A|$ has eigenvalues $|\sigma_i|$. Consequently, when A has singular values $\{\sigma_i\}_i$, then

$$||A||_p = \left(\sum_i |\sigma_i|^p\right)^{1/p}. \tag{2.1.11}$$

Since the p-norm of a vector converges to the maximum of the absolute value of the entries, the operator p-norm converges to the operator norm as $p \to \infty$ and the definition $||\cdot||_\infty := ||\cdot||_{\mathrm{op}}$ is convenient. The p-norms fulfil the following relations:

Proposition 2.1.3. *For all $A, B \in \mathcal{B}(\mathcal{H})$ and $p, q, r \in [1, \infty]$:*

(i) $||A||_1 \leq \mathrm{tr}[|A|]$

(ii) For all unitaries U, V on $\mathcal{H}$: $||UAV||_p = ||A||_p$. *(Unitary invariace)*

(iii) $||AB||_p \leq ||A||_p ||B||_p$ *(Submultiplicativity)*

(iv) $1 \leq p \leq q \leq \infty$: $||A||_p \geq ||A||_q$ *(Monotonicty)*

(v) If $\frac{1}{p} + \frac{1}{q} = \frac{1}{r}$, *then* $||AB||_r \leq ||A||_p ||B||_q$. *(Hölder)*

(vi) Let $p^* := \frac{p}{p-1}$ *be the Hölder conjugate to p, then* $||A||_p := \{|\langle A, B\rangle|_{HS} \mid ||B||_{p^*} = 1\}$ *(Duality)*

Proof. (i) can be readily proved with the SVD of A. For (ii) note that the singular values are invariant under said unitaries. (iii) follows since every unitarily invariant matrix norm is submultiplicative, see chapter IV.2 in [Bha97]. The special of (iv) for $r = 1$ can be found in Corollary IV.2.6 in [Bha97]. For the general case, substitue A by $|A|^r$ and B by $|B|^r$, then

$$||AB||_r = \mathrm{tr}[|AB|^r]^{1/r} = |||A|^r |B|^r||_1^{1/r} \leq |||A|^r||_p^{1/r} |||B|^r||_q^{1/r} = ||A||_{rp} ||B||_{rp}, \tag{2.1.12}$$

where $1/p + 1/q = 1$ and thus $1/(rp) + 1/(rq) = 1/r$. Lastly, for (iv) and (vi) see IV.2.11 in [Bha97]. □

The special case $p = 1$ is called the trace-norm and $p = 2$ is called the Hilbert-Schmidt norm. The Hilbert-Schmidt norm is induced by the Hilbert-Schmidt inner product of two operators $A, B \in \mathcal{B}(\mathcal{H})$:

$$\langle A, B\rangle := \mathrm{tr}[A^*B]. \tag{2.1.13}$$

2.2 Observables, states and measurements

The subject of interest will be some quantum system. The preparation of such a system is the set of all actions that determine the probability distribution of any measurement performed on the system. The *state* then specifies the effect that this preparation has on the system. A simple example for a quantum system is the light being emitted from a source, where an experimentalist can manipulate the polarization of said light, that is vertically, horizontally or any degree in between.

It has turned out that a good way to model observable quantities of quantum systems if by bounded operators on a complex Hilbert space $\mathcal{H}$. For the sake of mathematical simplicity this

Hilbert space will be assumed to be finite-dimensional. A state on a system is a functional $\langle\cdot\rangle$ that maps bounded operators to complex numbers. It fulfills $\langle\mathbb{1}\rangle = 1$ and $\langle A^*A\rangle \geq 0$ for all operators A. The space of bounded operators is equal to the Hilbert-Schmidt space, $\mathcal{B}(\mathcal{H}) = \mathcal{B}_2(\mathcal{H})$, which is conveniently a Hilbert space. Thus, the Riesz representation theorem 2.1.1 [8]can be applied so that for every state there exists an element $\rho \in B_2(\mathcal{H})$ such that $\langle O\rangle = \mathrm{tr}[\rho^* O]$ for all $O \in \mathcal{B}_2(\mathcal{H})$. Normalization then implies $\mathrm{tr}[\rho] = 1$ and positivity implies positive semidefiniteness of ρ since for all $A \in \mathcal{B}(\mathcal{H})$:

$$\langle A^*A\rangle = \mathrm{tr}[\rho^* A^* A] = \overline{\mathrm{tr}[\rho A^* A]} = \overline{\mathrm{tr}[A\rho A^*]} \geq 0 \iff \mathrm{tr}[A\rho A^*] \geq 0 \tag{2.2.1}$$

If $\rho = B^*B \geq 0$ for some B, then $\mathrm{tr}[A\rho A^*] = \mathrm{tr}[AB^*BA^*] = \mathrm{tr}[(BA^*)^*(BA^*)] \geq 0$. $\rho \geq 0$ in turn implies $\rho = \rho^*$. These ρ are called density operators and they represent the state uniquely.

Definition 2.2.1 (Density operator)**.** A density operator on a Hilbert space $\mathcal{H}$ is an Hermitian positive semidefinite bounded operator $\rho \in \mathcal{B}(\mathcal{H})$ such that $\mathrm{tr}[\rho] = 1$. The space of all density operators is written as $\mathcal{D}(\mathcal{H})$.

To simplify notation, also the density operator directly is called the state. The space $\mathcal{D}(\mathcal{H})$ is a convex space. By the Krein-Milman theorem every compact convex subset of a vector space has extremal points. The trace condition on density operators is closed since the trace is continuous (w.r.t. the operator topology) and $\rho = \rho^* \geq 0$ are also closed conditions since $\rho \mapsto \langle v, \rho v\rangle$ and $\rho \mapsto \rho^*$ are continuous maps for every $v \in \mathcal{H}$. Hence $\mathcal{D}(\mathcal{H})$ is closed and bounded since $||\rho||_\infty \leq \mathrm{tr}[\rho] = 1$. By Heine-Borel the space of density operators is compact and consequently there are extremal points. These extremal points are called pure states.

Proposition 2.2.1. *Pure states are of the form* $\rho = \psi\langle\psi,\cdot\rangle = |\psi\rangle\langle\psi|$

Proof. Let $\rho = |\psi\rangle\langle\psi|$ be pure and assume $\rho = \lambda\rho_1 + (1-\lambda)\rho_2$. Then $\mathrm{Ker}(|\psi\rangle\langle\psi|) \subset \mathrm{Ker}(\rho_i)$ for $i \in \{1,2\}$. This implies $\mathrm{Im}(\rho_i) = \mathrm{Ker}(\rho_i)^\perp \subset \mathrm{Ker}(|\psi\rangle\langle\psi|) = \mathrm{Im}(|\psi\rangle\langle\psi|) = \mathbb{C}|\psi\rangle\langle\psi|$, so that $\rho_i = c_i|\psi\rangle\langle\psi|$ for some $c_i \in \mathbb{C}$ for all $i \in \{1,2\}$. Therefore the convex combination is trivial. Now let ρ be an extremal point and suppose $\mathrm{rk}(\rho) \geq 2$, then the spectral decompositon of ρ yiels a non-trivial convex combination of ρ. $\mathrm{tr}[\rho] = 1$ implies then $\rho = |\psi\rangle\langle\psi|$ for some $\psi \in \mathcal{H}$. □

Let the physical system be in a state $\rho \in \mathcal{B}(\mathcal{H})$. When measuring an observable O there may be multiple measurement outcomes X_α which can occure with a probability $p(X_\alpha|\rho, O) \in [0,1]$. How these measurements behave in quantum mechanics is postulated to be the following: To each measurement outcome X_α there is an *effect operator* $M(X_\alpha) \in \mathcal{B}(\mathcal{H})$ such that $p_\alpha = \mathrm{tr}[M(X_\alpha)\rho]$. The effect operators depend on the observables one wants to measure so one also writes $p(X_\alpha|\rho, M)$ for the probability. Naturally, the possible outcomes should be elements of a sigma algebra $\mathbb{B}$ which is part of a measurable space $(X, \mathbb{B})$ with a sample space X. Since $p(X_\alpha|\rho, M)$ has to be non-negative for every state it follows that $M(X_\alpha) \geq 0$: Let $\rho = |\psi\rangle\langle\psi|$ be an arbitrary pure state, then $\mathrm{tr}[M(X_\alpha)\rho] = \langle\psi, M(X_\alpha)\psi\rangle \geq 0$. Normalization of the probabilites additionally implies that for any disjoint parition of the sample space $X = \bigsqcup_\alpha X_\alpha$ it must hold $1 = \mathrm{tr}[\rho M(X)]$ for all density operators ρ. From this follows $\mathbb{1} = M(\bigsqcup_\alpha X_\alpha) = \sum_\alpha M(X_\alpha)$, where the last equality sign follows from requiring $p(\bigsqcup_\alpha X_\alpha|\rho, M) = \sum_\alpha p(X_\alpha|\rho, M)$. A set of effect operators fulfilling these conditions is called a positive operator valued measurement, in short POVM:

Definition 2.2.2 (POVM)**.** A POVM on a measurable space $(X, \mathbb{B})$ is a map $M : \mathbb{B} \to \mathcal{B}(\mathcal{H})$ such that

(i) $\forall Y \in \mathbb{B} :\ M(Y) = M(Y)^* \geq 0$

(ii) Let $\{X_k\}_k \subset \mathbb{B}$ be a disjoint union of X, then $\sum_k M(X_k) = \mathbb{1}$

A POVM M is called a projective measurement or a von Neumann measurement or a sharp POVM if $M(Y)$ is an orthogonal projection for every $Y \in \mathbb{B}$.

By the definition of a POVM there is no inherent upper bound on the number of effect operators (i.e. the size of $\mathbb{B}$) a POVM can have. One POVM effect $M(Y)$ can always be split

[8]In infinite dimensional Hilbert spaces a similar result still holds due to a duality in the operator p-spaces, where the density operator needs to be trace-class.

up into $cM(Y) =: M(Y_1)$ and $(1-c)M(Y) =: M(Y_2)$ for a $c \in]0,1[$, thus creating one more outcome. However, when implementing these measurements experimentally, usually only sharp POVMs are used, where the splitting procedure of an effect does not work anymore because of the projection condition, unless zero effects are allowed. Indeed, in the case of a sharp POVM the number of non-zero effects is upper bounded by the dimension of the Hilbert space. To show this the following lemma is helpful:

Lemma 2.2.1. *Let M be a POVM on $(X, \mathbb{B})$ and $Y, J \in \mathbb{B}$, then*

(i) If $Y \subset J$, then $M(Y) + M(J/Y) = M(J)$ and $M(J) \leq M(Y)$.

(ii) $M(J \cup Y) \leq M(Y) + M(J)$ with equality iff $Y \cap J = \emptyset$.

(iii) If M is sharp and $Y \cap J = \emptyset$, then $M(Y)M(J) = 0$.

Now for a sharp POVM M it follows $d = \mathrm{tr}(\mathbb{1}) = \sum_k \mathrm{tr}[M(X_k)] = \sum_k \mathrm{rk}(M(X_k))$, which shows that the number of non-zero effects is upper bounded by d. In particular the sigma algebra is countable in this case and it is convenient to define $M_x := M(\{x\})$. One then often also says the POVM is the set $\{M_x\}$ of the effect operators. Note that POVMs only describe the probability with which each measurement result occurs, but do not describe a possible change of the state due to the measurement. This will be introduced in the chapter on the evolution of quantum systems.

Let it be mentioned here, that in measurement one often speaks of the expectation value of an observable. For this one needs to label each measurement outcome so that the expectation is the weighted average of these labels. This holds no intrinsic value since any labelling is valid. For example each POVM $(M_x)_x$ induces an operator $O = \sum_x x M_x$ so that $\mathrm{tr}[\rho O] = \sum_x x p_x$ yiels the expectation w.r.t. to this labeling.

2.3 Majorization and Mixedness

A state ρ was defined to be pure iff $\rho = |\psi\rangle\langle\psi|$ for some $\psi \in \mathcal{H}$. From the spectral decomposition one sees that such a pure state has only one non-zero eigenvalue, namely one. On the other hand a state ρ is called maximally mixed if all eigenvalues are the same, so $\rho = \mathbb{1}/d$. Stepping back from the spectrum, operationally a state should be called more mixed than another if it can be achieved via unitary operations and a classical mixing from the second state. More formally:

Definition 2.3.1. Let $\rho, \sigma \in \mathcal{D}(\mathcal{H})$ be two density operators. ρ is said to be more mixed than σ if there exists $N \in \mathbb{N}$, a set of unitaries $\{U_i\}_{i=1}^N$ and a probability distribution $\{p_i\}_{i=1}^N$ such that $\rho = \sum_{i=1}^N p_i U_i \sigma U_i^*$.

In turns out that this is related to the concept of majorization:

Definition 2.3.2. Let λ and μ be two finite sequences of non-negative real numbers summing up to one. $\lambda^\downarrow$ means that the sequence is ordered non-increasing, same for μ. Then λ is majorized by μ if

$$\sum_i \lambda_i^\downarrow \leq \sum_i \mu_i^\downarrow.$$

A density operator ρ is majorized by another density operator σ if their spectra are.

It can be shown that ρ is more mixed than σ if and only if $\rho\$$

2.4 Composite systems and entanglement

Suppose there are two systems, each in a state. One may ask in what space the joint state of both systems lies. In classical physics this is the direct sum of the state spaces of the subsystems[9]. However, in quantum physics this is more subtle. Firstly, there needs to be a distinction whether the two systems exclude each other or not. Think of a scattering process where one Hilbert space describes particles before and the other after the scattering as an example where the systems exclude each other. In this case it is straightforward to use the direct sum to couple the Hilbert spaces of the systems. Otherwise the tensor product has to be used:

[9] check that it is true

Definition 2.4.1. Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be Hilbert spaces and $\psi \in \mathcal{H}_1$, $\phi \in \mathcal{H}_2$. Their tensor product is a map $\psi \otimes \phi : \mathcal{H}_1 \otimes \mathcal{H}_2 \to \mathbb{C}$, $(\alpha, \beta) \mapsto \langle \alpha, \psi \rangle \langle \phi, \beta \rangle$. The space $\mathcal{H}_1 \otimes \mathcal{H}_2$ is then the completion of the space of all finite linear combinations of those maps w.r.t. the inner product $\langle \psi_1 \otimes \phi_1, \psi_2 \otimes \phi_2 \rangle := \langle \psi_1, \phi_1 \rangle \langle \phi_1, \phi_2 \rangle$

With these definitions the tensor prodcut of two Hilbert spaces is again a Hilbert space. The tensor product is not directly associative or commutative but it fulfils the following related equations:

$$(U \otimes V) \otimes W \cong U \otimes (V \otimes W) \text{ and } U \otimes V \cong V \otimes U \tag{2.4.1}$$

as well as linearity

$$\left(\sum_i c_i \psi_i \right) \otimes \left(\sum_j d_j \phi_j \right) = \sum_{ij} c_i d_j (\psi_i \otimes \phi_j) \ \forall \ c_i, d_j \in \mathbb{C}. \tag{2.4.2}$$

If $\{e_i\}_{i=1}^{d_1}$ and $\{f_i\}_{f=1}^{d_2}$ are orthonormal bases for Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$, respectively, then $\{e_i \otimes f_j\}_{i,j}$ is an orthonormal basis for $\mathcal{H}_A \otimes \mathcal{H}_B$. Thus, the dimension of $\mathcal{H}_A \otimes \mathcal{H}_B$ is $d_1 \cdot d_2$. The tensor product being used to build composite systems is usually postulated like the following:

Postulate 2.4.1. Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two Hilbert spaces that correspond to two distinguishable quantum systems, which do not exclude each other. Then the joint system is described via the Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$.

[10] Any two vector spaces over the same field that have the same finite dimension are isomorphic to each other. Thus, the spaces $\mathcal{B}(\mathcal{H}_A, \mathcal{H}_B)$ and $\mathcal{H}_A \otimes \mathcal{H}_B$ are isomorphic. In this case it will prove helpful to construct this isomorphism explicitly:

Theorem 2.4.1 (Hilbert-Schmidt isomorphism)**.** *The map*

$$\mathcal{I} : \mathcal{H}_1 \otimes \mathcal{H}_2 \to \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2), \ \psi \otimes \phi \mapsto |\phi\rangle\langle\overline{\psi}|$$

is a linear bijection fulfilling $\langle \Phi, \Psi \rangle = \mathrm{tr}[\mathcal{I}(\Phi)^* \mathcal{I}(\Psi)]$ *for all* $\Psi, \Phi \in \mathcal{H}_1 \otimes \mathcal{H}_2$.

Proof. Note that $\mathcal{I}$ is linear and hence it suffices to specify $\mathcal{I}$ on the basis elements of $\mathcal{H}_1$ and $\mathcal{H}_2$, so the map stated in the theorem is well-defined. Take bases $\{e_i\}$ and $\{f_j\}$ for $\mathcal{H}_1$ and $\mathcal{H}_2$, respectively. Also note that the conjugation of a vector conjugates each coefficient, so it depends on the chosen basis. So fix the basis mentioned above, so that in particular $\overline{e}_i = e_i$. As a consequence, $\mathcal{I}(e_i \otimes f_j) = |f_j\rangle\langle e_i|$ for all i, j, so that $\mathcal{I}$ maps an orthonormal basis of $\mathcal{H}_1 \otimes \mathcal{H}_2$ to one of $\mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$. This means $\mathcal{I}$ is an isomorphism. To show the inner product equation, expand Φ and Ψ in a basis: $\Phi = \sum_{ij} c_{ij} e_i \otimes f_j$ and $\Psi = \sum_{ij} d_{ij} e_i \otimes f_j$. Then

$$\mathrm{tr}[\mathcal{I}(\Phi)^* \mathcal{I}(\Psi)] = \sum_{ijkl} \overline{c}_{ij} d_{kl} \mathrm{tr}[|f_j\rangle\langle e_i|e_k\rangle\langle f_l|] = \sum_{ij} \overline{c}_{ij} d_{ij} = \langle \Phi, \Psi \rangle. \tag{2.4.3}$$

□

[11] With this isomophism the following theorem on the decomposition of elements in a tensor product space is easy to prove:

Theorem 2.4.2 (Schmidt decomposition)**.** *Let* $\psi \in \mathcal{H}_1 \otimes \mathcal{H}_2$ *and* $\{e_k\}_k, \{f_k\}_k$ *be two ONBs for* $\mathcal{H}_1$ *and* $\mathcal{H}_2$*, respectively. Then there is* $r \in \mathbb{N}$ *and a unique vector* $(s_i > 0) \in \mathbb{R}^r$*, such that*

$$\psi = \sum_{i=1}^{r} s_i e_i \otimes f_i$$

and $||\psi||^2 = \sum_{i=1}^{r} s_i^2$.

In this context, the s_i are called Schmidt coefficients of ψ and r is its Schmidt rank.

[10] put motivation via C star algebras or/and that physicist at the time already know this without aqm

[11] before this introduce tensor prodcut of operators, $||A \otimes B||$ and $\mathrm{tr}[A \otimes B]$

Proof. missing □

Note that the Schmidt decomposition holds for arbitrary Hilbert spaces, so an operator $\rho \in \mathcal{B}(\mathcal{H}_1) \otimes \mathcal{B}(\mathcal{H}_2)$ has a Schmidt decomposition $\rho = \sum_i s_i A_i \otimes B_i$, where $\{A_i\}$ and $\{B_i\}$ are ONBs w.r.t. the Hilbert-Schmidt inner product. However, in quantum information theory the Schmidt decomposition for operators is much less frequently used than the one for pure states because of its role in entanglement. This will become more clear below when the subject is introduced.[12]

Suppose there are two measurements which are described by two POVMs M_1 and M_2 acting on two systems described by $\mathcal{H}_1$ and $\mathcal{H}_2$. For each POVM M_i there is a measurable space $(X_i, \mathbb{B}_i)$, where $i \in \{1, 2\}$. The POVM that describes both measurements on the joint system $\mathcal{H}_1 \otimes \mathcal{H}_2$ has corresponding measurable space $(X_1 \times X_2, \sigma(\mathbb{B}_1 \times \mathbb{B}_2))$. Here, $\sigma(\mathbb{B}_1 \times \mathbb{B}_2))$ is the product sigma algebra known from classical probability theory. From the postulate on composite systems it follows that the joint POVM should fulfil $M(Y_1 \times Y_2) = M_1(Y_1) \otimes M_2(Y_2)$ for all $Y_i \in \mathbb{B}_i$, $i \in \{1, 2\}$. Extending this via unions and complements [13] onto the whole product sigma algebra defines the POVM. Now, starting from a joint measurement an observer in a subsystem, say $\mathcal{H}_1$, will 'see'[14] a corresponding submeasurement on his or her subsystem. This is just the marginal distribution: Fix a joint state ρ and the joint measurement M, then the probability distribution on system 1 is

$$p_1(Y_1) = p(Y_1 \times X_2) = \mathrm{tr}[\rho M(Y_1 \times X_2)] = \mathrm{tr}[\rho M_1(Y_1) \otimes \mathbb{1}_{\mathcal{H}_2}] \tag{2.4.4}$$

In turns out that there is a unique state on $\mathcal{H}_1$ such that the distribution p_1 corresponds to measuring M_1 in this state. This state is called the partial trace of ρ over the second system and recorded in the following theorem:

Theorem 2.4.3 (Partial trace)**.** *There is a unique map*

$$\mathrm{tr}_2 : \mathcal{B}(\mathcal{H}_1 \otimes \mathcal{H}_2) \to B(\mathcal{H}_1), \ \rho \mapsto \rho_1$$

s.t. for all $Y \in \mathcal{B}(\mathcal{H}_1)$, $\mathrm{tr}[\rho(Y \otimes \mathbb{1})] = \mathrm{tr}[\mathrm{tr}_2[\rho]Y]$ *holds true. The partial trace fulfils the following properties:*

(i) tr_2 *is continuous w.r.t. the standard operator topologies.*

(ii) tr_2 *is trace-preserving.*

(iii) Let $\{b\}_b$ *be an ONB for* $\mathcal{H}_2$*. For any* $\phi, \psi \in \mathcal{H}_1$*:*

$$\langle \phi, \rho_1 \psi \rangle = \sum_b \langle \phi \otimes b, \rho(\psi \otimes b) \rangle$$

(iv) If ρ *is positive-semidefinite, so is* ρ_1*.*

(v) $\forall A \in \mathcal{B}(\mathcal{H}_1), B \in \mathcal{B})(\mathcal{H}_2) : \ \mathrm{tr}_2[A \otimes B] = \mathrm{tr}[B]A.$

Proof. missing □

From (v) also follows that the partial trace is linear and that it is cyclic in the system that gets trace out. The partial trace tr_1 over the first system is defined completely analogously (or by the fact that $\mathcal{H}_1 \otimes \mathcal{H}_2 \cong \mathcal{H}_2 \otimes \mathcal{H}_1$). When a composite system consists of more than two partite systems, the partial trace over multiple systems is defined by grouping these together, since the tensor product of Hilbert spaces is again a Hilbert space. For example, consider $\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3$, then for all bounded operators O on this joint space, it holds that $\mathrm{tr}_{12}[O] = \mathrm{tr}_1 [\mathrm{tr}_2 [O]]$.[15]

So in eq. 2.4.4 one sees that $p_1(Y_1) = \mathrm{tr}[\rho_1 M_1(Y_1)]$. This defines the state that an observer on a partite system observes, it is called the reduced state ρ_1 of ρ. In the literature, notation like ρ_1 may not mean a reduced state from a joint state ρ, but simply any state ρ_1 on $\mathcal{H}_1$. This needs to be carefully distinguished. In this work, reduced states will always be depicted by a

[12] stimmt das wirklich?
[13] check
[14] exchange
[15] prove this!!!

subindex like $\rho_1 = \mathrm{tr}_2[\rho]$ (just as defined above) and the correspondence to a subsystem may be indicated by a superindex like ρ^1.

Interestingly, the two reduced states of a bipartite pure state have the same spectrum: Let $\rho = |\psi\rangle\langle\psi|$ be the pure state in question. It has the Schmidt decomposition $\psi = \sum_i s_i e_i \otimes f_i$ so that

$$\rho = \sum_{ij} s_i s_j |e_i\rangle\langle e_j| \otimes |f_i\rangle\langle f_j|. \tag{2.4.5}$$

The reduces states then read

$$\rho_1 = \sum_i s_i^2 |e_i\rangle\langle e_i| \text{ and } \rho_2 = \sum_i s_i^2 |f_i\rangle\langle f_j|. \tag{2.4.6}$$

1617

One of the most interesting and useful features of quantum mechanics is that partite systems of a joint system can exhibit correlations that go beyond the correlations known from classical statistics. This is called entanglement: A state $\langle\cdot\rangle$ on a joint state $\mathcal{H}_1 \otimes \mathcal{H}_2$ for which $\langle A \otimes B\rangle = \langle A \otimes \mathbb{1}\rangle\langle \mathbb{1} \otimes B\rangle$ holds for all $A \in \mathcal{B}(\mathcal{H}_1)$ and $B \in \mathcal{B}(\mathcal{H}_2)$ is called a product state. In terms of density operators:

$$\mathrm{tr}[\rho A \otimes B] = \mathrm{tr}[\rho_1 A]\mathrm{tr}[\rho_2 B] = \mathrm{tr}[\rho_1 A \otimes \rho_2 B] = \mathrm{tr}[(\rho_1 \otimes \rho_2)(A \otimes B)] \tag{2.4.7}$$

From this follows that $\rho = \rho_1 \otimes \rho_2$[18]. Product states are exactly the states that do not exhibit any kind of correlation between the subsystems since the probabilites always factorize.[19]. If one is interested in states that exhibit quantum entanglement, then classical correlation is still allowed. That is, states that are convex combinations of pure states exhibit classical but no quantum mechanical correlation. These states are called separable. This is summarized in the following definition:

Definition 2.4.2. Let $\rho \in \mathcal{D}(\mathcal{H}_1 \otimes \mathcal{H}_2)$ be a bipartite density operator. The state is called a product state if $\rho = \rho_1 \otimes \rho_2$ and it is called separable if it is a convex combination of product states. A state that is not separable is called entangled.

Product states are obviously separable. The other direction is true if the state is pure, i.e. pure states are separable if and only if they are product states. The relevant direction can be proven by considering a separable state $\rho = \sum_i p_i \rho_i^1 \otimes \rho_i^2$.[20]Then by the definition of pure states being the extremal points it follows $\rho = \rho_j^1 \otimes \rho_j^2$ for some j. [21]

Entanglement is a very valuable ressource[22], so it seems sensible to be able to quantify the amount of entanglement. Locc[23] Since entanglement is a quantum phenomenom of composite systems, entanglement should not increase when performing LOCC operations on the state in question. It turns out that there are states where neither one can be transformed into the other with means of LOCC. Thus, fixing an entanglement order on the set of states via LOCC is not possible (since there would be states where neither is bigger than the other w.r.t. this order). As a result, this order is a question of definition. Here, only bipartite entanglement will be considered[24].

Definition 2.4.3 (Entanglement monotone)**.** A map $E : \mathcal{D}(\mathcal{H}_1 \otimes \mathcal{H}_2) \to \mathbb{R}$ is called an entanglement monotone if it is non-increasing under LOCC[25].

For two states $\rho, \sigma \in \mathcal{D}(\mathcal{H}_1 \otimes \mathcal{H}_2)$ one then says that ρ is more entangled than σ if $E(\rho) > E(\sigma)$. [26]Consider bipartite entanglement between two Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$, both of dimension d.

[16]check if both partiete system have to have same dimension!!!
[17]maybe put quantum coupling theorem here if used later.
[18]why???
[19]why does factorizing imply no correlation
[20]Yes, if the indices i of the classical mixing take values in $\{1, 2\}$, then this could be confused with a reduced state, but this is not meant here.
[21]maybe add monogamy proposition if needed later.
[22]put proves/quellen of this here
[23]todo
[24]maybe this changes in the future???
[25]put this into equation also
[26]give examples of entanglement monotones

It can be shown[27] that the pure state $\psi \in \mathcal{H}_1 \otimes \mathcal{H}_2$, whose Schmidt coefficients are all equal and thus $1/\sqrt{d}$, maximizes every entanglement monotone over the space of density operators. This state is called maximally entangled

Sometimes the following two additional properties are added to the entanglement monotone:

(i) $E(\rho) = 0$ if and only if ρ is separable (Faithfulness)

(ii) The entanglement monotone maps the maximally entangled state to $\log d$ (Normalization)

This would then be called an entanglement measures[28] entanglement measures and maximall mixed staes why does that not work for mixed states. The notion of distribution of singular values as an indicator for the entanglement unfortunately does not transfer to mixed states. For example, the maximally mixed state $\rho = \frac{1}{d^2}\mathbb{1}_{\mathcal{H}_1} \otimes \mathbb{1}_{\mathcal{H}_2}$ has all Schmidt coefficients equal to $1/d^2$, but is clearly a separable state since it is the tensor product of two maximally mixed states.

Apart from simplicity, one good reason for considering bipartite entanglement specifically is the fact that if two systems are maximally entangled (remember that this is well-defined now) they can not be maximally entangled with a third system as well. This is known as monogamy of entanglemnt and o formalize it, one first needs the concept of a purification.

Definition 2.4.4 (Purification)**.** Let $\rho \in \mathcal{D}(\mathcal{H})$ be a possibly mixed state. If there exists a Hilbert space $\mathcal{H}_A$ (ancilla space) and a unit vector $\psi \in \mathcal{H} \otimes \mathcal{H}_A$, so that

$$\rho = \mathrm{tr}_A[|\psi\rangle\langle\psi|],$$

then ψ is called a purification of ρ.

Indeed, for every density operator there exists a purification where the dimension of the used ancilla space is the rank of the density operator.[29]

Proposition 2.4.1 (Monogamy)**.** *If a pure state is a reduced state of a composite state, then the composite system is in a product state.*

Proof. Let $\rho = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i| \in \mathcal{D}(\mathcal{H})$ be a density operator on $\mathcal{H}$. A purification is e.g. $|\psi\rangle = \sum_i \sqrt{\lambda_i}|\lambda_i\rangle \otimes |e_i\rangle$, where $\{e_i\}_i$ is an ONB for the ancilla space $\mathcal{H}_A$. If ρ is a product state, then the purification is a product state.[30] □

Consequently, if $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ itself is a composite system, then ρ can only[31] be maximally entangled if it is a pure state, namely the maximally entangled state. Hence, the monogamy proposition implies that if $\mathcal{H}$ is coupled with any other system, then $\mathcal{H}_1 \otimes \mathcal{H}_2$ and the third system can not be entangled as well.

naimark extension missing!!!

2.5 entropy stuff maybe

[27] cite this

[28] give examples of monotones from above who are measures and who are not.

[29] prove this of course!!! and say it is related/called uhlmanns theorem and state that purifications are related via local operations

[30] uniqueness?

[31] does not follow from above

Chapter 3

Evolution of quantum systems

The change of the state may be either viewed as an instantaneous one-shot operation or as a contintuous transformation in continuous time. In the former case, the evolution is described by channels and in the latter case by a semigroup structure. Naturally, when considering a continuous evolution of the state with the time t as a contiuous parameter, then at every point t there should be a one-shot description via a channel that models the evolution from a starting time to time t. It turns out that the other way does not hold true for every channel, but for a subclass. These are called embeddable channels[1][2]

3.1 Instantaneous evolution

somewhere in this chapter do evolution of closed systems as a special case and show from no information loss pircinciple that evolution has to be unitally.

The evolution of a system can either be seen as part of the preparation of the state (Schrödinger picture) or as part of the measurement process (Heisenberg picture) (or be split up into both corresponding to the interaction picture). In general, a evolution of a state will be some map $T : \mathcal{D}(\mathcal{H}) \to \mathcal{D}(\mathcal{H}')$ which maps a density operator to the evolved density operator. Such a map must be consistent with convex mixtures of the input, so it has to be affine. Any affine map can be uniquely extended to a linear map. Since T map density operators to density operators again, it needs to map positive semidefinite operators to positive semidefinite operators (positivity of the map) and be trace-preserving[3]. If a state $\rho \in \mathcal{D}(\mathcal{H})$ is coupled with an environment $\mathcal{H}_E$, then the joint evolution on $\mathcal{H} \otimes \mathcal{H}_E$ is described by the map $T \otimes \mathbb{1}_{\mathcal{B}(\mathcal{H}_E)}$ and this needs to positive map again. If this is indeed the case for every environment, then the map is called completely positive. These properties are recorded in the following definition:

Definition 3.1.1. Let $\mathcal{L} \subset \mathcal{B}(\mathcal{H})$ be a linear subspace including the identity $\mathbb{1} \equiv \mathbb{1}_{\mathcal{H}}$. A linear map $T : \mathcal{L} \to \mathcal{B}(\mathcal{H})$ is

(i) trace-preserving if $\forall A \in \mathcal{L} :\ \mathrm{tr}[T(A)] = \mathrm{tr}[A]$

(ii) unital if $T(\mathbb{1}) = \mathbb{1}$

(iii) positve if $A \in \mathcal{L} :\ T(A^*A) \geq 0$

(iv) completely positive if $T \otimes \mathbb{1}_n$ is positive for every $n \in \mathbb{N}$ and $\mathbb{1}_n := \mathbb{1}_{\mathcal{B}(\mathbb{C}^n)}$

It suffices to check complete positivity on the maximal entangled pure state:

Proposition 3.1.1. *todo*

The splitting of the evolution into preparation and measurement mentioned above can be formalized in the following:

[1] put this later.

[2] also put later thate a trafo is reversible if and only if unitary

[3] Strictly, it only needs to preserve the trace one, but by linearity this extends to preserving arbitrary traces

Theorem 3.1.1 (Heisenberg vs. Schrödinger)**.** *Let $T : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ be a bounded linear map, then there is a unique linear map $T^* : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ such that for all $\rho, A \in \mathcal{B}(\mathcal{H})$:*

$$\mathrm{tr}[AT(\rho)] = \mathrm{tr}[T^*(A)\rho].$$

T^* is called the dual map of T and the following properties hold:

(i) T is positive $\iff$ T^* is positive

(ii) T is completely positive $\iff$ T^* is completely positive

(iii) T is trace-preserving $\iff$ T^* is unital

The interpretation is the following: Let T describe the evolution of a state ρ in the Schrödinger picture. Then the probability of measuring outcome x is given by $p_x = \mathrm{tr}[M_x T(\rho)]$. However, the same probability distribution can be achieved by evolving the POVM effects according to T^* so that $p_x = \mathrm{tr}[T^*(M_x)\rho]$.

Proof. Consider $f : \mathcal{B}(\mathcal{H}) \to \mathbb{C},\ B \mapsto \mathrm{tr}[T(B)A]$ for $A \in \mathcal{B}(\mathcal{H})$. f is linear and bounded due to Hölder's inequality. Hence, by Riesz representation there is a unique $(T^*(A))^*$ such that $f(B) = \langle (T^*(A))^*, B\rangle_{\mathrm{HS}} = \mathrm{tr}[T^*(A)B]$. T^* is linear since $\mathrm{tr}[AT(\rho)]$ is linear in A [4] Property (i) can be proven by choosing $\rho = |\psi\rangle\langle\psi|$. Same argument is valid for $T \otimes \mathbb{1}_n$, so (ii) follows. Finally, (iii) is implied by $0 = \mathrm{tr}[T(B) - B] = \mathrm{tr}[(T^*(\mathbb{1}) - \mathbb{1})B]$. □

Proposition 3.1.2. *todo here on norms of dual map etc.*

Hence, every quantum channel is an element of $\mathcal{B}(\mathcal{B}(\mathcal{H}_{\mathrm{in}}), \mathcal{B}(\mathcal{H}_{\mathrm{out}}))$.

Definition 3.1.2. A map $T : \mathcal{B}(\mathcal{H}_{\mathrm{in}}) \to \mathcal{B}(\mathcal{H}_{\mathrm{out}})$ is called a quantum channel if it is linear, trace-preserving and completely positive.

The set of all quantum channels is convex. To define extremal quantum channels one needs to check if the set of quantum channels is compact in $\mathcal{B}(\mathcal{B}(\mathcal{H}_{\mathrm{in}}), \mathcal{B}(\mathcal{H}_{\mathrm{out}}))$. This is done the easiest when using specific representations of quantum channels.

3.1.1 Representations of quantum channels

The map $\rho \mapsto \rho \otimes \sigma$, where σ is a density operator, is a valid quantum channel. As well are the maps $\mathcal{D}(\mathcal{H}_1 \otimes \mathcal{H}_2) \ni \rho \mapsto \mathrm{tr}_2[\rho]$ and $\rho \mapsto U\rho U^*$ for a unitary U. Complete positivity and trace-preserving are preserved under composition[5]. It turns out that any quantum channel is a composition of the maps above:

Theorem 3.1.2. *todo*

Proposition 3.1.3 (Ambiguity of Kraus decomposition)**.** *todo*

[6] In turns out that there is another way to represent a quantum channel uniquely by a matrix, this is the Choi matrix.

Definition 3.1.3. Let $T : \mathcal{B}(\mathcal{H}_1) \to \mathcal{B}(\mathcal{H}_2)$ be a quantum channel and $\{e_i\}_i$ be an ONB for $\mathcal{H}_1$, then define $\Omega := \sum_i e_i \otimes e_i$. The Choi matrix of T is $C_T := (id \otimes T)(|\Omega\rangle\langle\Omega|)$.

That the channel can be recoverd from its Choi matrix is written down in the following theorem:

Theorem 3.1.3 (Choi)**.** *Let $T : \mathcal{B}(\mathcal{H}_1) \to \mathcal{B}(\mathcal{H}_2)$ and $\dim(\mathcal{H}_1) =: d_1$, then*

(i) $T \mapsto C_T$ is a bijection whose inverse is determined via

$$\forall A \in \mathcal{B}(\mathcal{H}_1) \forall B \in \mathcal{B}(\mathcal{H}_2) :\ \mathrm{tr}[T(A)B] = \mathrm{tr}[C(A^T \otimes B)]$$

(ii) $C = C^* \iff (\forall A \in \mathcal{B}(\mathcal{H}_1) : T(A^*) = T(A)^*)$

[4] boundedness of T^* missing

[5] prove this

[6] put matrix representation of quantum channels here also somewhere

(iii) $C \geq 0 \iff T$ *is completely positive.*

(iv) $\mathrm{tr}_2[C] = \mathbb{1} \iff T$ *is trace-preserving.*

(v) $\mathrm{tr}_1[C] = \mathbb{1} \iff T$ *is unital.*

Proof. todo □

With this characterisation the set of all quantum channels (in Schrödinger picture) from $\mathcal{B}(\mathcal{H}_1)$ to $\mathcal{B}(\mathcal{H}_2)$ is

$$\mathcal{C}(\mathcal{H}_1, \mathcal{H}_2) := \{T \in \mathcal{B}(\mathcal{B}(\mathcal{H}_1), \mathcal{B}(\mathcal{H}_2)) | C_T = C_T^*, C_T \geq 0, \mathrm{tr}_2[C] = \mathbb{1}\} \tag{3.1.1}$$

the conditions on C_T are all closed under limits and thus the space of quantum channels is closed. It is also bounded w.r.t. the operator norm since $||T|| = 1$ for every quantum channel T. Hence, by the Krein-Milman theorem there are extremal quantum channels. [7]

Finally, also the state change due to a measurement process will be visited. [8] Let $(X, \mathbb{B})$ be the measurable space corresponding to the measurement outcomes.

Definition 3.1.4 (Quantum instrument)**.** Let $\mathcal{CP}(\mathcal{H}_1, \mathcal{H}_2)$ be the set of completely positive linear maps from $\mathcal{B}(\mathcal{H}_1)$ to $\mathcal{B}(\mathcal{H}_2)$. An instrument is a map $I : \mathbb{B} \to \mathcal{CP}(\mathcal{H}_1, \mathcal{H}_2)$, $Y \mapsto I_Y$ so that

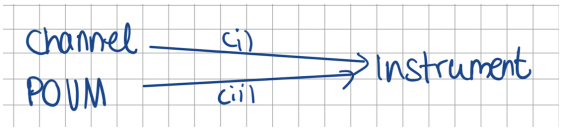
(i) I_X is trace-preserving.

(ii) Let $\{X_k\}_k \subset \mathbb{B}$ be a disjoint partition of X, i.e. $X = \bigsqcup_k X_k$, then $I_X = \sum_k I_{X_k}$ element-wise.

The interpretation of an instrument is as follows: Suppose the system is in a state ρ, then for each measurement outcome $Y \in \mathbb{B}$ the instrument provides the probability according to $\mathrm{tr}[I_Y(\rho)]$ and the post-measurement state as $I_Y(\rho)/\mathrm{tr}[I_Y(\rho)]$. Property (i) in definition 3.1.4 ensures that that getting any measurement result happens with unit probability. [9] Ignoring the (classical) measurement results[10] leads to a mixing of each of the outcomes, i.e. the post-measurement state in this case is

$$\sum_{Y \in \mathbb{B}} \mathrm{tr}[I_Y(\rho)] \frac{I_Y(\rho)}{\mathrm{tr}[I_Y(\rho)]} = \sum_{Y \in \mathbb{B}} I_Y(\rho) = I_X(\rho). \tag{3.1.2}$$

Ignoring the state change of the measurement should lead to a POVM-like description of the measurement process. Indeed, to each probability $\mathrm{tr}[I_Y(\rho)] = \mathrm{tr}[\rho I_Y^*(\mathbb{1})]$ one can assign the operator $I_Y^*(\mathbb{1})$, where then $\{I_Y^*(\mathbb{1})\}_{Y \in \mathbb{B}}$ is a POVM.[11] In this way an instrument generalizes quantum channels and POVMs. On the other hand, each channel and each POVM induce a quantum instrument themself. For the sake of simplicity, let now $X \subset \mathbb{N}$, so the measurement outcomes are countable.



(i) Given a quantum channel $T(\cdot) = \sum_{i \in X} K_i \cdot K_i^*$, then for $Y \subset \mathbb{N}$, $I_Y(\cdot) = \sum_{i \in Y} K_i \cdot K_i^*$ induces an instrument I.

(ii) Given a POVM $M = (M_i)_{i \in X}$, then for any $Y \subset X$

$$I_Y(\cdot) = \sum_{i \in Y} M_i^{1/2} \cdot M_i^{1/2} \tag{3.1.3}$$

defines an instrument, called the Lüders instrument corresponding to the POVM M.

[7] put some results on extremal channels here

[8] put this somewhere: Note that a measurment process can generally be viewed as an interaction of the system with an environment, the measurment apparatus.

[9] auf jeden fall hier noch state change durch projektive messungen einführen

[10] The read-out of the measurment will always be in some classical form.

[11] maybe add $I_{Y \cup J} = I_Y + I_J$ if disjoint

3.1.2 Spectral properties

For any linear map $T : V \to V$, where V is a complex vector space one can define the spectrum as the set

$$\sigma(T) := \{z \in \mathbb{C} : z\mathbb{1} - T \text{ is not invertible}\} \subset \mathbb{C}. \tag{3.1.4}$$

If T has an eigenvalue λ with eigenvector $v \neq 0$, then $T(v) = \lambda v \implies (\lambda\mathbb{1} - T)v = 0$ implies that $\lambda \in \sigma(T)$. An operator is invertible if and only if it is bijective. So $z\mathbb{1} - T$ not being invertible could go wrong at it not being injective or not surjective (or both). In finite dimensions, surjectiveness and injectiveness are equivalent for a linear map due to the rank theorem. So in this case $z \in \sigma(T)$ implies the existence of $x \neq y$ so that $(z\mathbb{1} - T)x = (z\mathbb{1} - T)y \implies T(y - x) = z(y - x)$. That is, the spectrum consists exactly of the eigenvalues of the map. Now think of $T \in \mathcal{B}(\mathcal{B}(\mathcal{H}))$ which can be viewed as a linear map between complex matrix spaces of the same dimension. The spectral radius, i.e. the supremum of the absolute values of the eigenvalues, will be denoted by $r(T)$. One can show that the spectrum is closed set, non-empty and $r(T) \leq ||T||$[12].

A very useful result is that of Russo and Dye[13]: In a unital C^*-algebra, the closure of the convex hull of the unitary elements is equal to the closed unit ball. In this work, the theorem will be used for the space of complex matrices, $\mathbb{C}^{n\times n}$, with the usual operator norm making it a unital C^*-algebra. The unitary matrices then form the group $U(n)$ which is already closed. Consequently, any complex matrix A with $||A|| = 1$ can be written as a convex combination of unitary matrices. This result is useful when computing norms of operators on $\mathbb{C}^{n\times n}$:

Proposition 3.1.4. *Let $T \in \mathcal{B}(\mathbb{C}^{d\times d}$ be a positive map and $d \in \mathbb{N}$, then*

3.1.3 maybe more on channels and functions on them

3.2 Evolution in continuous time

Considering the time evolution of a system, again there should be some kind of maps that transform the state or the observables. These operators will now depend on the time t as a continuous parameter. For the moment, assume the state or observable space is some Banach space X (think $\mathcal{B}(\mathcal{H}, \mathcal{H}')$). To describe the evolution operators, consider a map $T : \mathbb{R} \to \mathcal{L}(X)$, which maps each time $t \in \mathbb{R}$ to a valid evolution operator $T(t) =: T_t$. They should fulfill the property that (i) $T_t \circ T_s = T_{t+s}$ holds at all times t and s (semigroup property) and (ii) some kind of continuity, i.e. for all $x \in X$ and $s \in \mathbb{R}$: $T_t x \to T_s x$ in X as $t \to s$. The semigroup property implies that the time evolution from on point in time to another does not depend on the explicit points in time but only on the difference.[14] For convenience the gauge $T_0 = \mathbb{1}$ is used, so that time evolutions are described by strongly continuous one-parameter semigroups:

Definition 3.2.1 (C_0-semigroup)**.** A strongly continuous one-parameter semigroup or just C_0-semigroup on a Banach space X is a group homomorphism $T : (\mathbb{R}, +) \to (\mathcal{L}(X), \circ)$ such that $T(0) = \mathbb{1}$ and $\forall x \in X : \lim_{t\to 0} ||T(t)x - x|| = 0$. The semigroup is additionally said to be continuous in norm or uniformly continuous if $\lim_{t\to 0} ||T(t) - \mathbb{1}|| = 0$.

An interesting special case of this is the time evolution of a closed system that is not interacting with an environement. As described above, evolutions of a closed systems are invertible in which case the semigroup becomes a proper group.

Postulate 3.2.1. Every closed quantum system described by the Hilbert space $\mathcal{H}$ has an according Hermitian operator $H : \mathcal{H} \to \mathcal{H}$, called the Hamiltonian, that governs the time evolution of pure states $\psi \in \mathcal{H}$ by the Schrödinger equation

$$i\frac{\mathrm{d}}{\mathrm{d}t}\psi = H\psi, \tag{3.2.1}$$

where $i^2 = -1$ [15]

[12] citation
[13] citation
[14] why???
[15] expand massively on this also historically

Via the spectral decomposition this extends to mixed states $\rho \sum_i p_i |\psi\rangle\langle\psi| \in \mathcal{D}(\mathcal{H})$ through the von Neumann equation

$$i\frac{\mathrm{d}}{\mathrm{d}t}\rho = i\sum_i p_i \left(|\dot{\psi}\rangle\langle\psi| + |\psi\rangle\langle\dot{\psi}|\right) = \sum_i p_i \left(H|\psi\rangle\langle\psi| - |\psi\rangle\langle\psi|H^*\right) = H\rho - \rho H =: [H,\rho]. \quad (3.2.2)$$

Here, the derivative of $\langle\psi|$ follows from taking the Hermitian adjoint of the Schrödinger equation. It can be shown[16], that given an initial condition $\rho(t_0)$, the Neumann differential equation has the unique solution $\rho(t) = U(t-t_0)\rho_0 U(t-t_0)^*$ with $U(t) := \exp(-iHt)$. This indeed fulfills the properties of a C_0-semigroup, specifically for arbitrary $\psi \in \mathcal{H}$ it is

$$||U(t)\psi - \psi|| = \quad (3.2.3)$$

[17] Additionally, each $U(t)$ is invertible, $U(t)^{-1} = U(t)^*$. So every $U(t)$ is a unitary operator. As described above[18] one can show that every closed system is goes through a unitary time evolution, so in that sense the postulate does not clash with that statement. To furthermore validify (???) the postulate, Stone's theorem is interesting. It states that any unitary time evolution is generated by such an Hermitian operator[19]

In quantum mechanics, the Hamiltonian can additionally be used to define the energy of a closed system as the expectation value of the Hamiltonian: $E \coloneqq \mathrm{tr}[\rho H]$. It turns out that this definition converges (experimentally??) to the notion of energy known from classical physics when taking classical/macroscopic limits[20]

There is another reason why the evolution of closed systems is a good example. It presents the concept of an infinitesimal generator of the semigroup: The time-evolutions $\exp(-iHt)$ are completely determined from the Hamiltonian H.

Definition 3.2.2 (Infinitesimal generator)**.** The infinitesimal generator of a C_0-semigroup T over X is the map $A : D(A) \to X$ defined by

$$x \mapsto \lim_{t\to 0} \frac{T(t)x - x}{t}$$

and $D(A) := \{x \in X |$ the limit above exists$\}$

For the example $U_t = \exp(-iHt)$ the generator is

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \exp(-iHt) = -iH \quad (3.2.4)$$

It turns out that uniformly continuous C_0-semigroups have bounded generators and their domain is the whole space, $D(A) = X$.[21] In the quantum evolution setting the space X is fixed to that of the sates, i.e. $X = \mathcal{D}(\mathcal{H})$, and every element of the semigroup should additionally be a quantum channel. Such groups are called quantum dynamical semigroups (QDSs). In this work, only QDSs with bounded generators will be considered [22], so e.g. norm continuous QDSs. These bounded genertors will be called $\mathcal{L}$

In this work, only

[16] do this!!!
[17] prove this
[18] do this
[19] citation
[20] citation!!!
[21] citation!
[22] give examples other than schrödinger equation

Chapter 4

The quantum Zeno limit theorem

Consider a QDS $\{\exp(\mathcal{L}t) : \mathcal{D}(\mathcal{H}) \to \mathcal{D}(\mathcal{H})\}_{t\in R}$ where $\mathcal{L} : \mathcal{D}(\mathcal{H}) \to \mathcal{D}(\mathcal{H})$ is the bounded generator. Then up to a initial condition the time evolution is governed by the differential equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho(t) = \mathcal{L}(\rho(t)) \tag{4.0.1}$$

The easiest example is when $\mathcal{L} = -i[H, \cdot]$, which generates the von Neumann evolution. The idea of the Zeno effect is that this time evolution is intercepted by a projective measurement, say at times $t_1, t_2, t_3, \cdot$, and to perform these measurements with increasing frequency. So the time Δt between two measurements goes to zero. Intuitively, each measurment projects the state onto a subspace of the system and if each there if there is no time in between measurements, the evolution should be frozen onto such a subsystem. Mathematically, it seems interesting if this limit procedure yields another, possibly different, time evolution in the end. It turns out that, under some conditions on the measurement, it does. Even more, not only projective measurements are possible but also other quantum operations who have an appropriate eigenspace.

Definition 4.0.1. A quantum operation is a completely positive linear map $\mathcal{D}(\mathcal{H}) \to \mathcal{D}(\mathcal{H})$ that does not increase the trace.

[1] An example for a quantum operation is a projective measurement $\rho \mapsto P\rho P/\mathrm{tr}[P\rho]$ where $P = P^* = P^2$ or $\rho \mapsto P\rho P + (\mathbb{1} - P)H(1 - P)$. An operation that does not come from a projective measurement is e.g. the amplitude damping channel

$$\rho \mapsto \begin{pmatrix} \rho_{00} + \gamma\rho_{11} & \sqrt{1-\gamma}\rho_{01} \\ \sqrt{1-\gamma}\rho_{10} & (1-\gamma)\rho_{11} \end{pmatrix} \tag{4.0.2}$$

For the existence of the Zeno limit the following assumptions are needed: Let $M \in \mathcal{B}(\mathcal{B}(\mathcal{H}))$ be a quantum operation and $\mathbb{D}_\delta := \{z \in \mathbb{C} || z| \leq \delta\}$, then assume

$$\begin{aligned} 1 &\in \sigma(M) \\ \exists \delta < 1 : \sigma(M) &\subset \{1\} \cup \mathbb{D}_\delta \end{aligned} \tag{4.0.3}$$

where $\sigma(M)$ is the spectrum of M which consists of all eigenvalues of M in the case of finite Hilbert spaces that was fixed in the beginning.

In general, such a quantum operation will freeze the evolution in some eigenspace. Not all quantum operations are diagonalizable [2] so that the notions of eigenspace (geometric and generalized) do not necessarily coincide. In turns out that the more important one for the Zeno freezing is the generalized eigenspace

$$G_\lambda = \bigcup_{k\geq 1} \mathrm{Ker}((T - \lambda\mathbb{1})^k), \tag{4.0.4}$$

where λ is the corresponding eigenvalue. Now, assume the corresponding eigenvectors are v_i, then $\sum_i |v_i\rangle\langle v_i|$ is the projector onto the geometric eigenspace. For the projector onto the generalized

[1] say about relevance of quantum operations bc these do not map states to states???

[2] cite

eigenspace, the Riezs projector is needed. To define it, first note that when $f : G \subset \mathbb{C} \to \mathcal{A}$ is a continuous function from an open set G to a Banach algebra $\mathcal{A}$ (here always complex matrices), then the Bochner integral $\int_\gamma f(z)\,\mathrm{d}z$ is well-defined[3], where $\gamma : [a, b] \subset \mathbb{R} \to \mathbb{C}$ is a path of finite length. When $\mathcal{A} = \mathbb{C}^{n\times n}$, then the Bochner integral is simply the entrywise contour integration in $\mathbb{C}$.[4]

The resolvent mapping $R_\lambda = (\lambda\mathbb{1} - T)^{-1} : \rho(T) \to \mathcal{A}$ is holomorphic on the resolvent $\rho(T)$ and hence continuous[5].[6]As a consequence, the following is well-defined.

Definition 4.0.2. Let $\mathcal{B}$ be a finite-dimensional complex Banach space, e.g. $\mathcal{A} = \mathbb{C}^{n\times n}$. Let T be an operator on $\mathcal{B}$ and $\Gamma : [0, 1] \to \Gamma([0, 1]) \subset \rho(T)$ be a contour of finite length with positive orientation, then

$$\mathcal{B} \leftarrow \mathcal{B} : P_\Gamma = \frac{1}{2\pi i} \oint_\Gamma (z\mathbb{1} - T)^{-1}\,\mathrm{d}z \tag{4.0.5}$$

is called the Riesz projection operator.

To show that this is actually a projection, realize that $R_z - R_w = (w - z)R_zR_w$ since

$$\begin{aligned} (z\mathbb{1} - T)(\mathbb{1} - (z - w)(z - T)^{-1}) &= w\mathbb{1} - T \\ \implies R_z^{-1}(\mathbb{1} - (z - w)R_z) &= R_z^{-1} - (z - w)\mathbb{1} = R_w^{-1} \\ \implies (R_z^{-1} - R_w^{-1})^{-1} &= R_zR_w(R_w - R_z)^{-1} = (z - w)^{-1}\mathbb{1}. \end{aligned} \tag{4.0.6}$$

Now the square of the Riesz projector is [7]

$$\begin{aligned} P_\Gamma^2 &= \frac{1}{(2\pi i)^2} \oint_\Gamma \left(R_z \oint_\Gamma R_w\,\mathrm{d}w \right)\,\mathrm{d}z = \frac{1}{(2\pi i)^2} \oint_\Gamma \oint_\Gamma (w - z)^{-1}(R_z - R_w)\,\mathrm{d}w\,\mathrm{d}z \\ &= \frac{1}{(2\pi i)^2} \int_\Gamma R_z \int_\Gamma (w - z)^{-1}\,\mathrm{d}w\,\mathrm{d}z - \frac{1}{(2\pi i)^2} \end{aligned} \tag{4.0.7}$$

[8] Any other valid quantum operation with the same spectral projector for eigenvalue 1 yields the same Zeno dynamics and could be easier to implement experimentally explain why only non-selective operations allowed!!! [9]

[3] cite

[4] cite

[5] For the resolvent operator to actually lie in the Banach algebra, it needs to be bounded which is equivalent to T being closed. But, again, in the finite-dimensional case this is all superfluous.

[6] cite

[7] can you change order of integration???

[8] finite dimensional zeno case in [Bur+20] thm 1, möbus wolf paper [MW19] and same in [BZ18]

[9] why only non-selective

3. Check the quantum-operation properties

(a) Complete positivity (CP)

Each map $X \mapsto \Pi_\alpha X \Pi_\alpha$ is completely positive because it has a single Kraus operator $K_\alpha = \Pi_\alpha$
A convex sum of CP maps is CP, so M is completely positive.

(b) Trace preservation

For any density operator ρ,

$$\mathrm{tr}[M(\rho)] = \sum_\alpha \mathrm{tr}(\Pi_\alpha \rho \Pi_\alpha) = \mathrm{tr}\left(\rho \sum_\alpha \Pi_\alpha\right) = \mathrm{tr}(\rho),$$

so M is trace-preserving.
Hence M is **CPTP**, i.e. a bona fide *quantum channel* (and thus a valid quantum operation).

---

4. Check the spectral condition

Let's find the spectrum of M on $\mathcal{B}(\mathcal{H})$:

$$M(X) = \sum_\alpha \Pi_\alpha X \Pi_\alpha.$$

Every X decomposes as

$$X = \sum_{\alpha,\beta} \Pi_\alpha X \Pi_\beta.$$

Then

$$M(\Pi_\alpha X \Pi_\beta) = \begin{cases} \Pi_\alpha X \Pi_\alpha, & \text{if } \alpha = \beta, \\ 0, & \text{if } \alpha \neq \beta. \end{cases}$$

Hence:

- all *diagonal blocks* $\Pi_\alpha X \Pi_\alpha$ are eigenoperators with eigenvalue 1,
- all *off-diagonal blocks* $\Pi_\alpha X \Pi_\beta$ ($\alpha \neq \beta$) are eigenoperators with eigenvalue 0.

Thus:

$$\mathrm{spec}(M) = \{0, 1\}.$$

✅ This fits the spectral condition with $\delta < 1$: we can take $\delta = \frac{1}{2}$. sav.

Figure 4.1: pinching is a valid quantum operation

Two channels M_1, M_2 give the same Zeno dynamics **iff** their spectral projectors onto the peripheral eigenvalue 1 coincide, $P_{M_1} = P_{M_2}$, and both obey the spectral condition (no Jordan blocks at 1 and the rest of the spectrum $\subset \{|z| < 1\}$).

How to build many such M with the same P

Let E be the conditional expectation (pinching) onto the block-diagonal algebra
$\mathcal{A} = \bigoplus_\alpha \mathcal{B}(\Pi_\alpha \mathcal{H})$ (so $E(X) = \sum_\alpha \Pi_\alpha X \Pi_\alpha$).
Any CPTP map of the form

$$M \;=\; E + C, \quad \text{with} \quad EC = CE = 0, \;\; C(\mathcal{A}) = 0, \;\; \|C\| < 1,$$

has spectrum $\{1\} \cup \mathrm{spec}(C) \subset \{1\} \cup \{|z| < 1\}$ and **the same** projector $P = E$. Hence it yields the **same Zeno limit**.

Concrete examples (all share the same $P = E$)

- **Partial dephasing** (two-projector case P, Q):
  $M_\gamma(X) = PXP + QXQ + \gamma(PXQ + QXP)$ with $|\gamma| < 1$.
  Spectrum $\{1, \gamma\}$; Zeno limit unchanged.
- **Random phase twirling** over the block projectors:
  $M(X) = \int U(\vec{\theta}) X U(\vec{\theta})^\dagger \, d\mu(\vec{\theta})$ with
  $U(\vec{\theta}) = \sum_\alpha e^{i\theta_\alpha} \Pi_\alpha$.
  Any nontrivial distribution kills off-diagonals in expectation (strict contraction), same P.
- **"Pinch + damping" on off-diagonals:**
  Decompose $X = \sum_\alpha X_{\alpha\alpha} + \sum_{\alpha \neq \beta} X_{\alpha\beta}$ and set
  $M(X) = \sum_\alpha X_{\alpha\alpha} + \sum_{\alpha \neq \beta} \lambda_{\alpha\beta} X_{\alpha\beta}$ with $|\lambda_{\alpha\beta}| < 1$.
  Again $P = E$.

Figure 4.2: building a quantum operation with the same zeno dynamics

Chapter 5

Trade-offs

Chapter 6

Information/Correlation bounds

6.1 Lieb-Robinson

lieb-robinson bound cannot increase with zeno effect:

General LR bound (finite volume): For $A \in \mathcal{A}_X$, $B \in \mathcal{A}_Y$ with $X \cap Y = \varnothing$,

$$\|[\tau_t^\Lambda(A), B]\| \;\le\; 2\,\|A\|\,\|B\|\,\big(e^{v_F(\Phi)\,|t|}-1\big)\,\min_{x\in X,\,y\in Y}\frac{1}{F(d(x,y))},$$

with LR "velocity" (more precisely, the exponential rate)

$$v_F(\Phi) \;:=\; 2\,C_F\,\|\Phi\|_F.$$

(Variants with an explicit distance factor $e^{-\mu d(X,Y)}$ come from choosing $F(r) = e^{-\mu r}(1+r)^{-(D+1)}$, etc.; the conclusion below is unchanged.)

Zeno pinching on Z

Let $Z \subset \Gamma$ and let $\mathcal{E}_Z : \mathcal{A} \to \mathcal{A}$ be a **local conditional expectation** acting nontrivially only on Z (pinching): there are orthogonal projections $\{P_j\} \subset \mathcal{A}_Z$ with $\sum_j P_j = \mathbf{1}_Z$ such that

$$\mathcal{E}_Z(X) = \sum_j P_j X P_j \qquad \text{and} \qquad \mathcal{E}_Z|_{\mathcal{A}_{Z^c}} = \mathrm{id}.$$

Key properties (all standard): $\mathcal{E}_Z$ is unital, completely positive, idempotent, **contractive** ($\|\mathcal{E}_Z(X)\| \le \|X\|$), and **support non-expanding** ($\mathrm{supp}(\mathcal{E}_Z(X)) \subseteq \mathrm{supp}(X)$).

Define the **pinched interaction**

$$\Phi^{(Z)}(X) := \mathcal{E}_Z\big(\Phi(X)\big), \qquad H_\Lambda^{(Z)} := \sum_{X\subset\Lambda} \Phi^{(Z)}(X).$$

This is exactly the generator you get from the continuous Zeno limit for measurements/pinchings on Z.

Monotonicity of LR constants under local pinching

Lemma (interaction norm does not increase).
$\|\Phi^{(Z)}\|_F \le \|\Phi\|_F$.

Proof. For any $x \in \Gamma$,

$$\sum_{X\ni x} \frac{\|\Phi^{(Z)}(X)\|}{\min_{y\in X} F(d(x,y))} \;\le\; \sum_{X\ni x} \frac{\|\Phi(X)\|}{\min_{y\in X} F(d(x,y))}$$

by contractivity of $\mathcal{E}_Z$ and preservation of supports. Taking the supremum over x gives the claim. □

Theorem (LR velocity cannot increase under local pinching).
Let $v_F(\Phi) = 2C_F\|\Phi\|_F$ and $v_F(\Phi^{(Z)}) = 2C_F\|\Phi^{(Z)}\|_F$. Then

$$v_F(\Phi^{(Z)}) \;\le\; v_F(\Phi).$$

Proof. Immediate from the lemma and the general L ↓ ɔund applied to $\Phi^{(Z)}$. □

Figure 6.1: lr velocity

Chapter 7

Many body Hamiltonians

mention anderson tight binding model, spin glasss, heisenberg, xxx, xy etc

A crucial point when considering a multi-particle systems is that of distinguishability. It turns out that physics does not let one label particles/states that have exactly the same properties[1]. For example, think of two particles with exactly the same mass, charge, spin and so on.

7.1 Second quantization

As already mentioned, to each closed system there is an operator, called the Hamiltonian, that describes the time evolution. Historically, the Hamiltonian was gained from the classical energy function E by a procedure known as quantization. Quantization, however, is more of a historically useful tool and not rigorous. Today, the Hamiltonian is more of a modelling parameter for the experiment, where a suitable Hamiltonian must be chosen to describe the experiment accordingly.

When considering multiple particles the tensor product notation of states gets a bit messy and

7.2 Quadratic Hamiltonians

7.3 trafos

the σ^+ and σ^- operator fulfil the anti-commutation relation $\{\sigma^+, \sigma^-\} = \mathbb{1}$. When there is a lattice this becomes $\{\sigma_j^+, \sigma_j^-\} = \mathbb{1}$ at a fixed site j. To view the σ^+ and σ_- as annihilation or creation operators, they still need to fulfil the anti-commutation relation on different sites, i.e. $\{\sigma_j^+, \sigma_l^-\} = 0$ whenever $j \neq l$.

$$\sigma_j^- \mapsto f_j = \exp(-i\pi \sum_{k<j} \sigma_k^+ \sigma_k^-)\sigma_j^- = \prod_{k<j} \exp(-i\pi \sigma_k^+ \sigma_k^-)\sigma_j^- \tag{7.3.1}$$

note that then

$$\exp(-i\pi \sum_{k<j} \sigma_k^+ \sigma_k^-) = \mathbb{1} - 2\sigma_k^+ \sigma_k^- = |0\rangle\langle 0| - |1\rangle\langle 1| \tag{7.3.2}$$

It follows $n_j \coloneqq f_j^* f_j = \sigma_j^+ \sigma_j^-$ and $\{\exp(-i\pi n_j), f_j\} = \{\exp(+i\pi n_j), f_j^*\} = 0$. Finally also the anti-commutation relations are fulfilled: $\{f_j^*, f_l\} = \delta_{jl}$ and $\{f_j, f_l\} = \{f_j^*, f_l^*\} = 0$

The XX-Hamiltonian $H = \sum_i J_i(\sigma_x^i \sigma_x^{i+1} + \sigma_y^i \sigma_y^{i+1})$ is mapped the following:

$$\sigma_x^i \sigma_x^{i+1} + \sigma_y^i \sigma_y^{i+1} = \frac{1}{4}[(\sigma_i^+ + \sigma_i^-)(\sigma_{i+1}^+ + \sigma_{i+1}^-) \mapsto \tag{7.3.3}$$

[1] citation needed

Chapter 8

tries without classification yet

8.1 General Setup

For the sake of simplicity, it is natural to start by considering closed systems that are governed by a time-independent Hamiltonian H generating the group $(\exp(-iHt))_{t\geq 0}$. The quantum operation will be a non-selective projective measurement $(P_\alpha)_{\alpha\in\mathcal{I}}$, where $\mathcal{I}$ is a finite index set. The state change due to this measurment is $\rho \mapsto T(\rho) := \sum_\alpha P_\alpha \rho P_\alpha$. The general idea of this section is to consider a multipartite Hilbert space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_Z$, where A and B are the systems of primary interest and Z (for Zeno) is an ancillary system. Think of A and B as being spatially separated for example and system Z between them. Nevertheless, the order of the tensor product is $\mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_Z$ for the sake of convenience. Now all systems, A, B and Z, may themselves consist of multiple single-particle Hilbert spaces, which will be assumed to have the same dimension. As an example, think of N particles and the Hilbert spaces model their spin, that is $\mathbb{C}^2$. Then particle 1 and particle N are the particles in system A and B, respecitvely, and between them are $N-2$ spins. The whole system is assumed to be in a starting state $\rho(0) \in \mathcal{D}(\mathcal{H})$. Intuitively, if the initial state is separable and the Hamiltonian does not include interaction between A and B, neither direct nor indirect through Z, then the state $\rho(t) = \exp(-iHt)\rho(0)\exp(iHt)$ stays separable. To formalize this, the Zeno subspace will be split up into three subspaces $Z_A, Z_B, Z_0 \subseteq Z = Z_A \otimes Z_B \otimes Z_0$. These spaces are chosen, such that particles in $Z_{A(B)}$ have non-zero interaction with those from $A(B)$ and particles in Z_0 have no non-zero interaction to particles outside of Z_0[1]. Consequently the Hamiltonian is of the form[2]

$$H = H_{AZ_A} + H_{BZ_B} + H_{Z_0}. \tag{8.1.1}$$

Such a Hamiltonian will be called non-bridging between A and B. Note that some of the Zeno subspaces may be empty.[3] If the initial state is a product state w.r.t. A, B, $\rho = \rho_{AZ} \otimes \rho_{BZ}$ then $\rho(t)$ is not entangled for all times t: Consider

$$\exp(-iHt) = \exp(-iH_{AZ_A}t)\exp(-iH_{BZ_B}t)\exp(-iH_{Z_0}t), \tag{8.1.2}$$

hence

$$\rho(t) = \exp(-iH_{AZ_A}t)\rho_{AZ} \tag{8.1.3}$$

The general idea is to perform the Zeno measurements on the Zeno subsystem without interacting with systems A or B and creating entanglement between the two. This is known as measurement-induced entanglement although there the measurement are usually selectice[4]. The Zeno dynamics is[5]

$$\tilde{\rho}(t) := \exp(-iT(H)t)T(\rho(0))\exp(iT(H)t) \tag{8.1.4}$$

where Ψ_t is a quantum channel describing the whole time evolution.

Since the map T seems of great importance, it is worthwile to investigate it a bit closer.

[1] formalize this (maybe earlier with summable potentials)
[2] identities on the other spaces are omitted and would need some weird reordering???
[3] show it does not create entanglement
[4] cite!!! and mention entanglement swapping and repeaters
[5] cite from above

Definition 8.1.1 (Pinching)**.** Let $\{P_\alpha\}_\alpha$ be a set of mutually orthogonal projections fulfilling $\sum_\alpha P_\alpha = \mathbb{1}$, then the map $X \mapsto \sum_\alpha P_\alpha X P_\alpha$ is called a pinching.

Note that the map T from above is such a pinching coming from the properties of a projective measurement[6] A pinching channel is a c.p. map that is idempotent and unital[7]. One property of a pinching T is that $T(\rho)$ is majorized by ρ whenever ρ is a density operator[8], meaning that the pinched operator is more mixed and its eigenvalues are more random in this sense, see [9]. Additionally, a pinching can not increase the operator norm: $||T(X)|| \leq ||X||$, see [Bha97] IV.2.12[10].

(4.4 in [Car10], pinching an operator B wrt the spectral projectors of A gives the operator that only keeps the eigenspaces of A and all off-diagonal blocks are set to zero.

in [Tom16] you find pinching inequality $T(H) = \sum_{i=1}^m P_i H P_i$ dann ist $A \geq T(A) \geq A/m$ for all A and if f is operator concave then $f(T(A)) \geq T(f(A))$ (thm 5.2.1 in [Bha97]). additionally ex. 10.6.6 in [Bha97]: $S(\mathcal{P}(A)||\mathcal{P}(B)) \leq S(A||B)$ and also in [Tom16]:

(in [Bha97] problem II.5.6 there is fischers inequality for pinching $\mathcal{P}$: $\det A \leq \det \mathcal{P}(A)$)

Let $\mathcal{P} = \sum_i P_i \cdot P_i$ be a pinching then $\mathcal{P}(|\psi\rangle\langle\psi|$ is pure iff there exists i s.t. $\psi \in \text{range}(P_i)$ [11]

In quantum physics, dephasing is a very important concept: If a state is written in a superposition $|\psi\rangle(t) = \sum_j c_j(t)|j\rangle$, the localization or dispersion of the state is governed by the phases between the coefficients $c_j(t) \in \mathbb{C}$. They may constructivly or desctructivly interfere at some points and times. The process of randomizing these phases between the coefficients is called dephasing, for example by external influence from the environment. A pinching channel is just such a dephasing[12]

If $P_\alpha = \mathbb{1}$ and an effect from a projective measurement, then every other effect is the zero operator[13]. As a result, the corresponding pinching map is just the identitiy mapping and the Zeno dynamics do not differ from the intial ones.

The analytical search for necessary or sufficient conditions for the effects P_α does not seem to be easy since in general detecting entanglement is not possibe efficiently [14]. If the state ${}_Z[\tilde{\rho}(t)]$ is supposed to be entangled, then (a much too strong) condition is to demand that the map $\Psi_t(\cdot) :=_Z [\exp(-iT(H)t)T(\cdot)\exp(-iT(H)t)]$ is at least not entanglement-breaking [15] since then it would map every state to a separable state. On this necessary condition to create entanglement, there is the following proposition:

Lemma 8.1.1. *Suppose all the $P_\alpha = |\alpha\rangle\langle\alpha|$ for some suitably chosen ONB $|\alpha\rangle$ of Z, then*

$$\Psi_t(\rho) = \sum_{\alpha : p_\alpha > 0} p_\alpha U_t^{(\alpha)} \sigma_\alpha U_t^{(\alpha)*}$$

, where p_α, $U_t^{(\alpha)}$ and σ_α can be found below.

Proof. and X is an arbitrary bounded operator on $\mathcal{H}$, then

$$T(X) = \sum_\alpha (\mathbb{1}_{AB} \otimes |\alpha\rangle\langle\alpha|) X (\mathbb{1}_{AB} \otimes |\alpha\rangle\langle\alpha|) f = \sum_\alpha X_{AB}^{(\alpha)} \otimes |\alpha\rangle\langle\alpha|, \tag{8.1.5}$$

where $X^{(\alpha)} := \langle\alpha|X|\alpha\rangle \in \mathcal{B}(\mathcal{H}_{AB})$. Consequently, $T(H) = \sum_\alpha H^{(\alpha)} \otimes |\alpha\rangle\langle\alpha|$. Now consider the

[6] prove this above.
[7] prove this
[8] introduce majoriaztion above
[9] citation !!!
[10] Actually, this holds true for every unitarily invariant norm.
[11] maybe use this later
[12] prove this 'kills cohrence between blocks'
[13] prove this
[14] elaborate
[15] introduce this above

propagator $\exp(-iT(H)t)$:

$$\begin{aligned}\exp(-iT(H)t) &= \sum_{n\geq 0}\frac{(-it)^n}{n!}\left(\sum_\alpha H^{(\alpha)}\otimes|\alpha\rangle\langle\alpha|\right)^n \\ &= \sum_{n\geq 0}\frac{(-it)^n}{n!}\sum_\alpha (H^{(\alpha)})^n\otimes|\alpha\rangle\langle\alpha| \\ &= \sum_\alpha \underbrace{\exp(-iH^{(\alpha)}t)}_{=:U_t^{(\alpha)}}\otimes|\alpha\rangle\langle\alpha|\end{aligned} \tag{8.1.6}$$

Now onto computing the partial trace over Z of the Zeno-evolved state. Remember that the act only on $\mathcal{H}_{AB}$, such that

$$\begin{aligned}\tilde{\rho}(t) &= \sum_{\alpha,\beta,\gamma}(U_t^{(\alpha)}\otimes|\alpha\rangle\langle\alpha|)(|\beta\rangle\langle\beta|\rho(0)|\beta\rangle\langle\beta|)(U_t^{(\gamma)*}\otimes|\gamma\rangle\langle\gamma|) \\ &= \sum_\alpha U_t^{(\alpha)}\rho(0)^{(\alpha)}U_t^{(\alpha)*}\otimes|\alpha\rangle\langle\alpha|,\end{aligned} \tag{8.1.7}$$

where for arbitrary $X\in\mathcal{B}(\mathcal{H})$ it was defined $X^{(\alpha)} = (\mathbb{1}_{AB}\otimes\langle\alpha|)X(\mathbb{1}_{AB}\otimes|\alpha\rangle)$[16]. Taking the partial trace over Z yields[17]

$$\Psi_t(\rho(0)) = \sum_\alpha U_t^{(\alpha)}\rho(0)^{(\alpha)}U_t^{(\alpha)*} \tag{8.1.8}$$

It will prove useful to write this as a convex mixture with

$$p_\alpha := \mathrm{tr}_{AB}[\rho_{\alpha\alpha}] = \langle\alpha|\mathrm{tr}_{AB}[\rho]|\alpha\rangle = \langle\alpha|\rho_Z|\alpha\rangle = \mathrm{tr}[P_\alpha\rho] \tag{8.1.9}$$

and if $p_\alpha > 0$, then $\sigma_\alpha := \frac{1}{p_\alpha}\rho_{\alpha\alpha}$
so that

$$\Psi_t(\rho) = \sum_{\alpha:p_\alpha>0} p_\alpha U_t^{(\alpha)}\sigma_\alpha U_t^{(\alpha)*} \tag{8.1.10}$$

and p_α are probabilities and σ_α is a valid density operator. □

If the P_α are not all rank-one (remember that their ranks need to add up to the dimension of $\mathcal{H}_Z$), then

Lemma 8.1.2. *Let the P_α be a projective measurement, then*

$$\Psi_t(\rho) = \sum_{\alpha:p_\alpha>0} p_\alpha U_t^{(\alpha)}\sigma_\alpha U_t^{(\alpha)*}$$

, where p_α, $U_t^{(\alpha)}$ and σ_α can be found below.

Proof. and X is an arbitrary bounded operator on $\mathcal{H}$, then

$$T(X) = \sum_\alpha(\mathbb{1}_{AB}\otimes P_\alpha)X(\mathbb{1}_{AB}\otimes P_\alpha) = \sum_\alpha X_{AB}^{(\alpha)}\otimes|\alpha\rangle\langle\alpha|, \tag{8.1.11}$$

where $X^{(\alpha)} := \langle\alpha|X|\alpha\rangle\in\mathcal{B}(\mathcal{H}_{AB})$. Consequently, $T(H) = \sum_\alpha H^{(\alpha)}\otimes|\alpha\rangle\langle\alpha|$. Now consider the propagator $\exp(-iT(H)t)$:

$$\begin{aligned}\exp(-iT(H)t) &= \sum_{n\geq 0}\frac{(-it)^n}{n!}\left(\sum_\alpha H^{(\alpha)}\otimes|\alpha\rangle\langle\alpha|\right)^n \\ &= \sum_{n\geq 0}\frac{(-it)^n}{n!}\sum_\alpha (H^{(\alpha)})^n\otimes|\alpha\rangle\langle\alpha| \\ &= \sum_\alpha \underbrace{\exp(-iH^{(\alpha)}t)}_{=:U_t^{(\alpha)}}\otimes|\alpha\rangle\langle\alpha|\end{aligned} \tag{8.1.12}$$

[16] defined above already??? okay write all for inital state ρ and $\rho_{\alpha\alpha}$
[17] defined above???

Now onto computing the partial trace over Z of the Zeno-evolved state. Remember that the act only on $\mathcal{H}_{AB}$, such that

$$\begin{aligned}\tilde{\rho}(t) &= \sum_{\alpha,\beta,\gamma} (U_t^{(\alpha)} \otimes |\alpha\rangle\langle\alpha|)(|\beta\rangle\langle\beta|\rho(0)|\beta\rangle\langle\beta|)(U_t^{(\gamma)*} \otimes |\gamma\rangle\langle\gamma|) \\ &= \sum_{\alpha} U_t^{(\alpha)} \rho(0)^{(\alpha)} U_t^{(\alpha)*} \otimes |\alpha\rangle\langle\alpha|,\end{aligned} \tag{8.1.13}$$

where for arbitrary $X \in \mathcal{B}(\mathcal{H})$ it was defined $X^{(\alpha)} = (\mathbb{1}_{AB} \otimes \langle\alpha|)X(\mathbb{1}_{AB} \otimes |\alpha\rangle)$[18]. Taking the partial trace over Z yields[19]

$$\Psi_t(\rho(0)) = \sum_{\alpha} U_t^{(\alpha)} \rho(0)^{(\alpha)} U_t^{(\alpha)*} \tag{8.1.14}$$

It will prove useful to write this as a convex mixture with

$$p_\alpha := \mathrm{tr}_{AB}[\rho_{\alpha\alpha}] = \langle\alpha|\mathrm{tr}_{AB}[\rho]|\alpha\rangle = \langle\alpha|\rho_Z|\alpha\rangle = \mathrm{tr}[P_\alpha \rho] \tag{8.1.15}$$

and if $p_\alpha > 0$, then $\sigma_\alpha := \frac{1}{p_\alpha}\rho_{\alpha\alpha}$
so that

$$\Psi_t(\rho) = \sum_{\alpha : p_\alpha > 0} p_\alpha U_t^{(\alpha)} \sigma_\alpha U_t^{(\alpha)*} \tag{8.1.16}$$

and p_α are probabilities and σ_α is a valid density operator. □

[No product measurements] Let H be the non-bridging Hamiltonian from [20] and let $\{P_\alpha^{(A)} \otimes P_\alpha^{(A)} \otimes P_\alpha^{(0)}\}_\alpha$ be a projective measurement on $\mathcal{H}_Z$ and $P_{\alpha^{(A,B,0)}} \in \mathcal{B}(\mathcal{H}_{Z_{A,B,0}})$ for all α.

8.2 conditions for creating entanglement

note that sp conbdition 1 is not in contradiction to my numerics since $H^(\alpha)$ factorizing is not equivalent to H factorizing, so you could have factorizing H where $H^(\alpha)$ is not (give example!!!)

8.3 see MSP: transfer matrices for rigorous results for thermodynamic properties

8.4 numerics

An simple first example of a many-body quantum system is that of 4 particles which have a 2-dimensional (pure) state space, so qubits. The particles are labeled 1 2 3 4 and there is a Heisenberg type interaction between 1 and 2 and between 3 and 4, more specifically the two-particle interactions are XX, that is, the Hamiltonian reads $H = H_{12} + H_{34}$ with $H_{12} = \sigma_1^x \otimes \sigma_2^x + \sigma_1^y \otimes \sigma_2^y$ and same for H_{34}. So, a simple time evolution with this Hamiltonian will not create any correlation between particles 1 and 4. The idea now is to measure particles 2 and 3 without using any of the measurement results, so that the post-measurement state is $T(\rho) = P\rho P + (1-P)\rho(1-P)$. Intertwining this measurement with time evolutions from the Hamiltonian according to the Zeno theorem then yields an effective Hamiltonian $T(H)$ and an effective starting state $T(\rho(0))$, so that $\tilde{\rho}(t) = \exp(-iT(H)t)T(\rho(0))exp(iT(H)t)$. The idea is to see if for a separable (regarding 1 and 4) starting state $\rho(0)$, $\mathrm{tr}_{23}[\tilde{\rho}(t)]$ is entangled. The entanglement of two qubits can be checked efficiently via the Peres criterion, i.e. the state is separable if and only if the partial transpose is again positive semidefinite, such that a state is entangled if and only if the negativity is positive.

As a first example $\rho(0) = |0\rangle\langle 0|^{\otimes 4}$ is chosen where the reduced state on systems 1 and 4 is a product state and thus not entangled. Figure 8.4 shows the negativity between system 1 and 4 for the starting state where every qubit is in the ground state, once for the usual time evolution and

[18] defined above already??? okay write all for inital state ρ and $\rho_{\alpha\alpha}$
[19] defined above???
[20] ref eq.

We are given

$$\Phi_t(\rho) \;=\; \sum_{\alpha} \left(U_t^{(\alpha)} \otimes |\alpha\rangle\langle\alpha|\right) \left(P_\alpha \rho P_\alpha\right) \left(U_t^{(\alpha)\,\dagger} \otimes |\alpha\rangle\langle\alpha|\right), \qquad P_\alpha = \mathbb{I}_{AB} \otimes |\alpha\rangle\langle\alpha|.$$

Because $P_\alpha^2 = P_\alpha$ and it acts only on Z,

$$(P_\alpha \rho P_\alpha) = (\mathbb{I} \otimes |\alpha\rangle\langle\alpha|)\, \rho\, (\mathbb{I} \otimes |\alpha\rangle\langle\alpha|).$$

Key identity for the partial trace over Z

For any operator Y on ABZ and any ONB $\{|j\rangle\}$ of $\mathcal{H}_Z$,

$$\mathrm{Tr}_Z[Y] = \sum_j (\mathbb{I}_{AB} \otimes \langle j|)\, Y\, (\mathbb{I}_{AB} \otimes |j\rangle).$$

We'll use this with the basis $\{|j\rangle\}$ that contains the $|\alpha\rangle$'s.

Apply Tr_Z to each summand

For a fixed α,

$$\begin{aligned}\mathrm{Tr}_Z\Big[\left(U_t^{(\alpha)} \otimes |\alpha\rangle\langle\alpha|\right)(\mathbb{I} \otimes |\alpha\rangle\langle\alpha|)\, \rho\, (\mathbb{I} \otimes |\alpha\rangle\langle\alpha|)\left(U_t^{(\alpha)\,\dagger} \otimes |\alpha\rangle\langle\alpha|\right)\Big] &= \sum_j (\mathbb{I} \otimes \langle j|)\, [\cdots]\, (\mathbb{I} \otimes |j\rangle) \\ &= \; (\mathbb{I} \otimes \langle\alpha|)\, \left(U_t^{(\alpha)} \otimes |\alpha\rangle\langle\alpha|\right) (\mathbb{I} \otimes \\ &\quad \text{(only } j = \alpha \text{ survives)} \\ &= U_t^{(\alpha)}\, (\mathbb{I} \otimes \langle\alpha|)\, \rho\, (\mathbb{I} \otimes |\alpha\rangle)\, U_t^{(\alpha)}\end{aligned}$$

Define the **block** of ρ in the $|\alpha\rangle$-sector:

$$\rho_{\alpha\alpha} := (\mathbb{I} \otimes \langle\alpha|)\, \rho\, (\mathbb{I} \otimes |\alpha\rangle) \in \mathcal{B}(\mathcal{H}_{AB}).$$

Hence

$$\mathrm{Tr}_Z\left[\Phi_t(\rho)\right] = \sum_\alpha U_t^{(\alpha)}\, \rho_{\alpha\alpha}\, U_t^{(\alpha)\,\dagger}. \qquad (\bigstar)$$

Writing it as a convex mixture

Let

$$p_\alpha := \mathrm{Tr}(\rho_{\alpha\alpha}) = \langle\alpha|\rho_Z|\alpha\rangle = \mathrm{Tr}\left[\Pi_\alpha\, \rho_Z\right], \quad \rho_{AB}^{(\alpha)} := \frac{\rho_{\alpha\alpha}}{p_\alpha} \;\; (\text{when } p_\alpha > 0).$$

Then $\rho_{AB}^{(\alpha)}$ is the **post-measurement conditional state on** AB given outcome α, and p_α is its probability. Using ($\bigstar$),

$$\mathrm{Tr}_Z\left[\Phi_t(\rho)\right] = \sum_\alpha p_\alpha\, \mathrm{Ad}_{U_t^{(\alpha)}}\left(\rho_{AB}^{(\alpha)}\right), \qquad \mathrm{Ad}_U(X) := UXU^\dagger.$$

This is exactly the "convex mixture" form shown: you get a classical mixture over outcomes α, with weights $p_\alpha = \mathrm{Tr}[\Pi_\alpha \rho_Z]$, of the unitary evolutions $U_t^{(\alpha)}$ acting on the corresponding conditional states on AB.

Excellent question — yes, essentially, **the factorization of $U_t^{(\alpha)}$ into $U_A^{(\alpha)} \otimes U_B^{(\alpha)}$** is equivalent (and in f *if and only if* equivalent) to the corresponding Hamiltonian block $H^{(\alpha)}$ being *non-entangling*, i.e. of the form

$$H^{(\alpha)} = H_A^{(\alpha)} \otimes \mathbf{1}_B + \mathbf{1}_A \otimes H_B^{(\alpha)}.$$

Let's unpack that rigorously.

Step 1. Given structure

We have:

$$U_t^{(\alpha)} = e^{-iH^{(\alpha)}t}.$$

This operator acts on $A \otimes B$.

Step 2. Necessary and sufficient condition for factorization

The claim $U_t^{(\alpha)} = U_A^{(\alpha)} \otimes U_B^{(\alpha)}$ for **all** t is equivalent to the generator $H^{(\alpha)}$ being a *sum of local terms*:

$$H^{(\alpha)} = H_A^{(\alpha)} \otimes \mathbf{1}_B + \mathbf{1}_A \otimes H_B^{(\alpha)}.$$

Proof sketch:

- **(⇒)**
  If $U_t^{(\alpha)} = U_A^{(\alpha)} \otimes U_B^{(\alpha)}$, take the derivative at $t = 0$:

  $$\frac{d}{dt} U_t^{(\alpha)}|_{t=0} = -iH^{(\alpha)} = \frac{d}{dt}(U_A^{(\alpha)} \otimes U_B^{(\alpha)})|_{t=0} = (-iH_A^{(\alpha)}) \otimes \mathbf{1}_B + \mathbf{1}_A \otimes (-iH_B^{(\alpha)}),$$

  giving exactly the desired decomposition.
- **(⇐)**
  Conversely, if $H^{(\alpha)} = H_A^{(\alpha)} \otimes \mathbf{1}_B + \mathbf{1}_A \otimes H_B^{(\alpha)}$, the two terms commute, and the exponential factorizes:

  $$e^{-itH^{(\alpha)}} = e^{-it(H_A^{(\alpha)} \otimes \mathbf{1}_B + \mathbf{1}_A \otimes H_B^{(\alpha)})} = e^{-itH_A^{(\alpha)} \otimes \mathbf{1}_B}\, e^{-it\mathbf{1}_A \otimes H_B^{(\alpha)}} = e^{-itH_A^{(\alpha)}} \otimes e^{-itH_B^{(\alpha)}}.$$

So indeed, **the local factorization of $U_t^{(\alpha)}$** is *equiva* ↓ . to $H^{(\alpha)}$ being a sum of commuting local generators on A and B.

Figure 8.1: just look at the pictures

> **Theorem.** Φ_t is EB for some/all t **iff** the pinching measurement is **rank-1**, i.e. every projector in the PVM has rank 1.
> In the present two-outcome case this means $\operatorname{rank} P = 1$ and $\operatorname{rank}(\mathbf{1} - P) = 1$ (so $\dim \mathcal{H}_S = 2$).

The proof splits into three lemmas.

Lemma A (unitary invariance of EB)

If Ψ is EB and U, V are unitaries on S, then $\mathrm{Ad}_U \circ \Psi \circ \mathrm{Ad}_V$ is EB.
Equivalently, EB-ness is invariant under pre/post composition with unitaries.

Proof. Let R be an arbitrary ancilla. For any ρ_{SR},

$$\big((\mathrm{Ad}_U \circ \Psi \circ \mathrm{Ad}_V) \otimes \mathrm{id}_R\big)(\rho_{SR}) = (U \otimes \mathbf{1})\,(\Psi \otimes \mathrm{id})((V \otimes \mathbf{1})\rho_{SR}(V^\dagger \otimes \mathbf{1}))\,(U^\dagger \otimes \mathbf{1}).$$

If Ψ is EB, then $(\Psi \otimes \mathrm{id})(\cdot)$ is always separable across $S|R$; conjugating by local unitaries $U \otimes \mathbf{1}$ preserves separability. ∎

Consequence for us. Since $\Phi_t = \mathrm{Ad}_{U_t} \circ T$, Lemma A gives:

$$\boxed{\Phi_t \text{ is EB} \iff T \text{ is EB.}}$$

Hence it suffices to characterize **when the pinching T is EB.**

Lemma B (rank-1 pinching is EB)

If the PVM is rank-1 (each projector has rank 1), then the pinching channel T is EB.

Proof. Pick an orthonormal basis $\{|i\rangle\}_i$ consistent with the rank-1 PVM so that
$T(X) = \sum_i |i\rangle\langle i|X|i\rangle\langle i|$, i.e. **complete dephasing** in that basis.
Realize T as a **measure-and-prepare** channel with POVM $\{E_i = |i\rangle\langle i|\}$ and preparations $\{\sigma_i = |i\rangle\langle i|\}$:

$$T(X) = \sum_i \mathrm{Tr}(E_i X)\,\sigma_i.$$

Measure-and-prepare channels are EB by definition (they output a convex combination of fixed states depending only on the classical outcome), or equivalently their Choi matrices are separable. ∎

Lemma C (if some outcome has rank ≥ 2, pinching is not EB)

If the PVM contains a projector of rank ≥ 2 (in our two-outcome case: $\operatorname{rank} P \geq 2$ or $\operatorname{rank}(\mathbf{1} - P) \geq 2$), then T is **not** EB.

Let the measurement act only on Z: $P_\alpha = \mathbf{1}_{AB} \otimes \Pi_\alpha$.
Let $T(H) = \sum_\alpha H^{(\alpha)} \otimes \Pi_\alpha$ and $U_t = \sum_\alpha U_t^{(\alpha)} \otimes \Pi_\alpha$ with $U_t^{(\alpha)} = e^{-iH^{(\alpha)}t}$ on AB.
Define $\Psi_t = \mathrm{Tr}_Z \circ \mathrm{Ad}_{U_t} \circ T$.

Theorem (SP vs. entangling across $A|B$).

$$\boxed{\Psi_t \text{ is separability-preserving across } A|B \iff \forall \alpha:\ U_t^{(\alpha)} = U_A^{(\alpha)} \otimes U_B^{(\alpha)}.}$$

Equivalently, Ψ_t is **entangling** (not SP) iff there exists at least one branch α with $U_t^{(\alpha)}$ **non-local** on $A|B$.

Proof

(⇒) Suppose $\forall \alpha,\, U_t^{(\alpha)} = U_A^{(\alpha)} \otimes U_B^{(\alpha)}$.
Take any separable $\rho_{AB} = \sum_k q_k \rho_A^{(k)} \otimes \rho_B^{(k)}$. Then

$$\Psi_t(\rho_{AB}) = \sum_{\alpha,k} p_\alpha q_k \left(U_A^{(\alpha)} \rho_A^{(k)} U_A^{(\alpha)\dagger}\right) \otimes \left(U_B^{(\alpha)} \rho_B^{(k)} U_B^{(\alpha)\dagger}\right),$$

a convex combination of product states ⇒ separable. Hence Ψ_t is SP.

(⇐) Conversely, assume $\exists \alpha_0$ with $U_t^{(\alpha_0)}$ **entangling** on AB (i.e. $\nexists U_A \otimes U_B$ equal to it). Then there exists a product pure state $|\phi_A\rangle \otimes |\phi_B\rangle$ such that

$$|\psi_{AB}\rangle := U_t^{(\alpha_0)}\big(|\phi_A\rangle \otimes |\phi_B\rangle\big)$$

is entangled. Choose the Z-state to select that branch: take $\rho_Z = \Pi_{\alpha_0}/\mathrm{Tr}\,\Pi_{\alpha_0}$ if Π_{α_0} is rank-1, or any state supported in $\operatorname{ran}\Pi_{\alpha_0}$ so that $p_{\alpha_0} = 1$ and $p_{\alpha \neq \alpha_0} = 0$. Then

$$\Psi_t\big(|\phi_A\phi_B\rangle\langle\phi_A\phi_B|\big) = U_t^{(\alpha_0)}\,|\phi_A\phi_B\rangle\langle\phi_A\phi_B|\,U_t^{(\alpha_0)\dagger} = |\psi_{AB}\rangle\langle\psi_{AB}|,$$

which is entangled. Hence Ψ_t is **not** SP. ∎

What stays true from my earlier (ii)

If some measurement block on Z has **rank** ≥ 2, you can choose H so that within that block the induced $U_t^{(\alpha)}$ on AB is non-local (e.g. via mediated interactions), and then Ψ_t is entangling by the criterion above. So the "rank-≥ 2 enables entanglement creation (for suitable H, t)" part remains correct.

Yes. Here's a clean, self-contained proof.

Claim

Let U be a unitary on $\mathcal{H}_A \otimes \mathcal{H}_B$.
If U is **not** a local product (nor a local product composed with the swap), then there exists a product vector $|a\rangle \otimes |b\rangle$ such that $U(|a\rangle \otimes |b\rangle)$ is **entangled**.

Equivalently (contrapositive):
If U maps **every** product vector to a product vector, then

$$U = (V_A \otimes V_B) \quad \text{or} \quad U = (V_A \otimes V_B)\,S,$$

with V_A, V_B unitaries and S the swap. In particular, such a U cannot create entanglement from any product input.

Proof (product–preservers are local up to swap)

Assume U maps all product vectors to product vectors. Fix nonzero vectors $|a\rangle \in \mathcal{H}_A$ and choose two linearly independent $|b_0\rangle, |b_1\rangle \in \mathcal{H}_B$. Write

$$U(|a\rangle \otimes |b_j\rangle) = |\alpha_j\rangle \otimes |\beta_j\rangle, \qquad j = 0, 1.$$

Now consider the third product input $|a\rangle \otimes (|b_0\rangle + |b_1\rangle)$. By linearity,

$$U\big(|a\rangle \otimes (|b_0\rangle + |b_1\rangle)\big) = |\alpha_0\rangle \otimes |\beta_0\rangle + |\alpha_1\rangle \otimes |\beta_1\rangle,$$

and by assumption this sum is a product vector.

Fact: A sum of two simple tensors $x_0 \otimes y_0 + x_1 \otimes y_1$ is simple iff either x_0 and x_1 are proportional or y_0 and y_1 are proportional.

Hence, for our pair either $|\alpha_0\rangle \parallel |\alpha_1\rangle$ or $|\beta_0\rangle \parallel |\beta_1\rangle$.

Varying $|b_1\rangle$ while keeping $|a\rangle, |b_0\rangle$ fixed, linear independence can be arranged for infinitely many choices; the above dichotomy can switch only if one of the two factors is actually **independent** of j.
Therefore, for this fixed $|a\rangle$ there are two possibilities:

- (A) There exists a vector $|\alpha(a)\rangle$ such that

  $$U(|a\rangle \otimes |b\rangle) = |\alpha(a)\rangle \otimes T_B|b\rangle \quad \text{for all } |b\rangle,$$

  for some linear map $T_B : \mathcal{H}_B \to \mathcal{H}_B$; or

Figure 8.2: sp condition

Theorem A (SP under product-block pinching + no bridge)

Setup.
Let A, B be finite–dimensional systems and let $Z = Z_A \otimes Z_B \otimes Z_0$.
Assume the Hamiltonian has **no bridge between the A-side and B-side**, i.e.

$$H \;=\; H_{AZ_A} \otimes \mathbf{1}_{Z_B Z_0} \;+\; \mathbf{1}_{Z_A Z_0} \otimes H_{BZ_B} \;+\; H_{Z_0}\,,$$

with no term acting nontrivially on both Z_A and Z_B.
Let the pinching be defined by a PVM on Z with **product blocks**

$$P_\alpha = \mathbf{1}_{AB} \otimes \Pi_\alpha, \qquad \Pi_\alpha = \Pi_\alpha^{(A)} \otimes \Pi_\alpha^{(B)} \otimes \Pi_\alpha^{(0)}.$$

Let

$$T(X) = \sum_\alpha P_\alpha X P_\alpha, \qquad U_t = e^{-iT(H)t}, \qquad \Phi_t = \mathrm{Ad}_{U_t} \circ T, \qquad \Psi_t = \mathrm{Tr}_Z \circ \Phi_t.$$

Claim.
For every t and every input ρ_{ABZ},

$$\Psi_t(\rho_{AB}) = \sum_\alpha p_\alpha \left(U_A^{(\alpha)} \otimes U_B^{(\alpha)}\right) \rho_{AB} \left(U_A^{(\alpha)} \otimes U_B^{(\alpha)}\right)^\dagger, \quad p_\alpha = \mathrm{Tr}(\Pi_\alpha \rho_Z),$$

for some unitaries $U_A^{(\alpha)}$ on A and $U_B^{(\alpha)}$ on B.
Hence Ψ_t is a **random product–unitary channel** (LO), therefore **separability-preserving across $A|B$.**

Proof.
Because each P_α factorizes across $Z_A|Z_B|Z_0$, we have

$$T(H) = \sum_\alpha P_\alpha H P_\alpha = \sum_\alpha \left(H_{AZ_A}^{(\alpha)} \otimes \mathbf{1}_{Z_B Z_0} + \mathbf{1}_{Z_A Z_0} \otimes H_{BZ_B}^{(\alpha)} + \; H_{Z_0}^{(\alpha)}\right),$$

where $H_\bullet^{(\alpha)}$ denotes the restriction to the corresponding block of Π_α.
All three summands act on **disjoint tensor factors**, hence commute, so within each block

$$U_t^{(\alpha)} = \exp\big(-\,i\,T(H)\,t\big)\big|_\alpha = \underbrace{e^{-iH_{AZ_A}^{(\alpha)}t}}_{=:U_{AZ_A}^{(\alpha)}} \otimes \underbrace{e^{-iH_{BZ_B}^{(\alpha)}t}}_{=:U_{BZ_B}^{(\alpha)}} \otimes e^{-iH_{Z_0}^{(\alpha)}t}.$$

Consequently, for any ρ_{ABZ},

$$\Phi_t(\rho) = \sum_\alpha \left(U_{AZ_A}^{(\alpha)} \otimes U_{BZ_B}^{(\alpha)} \otimes U_{Z_0}^{(\alpha)}\right) \left(P_\alpha \rho P_\alpha\right) \left(U_{AZ_A}^{(\alpha)} \otimes U_{BZ_B}^{(\alpha)} \otimes U_{Z_0}^{(\alpha)}\right)^\dagger.$$

Tracing out Z yields

$$\Psi_t(\rho_{AB}) = \sum_\alpha \mathrm{Tr}(\Pi_\alpha \downarrow_Z)\;\; \mathrm{Ad}_{U_A^{(\alpha)} \otimes U_B^{(\alpha)}}(\rho_{AB}),$$

Theorem B (Entanglement creation by pinching with an entangled block even with "no-bridge" H)

Statement.
There exist finite–dimensional systems A, B, Z_1, Z_2, a Hamiltonian

$$H = g\,\sigma_x^A \otimes \sigma_x^{Z_1} \otimes \mathbf{1}_{Z_2} \;+\; g\,\mathbf{1}_{Z_1} \otimes \sigma_x^{Z_2} \otimes \sigma_x^B \quad (g > 0),$$

which has **no term** acting nontrivially on both the A-side and B-side (no bridge), a two–outcome pinching on $Z_1 Z_2$ with an **entangled rank-2** projector

$$\Pi \;=\; |\Phi^+\rangle\langle\Phi^+| + |\Psi^+\rangle\langle\Psi^+|\,, \quad |\Phi^+\rangle = \tfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle),\; |\Psi^+\rangle = \tfrac{1}{\sqrt{2}}(|01\rangle + |10\rangle),$$

and a separable input $\rho_{AB} = |00\rangle\langle 00|$, $\rho_Z = \Pi/2$, such that for all sufficiently small $t > 0$ the reduced Zeno map

$$\Psi_t = \mathrm{Tr}_{Z_1 Z_2} \circ \mathrm{Ad}_{e^{-iT(H)t}} \circ T$$

produces an **entangled** state $\Psi_t(\rho_{AB})$ across $A|B$.

Proof.
(1) *Pinched Hamiltonian on the entangled block.*
On the 2-dimensional subspace $\mathcal{L} := \mathrm{ran}\,\Pi = \mathrm{span}\{|\Phi^+\rangle, |\Psi^+\rangle\}$ we have the identities

$$\Pi\,\sigma_z^{Z_1}\,\Pi = \Pi\,\sigma_x^{Z_2}\,\Pi =: X_L,$$

where X_L is the Pauli X on $\mathcal{L}$ with basis $\{|\Phi^+\rangle, |\Psi^+\rangle\}$.
Therefore the Π-block of the pinched Hamiltonian is

$$H^{(\Pi)} \;=\; \left(g\,\sigma_x^A + g\,\sigma_x^B\right) \otimes X_L.$$

(No term of the original H touched both sides, yet the **restriction** ties both couplings to the **same logical operator** X_L.)

(2) *State supported in the block and Zeno unitary.*
Choose $\rho_Z = \Pi/2$ so the pinching leaves the state in $\mathcal{L}$ with probability 1. Then in that block

$$U_t^{(\Pi)} = e^{-iH^{(\Pi)}t} = e^{-i(g\sigma_x^A + g\sigma_x^B) \otimes X_L\, t}.$$

(3) *Second–order reduced dynamics on AB.*
Using $\mathrm{Tr}_{\mathcal{L}}(X_L) = 0$ and $\mathrm{Tr}_{\mathcal{L}}(X_L^2) = 1$,

$$\mathrm{Tr}_Z\big([H^{(\Pi)}, [H^{(\Pi)}, \rho_{AB} \otimes \tfrac{\Pi}{2}]]\big) = g^2\Big(\{\sigma \downarrow_x^{B}, \rho_{AB}\} - 2\,\sigma_x^A \rho_{AB} \sigma_x^B - 2\,\sigma_x^B \rho_{AB} \sigma_x^A\Big).$$

Theorem B′ (Universal entangling pinching for any non-bridge H)

Setting.
Let A, B be finite–dimensional systems and $Z = Z_A \otimes Z_B \otimes Z_0$.
Assume the Hamiltonian has **no direct $A-B$ term** and **no term touching both Z_A and Z_B**:

$$H = \sum_j A_j \otimes X_j \otimes \mathbf{1}_{Z_B Z_0} \;+\; \sum_k B_k \otimes \mathbf{1}_{Z_A Z_0} \otimes Y_k \;+\; H_{Z_0},$$

with Hermitian A_j on A, B_k on B, X_j on Z_A, Y_k on Z_B.

Nondegeneracy (minimal) couplings.
Assume there exist indices $j_\star, k_\star$ such that

- $X := X_{j_\star}$ has at least two distinct eigenvalues (non-scalar on Z_A),
- $Y := Y_{k_\star}$ has at least two distinct eigenvalues (non-scalar on Z_B),
- $A := A_{j_\star}$ and $B := B_{k_\star}$ are not proportional to the identity on A, B, respectively.

Claim.
There exist

- a **rank-2 projector** Π on $Z_A \otimes Z_B$ (tensored with $\mathbf{1}_{Z_0}$),
- a **two-outcome pinching** $\{\,\Pi,\ \Pi^\perp\,\}$ on Z (outcomes forgotten),
- a **separable input** state $\rho_{ABZ} = |a\rangle\langle a| \otimes |b\rangle\langle b| \otimes \sigma_Z$ with σ_Z supported in $\mathrm{ran}\,\Pi$,
- and a time $t > 0$,

such that the reduced Zeno map

$$\Psi_t = \mathrm{Tr}_Z \circ \mathrm{Ad}_{e^{-iT(H)t}} \circ T$$

produces an **entangled** state $\Psi_t(|a\rangle\langle a| \otimes |b\rangle\langle b|)$ across $A|B$.

Proof

Step 1. Construct a "logical" 2-dimensional block on $Z_A Z_B$

Let X have eigenpairs $(\lambda_0, |x_0\rangle), (\lambda_1, |x_1\rangle)$ with $\lambda_0 \neq \lambda_1$, and Y have $(\mu_0, |y_0\rangle), (\mu_1, |y_1\rangle)$ with $\mu_0 \neq \mu_1$.

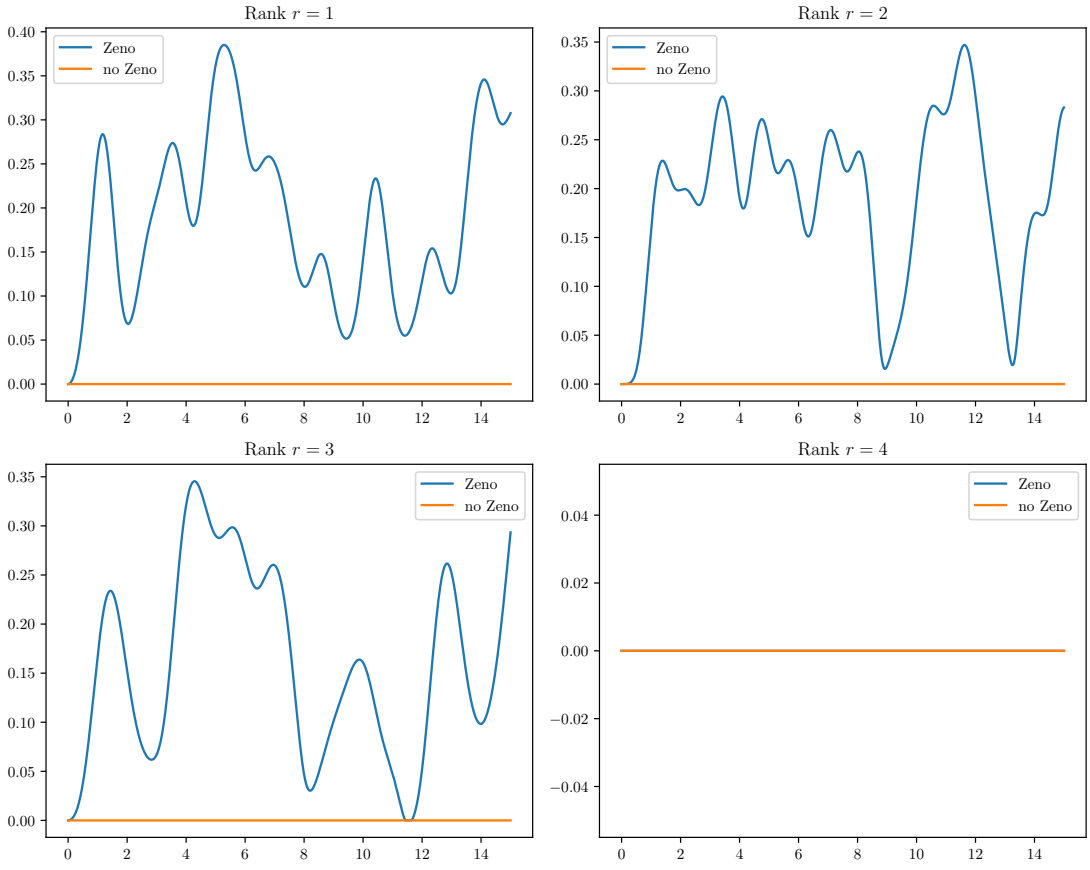


Figure 8.4: The negativity of systems 1 and 4 with the usual time evolution from the XX Hamiltonian for 12 and 34 (blue) and Zeno time evolution with projectors of different rank (orange) all for the starting state $\rho(0) = |0\rangle\langle 0|^{\otimes 4}$

Rank r	1	2	3
Negativity at $t = 20$	0.4991	0.4975	0.4706
Negativity at $t = 1000$	0.4931	0.4987	0.4968

Table 8.1: negativity of system 14 for optimized projector of rank r

once for the Zeno evolution with randomly created projectors of different rank. The figure shows that when the projector acting on the systems 23 has rank 1,2 or 3 then entanglement is created and only for rank 4 the negativity with zeno dynamics is still zero (not considering numerical errors). The case $r = 4$ is also immediaetly clear analytically: The only orthogonal projector from $\mathbb{C}^4$ to $\mathbb{C}^4$ is the identity for which $T(H) = H$ and thus the Zeno case is the same as the non-Zeno case. In the other cases one sees that the negativity needs some time to build up and then changes over time which suggests that choosing a projector that is time dependent may be helpfull if a constant high negativity is desired. Note that the time-span $t = 15$ of the simulation is not relevant for the problem since a change of the timescale can always be incorporated by a change of the Hamiltonian with the inverse factor.

It is also possible to numerically find the projections that yield this maximal entanglement of the reduced bipartite state 14, see listing A.3 and table 8.1 . The following lemma is useful to characterize orthogonal projections:

Lemma 8.4.1. *Let $P = P^* = P^2 : \mathbb{C}^n \to \mathbb{C}^n$ be an orthogonal projector of rank r, then it can be written as $P = UU^*$ where $U : \mathbb{C}^r \to \mathbb{C}^n$ is an isometry, i.e. $U^*U = \mathbb{1}$.*

[21]

Proof. $\mathrm{Im}(P)$ is an r-dimensional subspace of C^n, let $\{u_1, \ldots, u_r\}$ be an ONB for it. Let $v \in \mathbb{C}^n$ be arbitrary, then $Pv = \sum_{i=1}^r \langle u_i, v\rangle u_i$ since $Pv \in \mathrm{Im}(P)$, so $P = \sum_{i=1}^r |u_i\rangle\langle u_i|$. Define $U = \{u_1, \ldots, u_r\}$, then $P = UU^*$ □

[22]

Things one could vary and investigate the effects:

- size of the system, i.e. add more particles in one dimension, or look at lattices in 2D or 3D
- impact of the starting state: are there starting states for which the Zeno time evolved state never becomes entangled
- investigate the role of the projective measurement: find the optimal measurement for which the entanglement increases the fastest/the most. In particular the rank of the measurement could be interesting and one might conjecture that projections of rank 1 and rank 4 do not create the desired effect. The reason for this being that a continuously applied rank 1 projection freezes the dynamics onto a one dimensional space which might not allow the correlations to travel between systems 2 and 3 which is needed to create entanglement between 1 and 4 (heuristically speaking). A rank 4 projection however might not 'speed up' the dynamics enough to create effects[23]. Next, is it useful to change the measurement with time?[24]

So, now the goal is for a given separable state and a given time t to find the orthogonal projection $P \in \mathcal{B}(\mathcal{H}_{23})$ such that $\mathrm{tr}_{23}[\tilde{\rho}(t)]$ is maximally entangled. To define what maximally entangled is one needs to choose an entanglement measure[25]

I think i was already able to show analytically that if the projector is a product operator then no entanglement can be created:[26]

[21] is orthogonal needed for the lemma?
[22] Important is that the space of projections of rank r is the complex grassmannian $G(r, n)$ explain and use this
[23] try to show analytically that these cases are indeed not creating entanglement
[24] is this is even a case that is included in the zeno theorem???
[25] Here or before put definition of ent measure and negativity
[26] show this here

8.5 Majorization of spectra

In certain cases the Zeno effect yields an effective Hamiltonian $T(H)$ [27], then it can be interesting to see if this new Hamiltonian is majorized by the old one or if it is the other way around (also none of these can be true). In the study of thermal ressource theory the question is what states can be aquired from a given system with only thermal operations. If P commutes with H then then the projection is thermally allowed and the Zeno procedure is allowed, but then $T(H) = H$, so boring but look for other qunatum operations that are allowed.

8.6 Transport of excitation

The transportation of an excitation (or more generally any state) through a quantum system realizes transmission of information without additionally having to exchange classical information, as e.g. in the case of teleportation.

It was shown in [Chr+04] that perfect state transfer is possible in the XY model with interaction coefficients $J_n = \sqrt{n(N-n)}/2$ [28]. Tuning the interaction strength differently at each site n is currently not possible and the fidelity of the state transmission decreases rapdily with deviations from the optimal coefficients J_n [Zwi+11]. There is another way to create the optimal Hamiltonian: choose an experimentally implementable Hamiltonian H and choose measurements (that are experimentally implementable, [29]) such that $T(H) = PHP + (1-P)H(1-P)$ is the desired Hamiltonian.

As an example take the XX-model on a one-dimensional spin chain with possibly site-dependent interaction strenghts:

$$H = \sum_i J_i \underbrace{(\sigma_x^i \sigma_x^{i+1} + \sigma_y^i \sigma_y^{i+1})}_{=:B_i}. \tag{8.6.1}$$

Note that B_i is an operator on $\mathcal{H}_i \otimes \mathcal{H}_{i+1}$ A first idea is to choose projective measurements P such that $T(H)$ has again the form of equation (8.6.1)[30], but with different interaction strengths.

Define $U := 2P - 1$, this is a Hermitian unitary, then $T(H) = (H + UHU)/2$. Note that U is an involution, i.e. $U^2 = \mathbb{1}$, and Hermitian and unitary (which both follows from involution). The question is for which U $T(U)$ is again an XX-Hamiltonian but with different interaction strengths. For this define $S := \mathrm{span}\{B_i\}_{i=1,\dots,N}$. First, take $H = B_j$ for some j, then we deman $T(H) = (B_j + UB_jU)/2 \in S$ and since $B_j \in S$ it follows $UB_jU \in S$. Note that

$$\langle B_i, B_j \rangle_{\mathrm{HS}} = \mathrm{tr}[B_i B_j] = 8\delta_{ij} \tag{8.6.2}$$

Lemma 8.6.1. *Let $H = \sum_i J_i B_i$ as defined above[31] and U be a Hermitian unitary involution so that $H + UHU \in S := \{B_i\}_{i=1}^N$, then there is a permutation π and $z \in \{\pm 1\}^N$, so that*

$$UB_iU = s_i B_{\pi(i)} \forall i \text{ and } s_i = z_i z_{i+1} \tag{8.6.3}$$

Proof. [32] The Jordan-Wigner transformation of the Hamiltonian reads: $\sigma_j^- \mapsto f_j = \exp(-i\pi \sum_{k<j} \sigma_k^+ \sigma_k^-)\sigma_j$ then $B_i = \sigma_x^i \sigma_x^{i+1} + \sigma_y^i \sigma_y^{i+1} \mapsto$

there is a mistake that not all automorphism on u(N) are inner or transposition because that is only valid, apparently the result still holds. see chatgpt closed subgroup frage □

As a consequence $J_k' = (J_k + s_{\pi^{-1}(k)} J_{\pi^{-1}(k)})/2$. If we start with uniform interaction then $J_k' = J(1 + s_{\pi^{-1}(k)})/2 \in \{0, 1/2, 1\}J$. With a single Zeno channel the desired J_n for perfect state transfer is thus not possible [33], but with mixing of channels it is:[34]:

Proposition 8.6.1. *With mixing of channels, starting from a uniform XX model any interaction profile J_n can be realized: The explicit projections and mixture probabilities can be calculated the following:*

[27] check this with the general theorem

[28] with what boudary conditions and check if true

[29] quantum operations according to Zeno

[30] of course one could start with a different implementable hamiltonian and choose P so that T(H) is the desired XX one.

[31] write out here

[32] introduce jordan wigner trafo earlier and prove it is uniquely invertible

[33] why

[34] explain mixing of channels earlier

Schritt 1 — Jordan–Wigner und Einteilchen-Bild

Führe Jordan–Wigner ein. Schreibe H als

$$H = \sum_{i=1}^{N-1} \frac{J_i}{2} \left(c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i \right) = c^\dagger \, T \, c, \quad T = \sum_{i=1}^{N-1} \tfrac{J_i}{2} \, (E_{i,i+1} + E_{i+1,i}),$$

mit $c = (c_1, \ldots, c_N)^T$ und E_{jk} den Matrixeinheiten.
Der **XX-Unterraum** S entspricht genau den quadratischen, teilchenzahlerhaltenden Operatoren $c^\dagger M c$ mit **rein nächster-Nachbar**-Matrizen M (nur Nebendiagonalen belegt).

Schritt 2 — Ad_U normalisiert die $\mathfrak{u}(N)$-Darstellung

Die Menge $\{ \, i \, c^\dagger M c : M = M^\dagger \}$ ist eine Lie-Algebra isomorph zu $\mathfrak{u}(N)$.
Aus $\mathrm{Ad}_U(S) \subseteq S$ folgt insbesondere, dass Ad_U die von S **erzeugte** Lie-Algebra (also ganz $\mathfrak{u}(N)$) **auf sich** abbildet.
Damit induziert U eine Automorphie α von $\mathfrak{u}(N)$.

Fakt: Jede Automorphie von $\mathfrak{u}(N)$ ist entweder **inner** $X \mapsto VXV^\dagger$ oder die Komposition mit **Transposition/Konjugation**.
Da unsere Generatoren $E_{i,i+1} + E_{i+1,i}$ **reell-symmetrisch** sind, wirkt Transposition trivial; wir können also annehmen:

Figure 8.5: Caption njaha

$$U \, (c^\dagger M c) \, U \;=\; c^\dagger (V M V^\dagger) c$$

für ein $V \in U(N)$. Das heißt (auf Operatoren erster Ordnung):

$$U \, c \, U = V \, c, \qquad U \, c^\dagger U = c^\dagger V^\dagger,$$

also ist U (bis auf evtl. eine Phasen-/Parity-Funktion) die **Zweitquantisierung** von V und erhält die Teilchenzahl.

Schritt 3 — Erhaltung der NäCHSTER-Nachbar-Struktur

Wir verlangen nun $V \, T_i \, V^\dagger \in \mathrm{span}\{T_1, \ldots, T_{N-1}\}$ **für alle** Kantenmatrizen $T_i := E_{i,i+1} + E_{i+1,i}$.

Schreibe V spaltenweise $V = [v_1 \; \ldots \; v_N]$ mit $v_j \in \mathbb{C}^N$.
Für T_i ist $\mathrm{im}(T_i) = \mathrm{span}\{e_i, e_{i+1}\}$.
Dann hat $VT_iV^\dagger$ Bild $\mathrm{span}\{v_i, v_{i+1}\}$.
Damit $VT_iV^\dagger$ wieder **nur** eine Nebendiagonal-Kante (oder Linearkombination solcher) ist, müssen **für alle** i die beiden Vektoren v_i, v_{i+1} jeweils (bis auf Phase) **Standardbasisvektoren** sein.
Folglich ist V eine **monomiale** unitäre Matrix:

$$V = D \, P, \qquad P \text{ Permutation}, \quad D = \mathrm{diag}(e^{i\theta_1}, \ldots, e^{i\theta_N}).$$

Außerdem muss P **Nachbarn auf Nachbarn** schicken, also eine **Graph-Automorphie** des Pfades sein (bei OBC: Identität oder Spiegelung).

Figure 8.6: Caption jaha2

We also have $||T(H)|| \leq ||H||$ see prop above so the Zeno process can not increase the overall interaction strength, only change the distribution of the interactions on the sites.[35]

8.7 Entropy change

The idea is that the Zeno effect yields a quantum channel $\tilde{\Phi} : \rho \mapsto \mathrm{e}^{-iT(H)t} T(\rho) \mathrm{e}^{iT(H)t}$ at each point in time t.[36]. Without the Zeno effect the time evolution would be $\Phi(\rho) = \mathrm{e}^{-iHt} \rho \mathrm{e}^{iHt}$. Theorem 8.8 in channels script tells us T is a unital quantum channel, then $T(\rho)$ is majorized by ρ, i.e. $T(\rho)$ is more mixed than ρ. In this case, the map $\tilde{\Phi}$ is unital since $\tilde{\Phi}(\mathbb{1}) = \mathbb{1}$. Hence, $\tilde{\Phi}(\rho)$ is more mixed than ρ, but this also true for the no-Zeno time evolution since closed system evolve unitally. The idea is to see if there is a majorization relation between $\tilde{\Phi}(\rho)$ and $\Phi(\rho)$. Idea: Write $\tilde{\Phi}(\rho) = \Phi'(\Phi(\rho))$ for a map Φ' that is unital. (this is indeed iff). Since Φ is a unitary map, it has an inverse Φ^{-1} so that $\tilde{\Phi}(\rho) = (\tilde{\Phi} \circ \Phi^{-1})(\Phi(\rho))$. It remains to show that $\tilde{P}hi \circ \Phi^{-1}$ is positive unital and trace-preserving. Since Φ^{-1} is linear, so is $\tilde{\Phi} \circ \Phi^{-1}$. The composition is also positive and trace preserving and unital since both the involved maps are. So it is shown that

[35] put this above
[36] show $\tilde{\Phi}$ is indeed a quantum channel

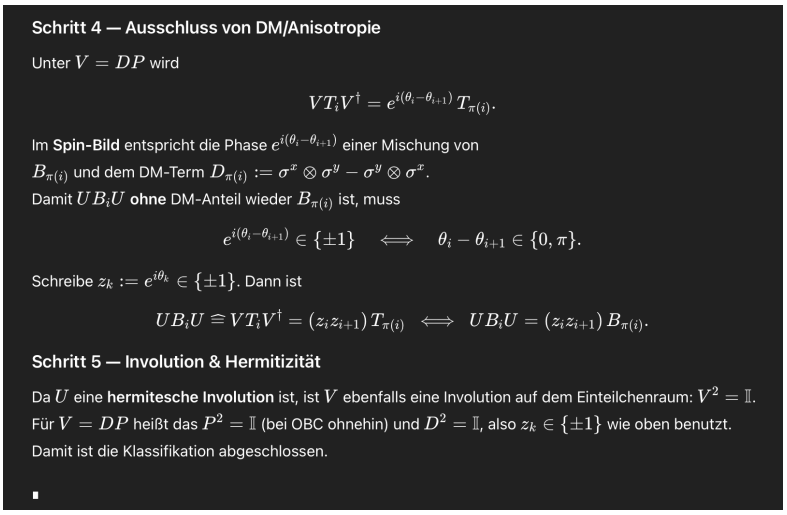


Figure 8.7: Caption jaha3

$\tilde{\Phi}(\rho) \preceq \Phi(\rho)$, e.g. $S(\tilde{\Phi}(\rho)) \geq S(\Phi(\rho))$. Next step: find lower bound on $S(\tilde{\Phi}(\rho))$ that is strictly greater than $S(\Phi(\rho))$.[37]

this was all unnecessary since unitary trafos preserve the spectrum so majorization only depends on if ρ and $T(\rho)$ are majorized by each other. apparently $T(rho)$ is called pinching and there is the pinching proposition (bhatia matrix analysis) that $T(\rho) \preceq \rho$.

Proposition 8.7.1. *for every c.p. unital projection T it is* [38] $S(\rho||T(\rho)) = S(T(\rho)) - S(\rho)$

Proof.

$$S(\rho||T(\rho)) = \mathrm{tr}[\rho(\log\rho - \log T(\rho))] = \mathrm{tr}[\rho \log T(\rho)] - S(\rho) \tag{8.7.1}$$

Now T as an operator on $\mathcal{B}(\mathcal{H})$ is self-adjoint (prove this), so that with $\log T(\rho) = T(\log T(\rho))$[39]:

$$S(\rho||T(\rho)) = S(T(\rho)) - S(\rho) \tag{8.7.2}$$

□

now use pinskers inequality to find that $S(T(\rho)) - S(\rho) \geq ||\rho - T(\rho)||_1^2/2$[40]. With

$$\begin{aligned}(1-2P)(\rho - T(\rho)) &= (1-2P)(\rho - P\rho P - \rho - P\rho P + P\rho + \rho P) \\ &= -2P\rho P + P\rho + \rho P + 4P\rho P - 2P\rho - 2P\rho P \\ &= P\rho - \rho P = [P, \rho]\end{aligned} \tag{8.7.3}$$

and realizing that $(1-2P)$ is a unitary one gets

$$S(T(\rho)) - S(\rho) \geq \frac{1}{2}\,|||[\rho, P]|||_1^2 \tag{8.7.4}$$

there is an upper bound

$$S(T(\rho)) - S(\rho) \leq \log d \tag{8.7.5}$$

oder $\leq \log \mathrm{rk} T(\rho)$ [41]

8.8 ausbreitungsgeschwindigkeit

figure 8.17 gives an upper bound on how fast $\rho(t)$ moves away from starting point $\rho(0)$ which is actually bigger for non-Zeno since the Hamiltonian has greater norm there. another upper

[37] maybe QDS description better
[38] show support relation is fulfilled in def of rel entropy
[39] prove this
[40] verwende hier auch andere ungleichung
[41] prove this

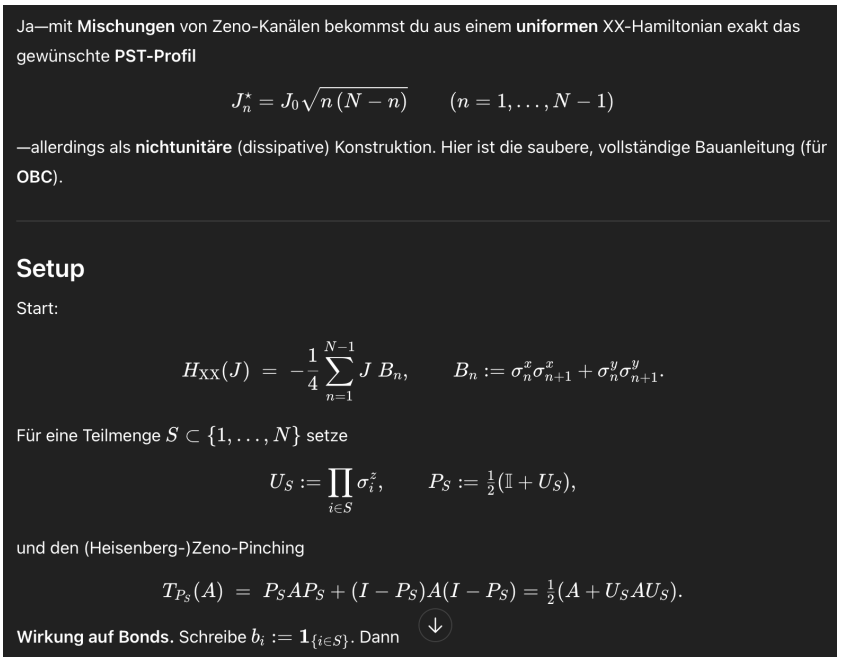


Figure 8.8: Enter Caption

bound uses the length of the spectrum, which also gets shorter[42] globally but can get bigger on a subspace of the spectrum, so if you consider propagation on a subspace, there may be an greater upper bound: a state that lives/starts in $P_i\mathcal{H}$ evolves as $\mathrm{e}^{-iH_it}\rho\mathrm{e}^{iH_it}$ where $H_i = P_iHP_i$ [43], so that then $\mathrm{d}/\,\mathrm{d}t||\rho(t)-\rho(0)|| \leq \Delta(H_i)$ which can be bigger than $\Delta(H)$, [44]

8.9 energy change

try to maximize oder minimize

$$\Delta E = \mathrm{tr}[\rho(t)] \tag{8.9.1}$$

8.10 removing anderson localization

A random system is a system whose Hamiltonian contains random parameters. An example is the tight-bining model

no distributin of corrlation due to destructive interference, maybe pinching can undo this. localization bothers people because transpoert is destroyed

8.11 maximizing mutual information and area law

Let $\rho \in \mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B)$ then the mutual information between A and B is

$$I(A:B) = S(\rho||\rho_A\otimes\rho_B) = S(\rho_A)+S(\rho_B)-S(\rho) \geq \frac{1}{2}||\rho-\rho_A\otimes\rho_B||_1^2 \tag{8.11.1}$$

If $\rho = \rho_A\otimes\rho_B$ is a product state then $I(A:B)=0$ (iff) and otherwise it is positive. In this sense the mutual information is a measure for how much information/correlation there is in a bipartite system. More specifically: Let A, B be any observables with norm less or equal than 1 then

$$I(A:B) \geq \frac{1}{2}\left(\langle A\otimes B\rangle_\rho - \langle A\rangle_{\rho_A}\langle B\rangle_{\rho_B}\right) \tag{8.11.2}$$

[42] prove this
[43] prove this!!!
[44] find examples where it is bigger at first, then find lower bound

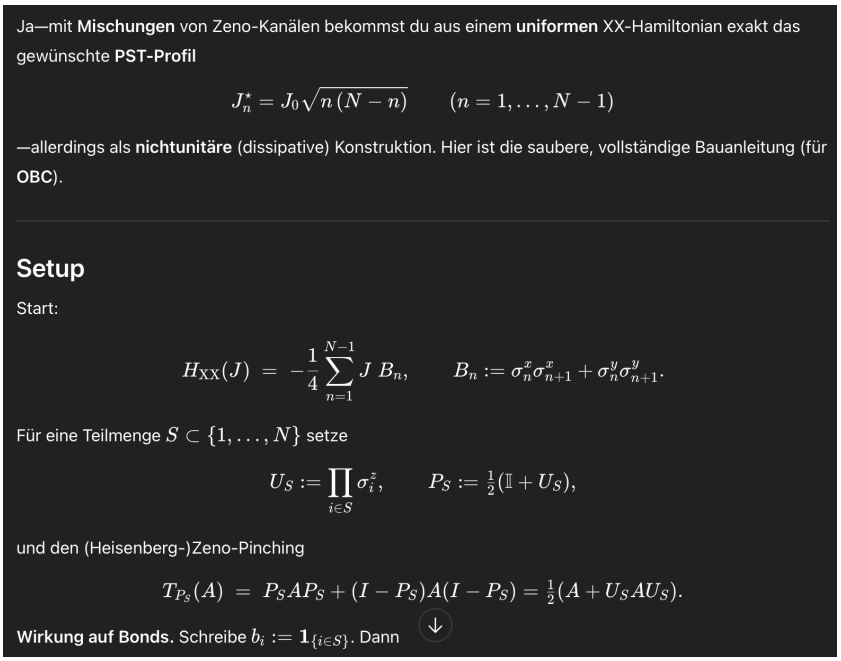


Figure 8.9: Enter Caption

The idea is to compare the mutual information of $\mathrm{e}^{-iHt}\rho(0)\mathrm{e}^{iHt}$ with that of $\mathrm{e}^{-iT(H)t}T(\rho(0))\mathrm{e}^{iT(H)t}$ and try maximizing the latter. We again take the one-dimensional spin chain as an example and will measure on qubits 2 to $n-1$ and consider the state $\rho_{1n} = \mathrm{tr}_{2\dots n-1}[\rho]$. Then

$$I(1:n) = S(\rho_1) + S(\rho_n) - S(\rho_{1n}) \tag{8.11.3}$$

Note that

$$S(\rho_1) = S(\mathrm{tr}_{\neg 1}[U_t\rho(0)U_t^*]) \tag{8.11.4}$$

Using Araki-Lieb $S(\rho_{AB}) \geq |S(\rho_A) - S(\rho_B)|$ we have

$$\begin{aligned} I(A:B) &\leq S(\rho_A) + S(\rho_B) - |S(\rho_A) - S(\rho_B)| \\ &= \left\{ \begin{array}{ll} 2S(\rho_B) & \text{if } S(\rho_A) \geq S(\rho_B) \\ 2S(\rho_A) & \text{else} \end{array} \right\} = 2\min\{S(\rho_A), S(\rho_A)\} \end{aligned} \tag{8.11.5}$$

- through pinching make 1d chain into blocks so fewer boundaries (why)
- hastings area law: entanglement across one boundary cut in 1D chain is O(1) if i increae number of cuts entanglement increases

look at mutual info for last entries very good.

8.12 creating bigger gap in spectrum and possible applications

8.13 free quadratic systems

propagation speed in some kind is known analytically 8.22: for literature see [CC06] or [AC18]

8.14 time-energy inequality

8.15 optimizing overlap with phi+

there are analytic necessary conditions for P, such as $P[G(P), T(H)](\mathbb{1} - P) = 0$. something with grassmannian manifold

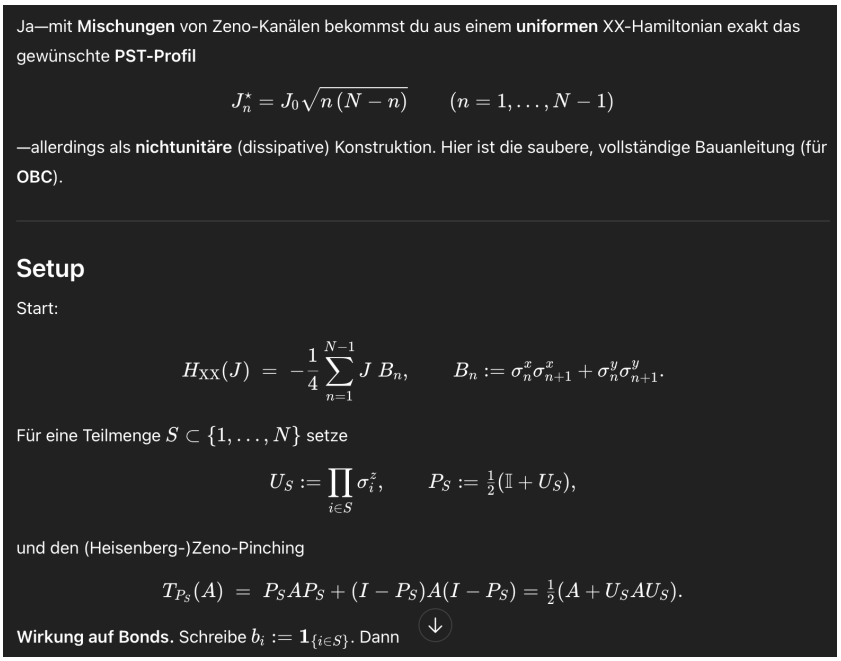


Figure 8.10: Enter Caption

8.16 lower lr bounds

also analytical optimization possible maybe

8.17 increasing lieb robinson velocity

Wirkung auf Bonds. Schreibe $b_i := \mathbf{1}_{\{i\in S\}}$. Dann

$$U_S B_n U_S = (-1)^{b_n + b_{n+1}}\, B_n.$$

Also

$$T_{P_S}(B_n) = \begin{cases} B_n, & b_n = b_{n+1}, \\ 0, & b_n \neq b_{n+1}. \end{cases}$$

Mit anderen Worten: Jeder T_{P_S} erzeugt eine **0/1-Maske** $s(S) \in \{0,1\}^{N-1}$ mit

$$s(S)_n = \mathbf{1}\{b_n = b_{n+1}\}, \qquad T_{P_S}(H_{\mathrm{XX}}(J)) = -\frac{1}{4}\sum_{n=1}^{N-1} J\, s(S)_n\, B_n.$$

Konvexe Hülle = voller Hyperwürfel

Für eine **Mischung** $\Phi = \sum_S p_S\, T_{P_S}$ gilt

$$\Phi\big(H_{\mathrm{XX}}(J)\big) = -\frac{1}{4}\sum_{n=1}^{N-1} J\, q_n\, B_n, \qquad q_n := \sum_S p_S\, s(S)_n \in [0,1].$$

Schlüsselpunkt: Die Menge $\{s(S) : S \subset \{1,\dots,N\}\}$ ist genau $\{0,1\}^{N-1}$.
Konstruktion: Gegeben $s \in \{0,1\}^{N-1}$, wähle $b_1 \in \{0,1\}$ beliebig und setze rekursiv

Figure 8.11: Enter Caption4

Konstruktion: Gegeben $s \in \{0,1\}^{N-1}$, wähle $b_1 \in \{0,1\}$ beliebig und setze rekursiv

$$b_{n+1} := \begin{cases} b_n, & s_n = 1, \\ 1 - b_n, & s_n = 0. \end{cases}$$

Dann liefert $S = \{i : b_i = 1\}$ genau diese Maske $s(S) = s$.

Folgerung: Die **konvexe Hülle** dieser 0/1-Masken ist der ganze **Hyperwürfel** $[0,1]^{N-1}$.
Damit kannst du **beliebige** Faktoren $q = (q_1, \dots, q_{N-1}) \in [0,1]^{N-1}$ realisieren:

$$\exists\, (p_S) \text{ mit } \sum_S p_S = 1,\ p_S \geq 0,\ \text{sodass } q = \sum_S p_S\, s(S).$$

(Carathéodory: In $\mathbb{R}^{N-1}$ reicht eine Mischung mit **höchstens** N Masken.)

Auf das PST-Profil abbilden

Wähle das Start-J groß genug, z. B.

$$J \;\geq\; \max_{1\leq n\leq N-1} J_0\sqrt{n(N-n)}.$$

Setze

Figure 8.12: Caption

Auf das PST-Profil abbilden

Wähle das Start-J groß genug, z. B.

$$J \;\geq\; \max_{1\leq n\leq N-1} J_0\sqrt{n(N-n)}.$$

Setze

$$q_n \;:=\; \frac{J_0\sqrt{n(N-n)}}{J} \;\in [0,1].$$

Da $q \in [0,1]^{N-1}$, gibt es Gewichte (p_S) mit $\sum p_S s(S) = q$. Für

$$\Phi = \sum_S p_S\, T_{P_S}$$

erhältst du

$$\Phi\big(H_{\mathrm{XX}}(J)\big) = -\frac{1}{4}\sum_{n=1}^{N-1} J\, q_n\, B_n = -\frac{1}{4}\sum_{n=1}^{N-1} J_0\sqrt{n(N-n)}\, B_n,$$

also **genau** die Zielkopplungen J_n^*.

Figure 8.13: Caption

Schritt A — Übergangswahrscheinlichkeiten festlegen.
Setze für $n = 1, \ldots, N-1$:

$$\Pr[b_{n+1} = b_n] \;=\; q_n, \qquad \Pr[b_{n+1} \neq b_n] \;=\; 1 - q_n.$$

Starte mit $\Pr[b_1 = 0] = \Pr[b_1 = 1] = \frac{1}{2}$.

Schritt B — Von Bits b zu einer Menge S.
Für eine Realisierung $b = (b_1, \ldots, b_N) \in \{0,1\}^N$ setze

$$S(b) \;:=\; \{\, i \in \{1, \ldots, N\} \,:\, b_i = 1 \,\}.$$

Schritt C — Die Gewichte p_S.
Die Verteilung auf den Mengen S ist die Bildverteilung von $b \mapsto S(b)$.
Explizit (für das S, dessen Indikatorfolge b ist):

$$\boxed{p_{S(b)} \;=\; \tfrac{1}{2} \prod_{n=1}^{N-1} \Big(q_n \, \mathbf{1}\{b_{n+1} = b_n\} \;+\; (1 - q_n)\, \mathbf{1}\{b_{n+1} \neq b_n\} \Big).}$$

Damit gilt automatisch

$$\sum_S p_S = 1, \qquad \mathbb{E}_S\big[\mathbf{1}\{b_n = b_{n+1}\}\big] = q_n,$$

und folglich (Heisenberg):

Figure 8.14: Caption

und folglich (Heisenberg):

$$\Phi(H) \;=\; \sum_S p_S \, T_{P_S}(H) \;=\; -\frac{1}{4} \sum_{n=1}^{N-1} (q_n J)\, B_n.$$

Wählst du J so, dass $q_n J = J_0 \sqrt{n(N-n)}$, bekommst du exakt dein Zielprofil.

Gut zu wissen: Das ist bereits eine **endliche** Mischung (über höchstens 2^N Mengen S). Du **musst** also nichts approximieren.

---

2) (Optional) Support klein halten: $\leq N$ Projektoren

Wenn du die Mischung **kompakt** haben willst, löse das kleine **LP**:

$$\text{Minimiere} \sum_S p_S \quad \text{sodass} \quad \sum_S p_S \, s(S)_n = q_n \;\; (n = 1, \ldots, N-1), \; \sum_S p_S = 1, \; p_S \geq 0.$$

Jede **basische** Lösung hat Stützmenge von Größe $\leq N$ (Carathéodory/LP-Theorie). Praktisch: nimm als Spalten die Binärmasken $s(S) \in \{0,1\}^{N-1}$ und löse mit Simplex; das liefert dir eine Mischung mit höchstens N verschiedenen S.

Figure 8.15: Caption

2. What a pinching does

Mathematically:

$$T(X) = \sum_i P_i X P_i$$

removes all **off-diagonal blocks** of X between the subspaces $P_i\mathcal{H}$.
That is, it sets to zero every component $P_i X P_j$ with $i \neq j$.

Hence, for any operator X,

$$T(X) = X \quad \Leftrightarrow \quad [X, P_i] = 0 \; \forall i,$$

i.e. X is block-diagonal in the measurement basis.

1. Intuitive picture: what is *dephasing*?

In quantum mechanics, everything about interference and localization comes from **relative phases** between components of a wavefunction.

If we write the state as

$$|\psi(t)\rangle = \sum_j c_j(t)\, |j\rangle,$$

then coherence means that the complex numbers $c_j(t)$ have **well-defined phase relations**, so that amplitudes from different paths can **interfere**.

- **Constructive interference** → enhances propagation
- **Destructive interference** → suppresses propagation (→ localization)

Now, if something randomizes those relative phases, interference disappears.
That process is called **dephasing**.

Figure 8.16: pinching does

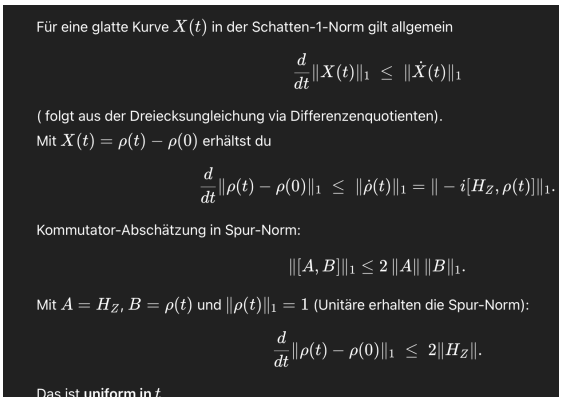


Figure 8.17: Caption

Set-up

Let $H = H^\dagger$ with eigenvalues $\lambda_{\min} \le \cdots \le \lambda_{\max}$ and **spectral diameter**

$$L(H) := \lambda_{\max} - \lambda_{\min}.$$

Write $U_t := e^{-itH}$ and $\rho(t) = U_t\,\rho\,U_t^\dagger$.

Step 1: operator-norm control of $U_t - I$ by $L(H)$

Center H at the mid-spectrum: set $c := (\lambda_{\max} + \lambda_{\min})/2$ and $K := H - c\mathbf{1}$.
Then $\operatorname{spec}(K) \subset [-L(H)/2,\ L(H)/2]$ and

$$U_t = e^{-itH} = e^{-itc}\,e^{-itK},$$

so $\|U_t - I\| = \|e^{-itK} - I\|$ (the global phase e^{-itc} drops out).

For Hermitian K one has the exact identity

$$\|e^{-itK} - I\| \;=\; 2 \max_{\mu \in \operatorname{spec}(K)} |\sin(t\mu/2)| \;\le\; 2\,\sin\!\Big(\frac{t\,L(H)}{4}\Big).$$

(We used that the spectrum of K lies in $[-L(H)/2, L(H)/2]$ and that $\sin$ is increasing on $[0, \pi/2]$.)

Step 2: convert to a trace-distance bound

Use the simple inequality (triangle + unitary invariance of $\|\cdot\|_1$)

$$\|\rho(t) - \rho\|_1 = \|(U_t - I)\rho U_t^\dagger + \rho(U_t^\dagger - I)\|_1 \le 2\,\|U_t - I\|.$$

Combining with Step 1 gives the **spectral-length bound**

$$\boxed{\;\|\rho(t) - \rho\|_1 \;\le\; 4\,\sin\!\Big(\frac{t\,L(H)}{4}\Big) \;\le\; \min\{4,\ t\,L(H)\}\,.\;}$$

(The last inequality uses $\sin x \le x$. For small t: $\|\rho(t) - \rho\|_1 \le L(H)\,t + O(t^3)$.)

Remarks:

- This improves the "derivative" bound $\|\rho(t) - \rho\|_1 \le 2\|H\|\,t$ whenever $L(H) < 2\|H\|$ (after centering, typically the case).
- Channel form: the same argument yields

$$\|\,\mathcal{U}_t - \mathrm{id}\,\|_\diamond \;\le\; 2\,\|U_t - I\| \;\le\; 4\,\sin\!\Big(\tfrac{t\,L(H)}{4}\Big).$$

Pinching (Zeno) cannot increase this bound

Let $\mathcal{P}$ be the Zeno pinching (a unital completely positive, positive map) on Z, and $H_Z := \mathcal{P}(H)$.
Order preservation of positive unital maps implies:

$$\lambda_{\min}(H)\,\mathbf{1} \;\le\; H \;\le\; \lambda_{\max}(H)\,\mathbf{1} \;\;\Longrightarrow\;\; \lambda_{\min}(H)\,\mathbf{1} \le H_Z \le \lambda_{\max}(H)\,\mathbf{1},$$

hence

$$L(H_Z) \;=\; \lambda_{\max}(H_Z) - \lambda_{\min}(H_Z) \;\le\; \lambda_{\max}(H) - \lambda_{\min}(H) = L(H).$$

Therefore the spectral-length bound **cannot get worse** under Zeno pinching:

$$\|\rho_Z(t) - \rho\|_1 \;\le\; 4\,\sin\!\Big(\frac{t\,L(H_Z)}{4}\Big) \;\le\; 4\,\sin\!\Big(\frac{t\,L(H)}{4}\Big).$$

So yes: besides the "speed" bound $\dot\rho(t)$ controlled by $\|H\|$, you have a **global, phase-optimized** upper bound controlled by the **length of the spectrum**, and pinching can only **reduce** that length, never increase it.

4. Explicit example

Take a three-qubit system A, B, Z with

$$H = J(\sigma_x^A\sigma_x^Z + \sigma_x^Z\sigma_x^B),$$

and pinch Z in the **computational (σ_z)** basis:

$$\mathcal{P}_Z(X) = \frac{1}{2}\big(X + \sigma_z^Z X \sigma_z^Z\big).$$

Compute $H_Z := (\mathrm{id}_{AB} \otimes \mathcal{P}_Z)(H)$.

Each term $\sigma_x^A\sigma_x^Z$ and $\sigma_x^Z\sigma_x^B$ anticommutes with σ_z^Z, so their pinching gives zero:

$$\mathcal{P}_Z(\sigma_x^A\sigma_x^Z) = 0, \quad \mathcal{P}_Z(\sigma_x^Z\sigma_x^B) = 0,$$

so $H_Z = 0$.

Now take the reduced "Hamiltonian" on AB:

$$H_{AB}^{(\mathrm{eff})} = \mathrm{tr}_Z\big(H_Z(\mathbf{1}_{AB} \otimes \sigma_Z)\big) = 0.$$

Nothing interesting here — total freezing.

But suppose instead we **pinch in a non-commuting basis**, e.g. the σ_x-basis on Z. Then σ_x^Z survives, and we get

$$H_Z = J(\sigma_x^A + \sigma_x^B) \otimes \sigma_x^Z.$$

Tracing out Z in some mixed state σ_Z with $\mathrm{tr}(\sigma_x^Z\sigma_Z) \neq 0$, we obtain an **effective Hamiltonian**

$$H_{AB}^{(\mathrm{eff})} = J\,\langle\sigma_x^Z\rangle\,(\sigma_x^A + \sigma_x^B),$$

whose eigenvalue spread (on AB) equals $2J|\langle\sigma_x^Z\rangle|$ — possibly **larger** than the pre-pinched effective Hamiltonian derived from the original H, which included cross-terms canceling partially between A and B.

Thus, **local pinching on Z** can *redistribute* the spectral weight and create larger local spectral differences after tracing out Z.

---

5. Mathematical statement (summary)

Let $H_Z = (\mathrm{id} \otimes \mathcal{P}_Z)(H)$. Then:

1. Globally: $L(H_Z) \le L(H)$.
2. For a subsystem $S \subset ABZ$, the reduced operator

$$H_S^{(\mathrm{eff})} := \mathrm{tr}_{S^c}\left(H_Z(\mathbf{1}_S \otimes \sigma_{S^c})\right)$$

   need not satisfy $L(H_S^{(\mathrm{eff})}) \le L(H_S^{(\mathrm{eff},orig)})$.
   In fact, depending on $\mathcal{P}_Z$ and the correlations in H, you can have $>$.

So **yes** — your intuition is absolutely correct:

> 🔷 *Global* pinching is always spectrum-contractive.
> 🔷 *Local* pinching on a subsystem, followed by reduction, can *increase* the spectral range of the **effective** Hamiltonian on the unmeasured part.
> 🔷 This happens because the contraction property is not preserved under partial trace — it's not monotone with respect to operator order.

↓

Figure 8.18: Caption

Setup and a key identity

Let $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_Z$.
A Zeno pinching on Z is $\Phi_{\{P_\alpha\}}(X) = \sum_\alpha (I_{AB} \otimes P_\alpha)\, X\, (I_{AB} \otimes P_\alpha)$.
The Zeno Hamiltonian is $H_Z = \Phi_{\{P_\alpha\}}(H)$.

For any state ρ on ABZ,

$$\boxed{\operatorname{tr}(\rho H_Z) - \operatorname{tr}(\rho H) = -2 \sum_{\alpha<\beta} \operatorname{Re}\, \operatorname{tr}\big(\rho\, (I\otimes P_\alpha)\, H\, (I \otimes P_\beta)\big).} \qquad (*)$$

So the energy change is precisely the negative twice of the *off–block* contributions of H in the chosen block structure on Z, weighted by ρ. If the summed off–block terms are negative, pinching increases the energy; if they are positive, it decreases.

Two immediate corollaries:

- **Bounds.** For every pinching,

$$\lambda_{\min}(H) \;\le\; \operatorname{tr}(\rho H_Z) \;\le\; \lambda_{\max}(H).$$

So you can never exceed the top eigenvalue of H nor go below the bottom one.

- **Commuting case.** If $[H, I_{AB} \otimes P_\alpha] = 0$ for all α (equivalently, you pinch in a spectral decomposition of H on Z), then the RHS of (*) vanishes and the energy is unchanged: $\operatorname{tr}(\rho H_Z) = \operatorname{tr}(\rho H)$.

Existence (explicit 2-level example)

Let AB be trivial and Z a qubit. Take $H = -\sigma_x$ and $\rho = |+\rangle\langle+|$ (the $+1$ eigenstate of σ_x). Then $\operatorname{tr}(\rho H) = -1$.
Pinch in the computational basis on Z, i.e. $P_0 = |0\rangle\langle 0|$, $P_1 = |1\rangle\langle 1|$. This kills the off-diagonal of H: $H_Z = 0$. Hence
$\operatorname{tr}(\rho H_Z) = 0 > -1$.
So Zeno dynamics strictly *raises* the energy expectation here.

How to maximize the energy increase (constructive recipes)

How to maximize the energy increase (constructive recipes)

It's convenient to write the two–outcome case first; it already captures the optimization idea and is often enough in practice.

(A) Two–projector pinching $\{P,\, I - P\}$ on Z

From (*),

$$\Delta_E(P) := \operatorname{tr}(\rho H_Z) - \operatorname{tr}(\rho H) = -2\, \operatorname{Re}\operatorname{tr}\big(\rho\, (I\otimes P)\, H\, (I \otimes (I-P))\big).$$

Introduce the Z-operator

$$K \;:=\; \operatorname{Re}\, \operatorname{tr}_{AB}\big(H\,\rho\big) \qquad (\text{partial trace over } AB).$$

Then a short cyclic-trace manipulation gives

$$\Delta_E(P) \;=\; -2\, \operatorname{tr}\big(PK(I-P)\big).$$

Because K is Hermitian, the right-hand side is nonpositive if P reduces K (block-diagonalizes it), and becomes *most negative* (i.e. Δ_E most *positive*) when P separates the positive and negative spectral parts of K. Concretely:

> **Recipe 1 (two blocks).** Let $P := \mathbf{1}_{\{K\ge 0\}}$ be the spectral projector of K onto its nonnegative eigenspaces (on Z). Then $\Delta_E(P)$ is locally maximal among all two-outcome pinchings $\{P, I-P\}$. Intuition: you cut Z along the sign structure of the Hermitian "energy-current" K, which minimizes the positive cross-term $\operatorname{tr}(PK(I-P))$.

This is easy to implement: compute K (an operator on Z only), diagonalize it, take P onto the span of eigenvectors with $\lambda \ge 0$.

(B) Multi–projector pinching $\{P_\alpha\}$ (rank-r_α blocks)

Write

$$\Delta_E(\{P_\alpha\}) = -2 \sum_{\alpha<\beta} \operatorname{tr}\big(P_\alpha K P_\beta\big).$$

The same logic says you want to choose an orthogonal decomposition of $\mathcal{H}_Z$ so that K is as close to block-diagonal as possible with its large positive and large negative directions segregated into *different* blocks—this kills positive cross-terms and turns negative ones into zeros, maximizing Δ_E.

> **Recipe 2 (many blocks).** Fix ranks r_α you are willing to use on Z. Diagonalize K on Z, sort its eigenvalues decreasingly, and pack the most positive eigenvectors together and the most negative ones together into separate blocks to define the P_α. This greedy spectral packing maximizes $-\sum_{\alpha<\beta} \operatorname{tr}(P_\alpha K P_\beta)$ under the given rank constraints.

Figure 8.19: Caption

Step 1 — A localized Hamiltonian that "bothers people"

Take the standard 1D Anderson tight-binding chain (single excitation / single particle):

$$H_S \;=\; \sum_{j\in\mathbb{Z}} \epsilon_j\, |j\rangle\langle j| \;+\; J \sum_j \big(|j\rangle\langle j{+}1| + |j{+}1\rangle\langle j|\big), \quad \text{with i.i.d. } \epsilon_j.$$

In 1D, any disorder (with mild assumptions) localizes all eigenstates: an initially local wavepacket does **not** spread (Anderson localization).

Step 2 — Add an ancilla "meter" per site and choose the pinching

Attach a two-level ancilla Z_j to each site j. Let the joint Hilbert space be $\mathcal{H} = \mathcal{H}_S \otimes \bigotimes_j \mathcal{H}_{Z_j}$.
Couple **only diagonally in the site basis** so that hopping on S survives:

$$H_{\rm int} = g \sum_j |j\rangle\langle j| \otimes \sigma_j^x, \qquad H_{\rm tot} = H_S \otimes \mathbf{1}_Z + H_{\rm int}.$$

Prepare each ancilla in the σ^x-basis mixture
$\rho_{Z_j}(0) = \frac{1}{2}(|+\rangle\langle+| + |-\rangle\langle-|)$
(or any state diagonal in σ^x).

Now perform **stroboscopic pinching on** Z every $\tau > 0$ in the σ^x **eigenbasis**, discarding the outcomes:

$$\mathcal{T}_Z(X) = \sum_{\vec{s}\in\{+,-\}^n} \Big(\mathbf{1}_S \otimes \Pi_{\vec{s}}\Big)\, X \,\Big(\mathbf{1}_S \otimes \Pi_{\vec{s}}\Big), \quad \Pi_{\vec{s}} = \bigotimes_j |s_j\rangle\langle s_j|.$$

The single cycle map (unitary for time τ under $H_{\rm tot}$ and then pinching) is

$$\Phi_\tau(\rho) = \mathcal{T}_Z\Big(e^{-iH_{\rm tot}\tau}\, \rho\, e^{+iH_{\rm tot}\tau}\Big).$$

Key property. Because $H_{\rm int}$ is diagonal in the $|j\rangle$ basis and the pinching kills **off-diagonals in** Z (not in S), the **system hopping terms** survive; what is removed is the **inter-trajectory coherence in** Z that would otherwise preserve phase memory between steps.

Step 3 — Effective reduced dynamics on S: pinching ⇒ site-dephasing

Track only the system S: $\rho_S = \mathrm{Tr}_Z\, \rho$.
For small τ, a second-order expansion of one cycle gives

$$\rho_S \;\mapsto\; \rho_S - i\tau[H_S, \rho_S] \;+\; \tau\,\gamma \sum_j \Big(|j\rangle\langle j|\,\rho_S\,|j\rangle\langle j| - \tfrac{1}{2}\{|j\rangle\langle j|, \rho_S\}\Big) \;+\; O(\tau^2),$$

with a **dephasing rate** $\gamma \sim g^2\tau$ (the exact prefactor depends on short-time details; the scaling in g, τ is robust).

For small τ, a second-order expansion of one cycle gives

$$\rho_S \;\mapsto\; \rho_S - i\tau[H_S, \rho_S] \;+\; \tau\,\gamma \sum_j \Big(|j\rangle\langle j|\,\rho_S\,|j\rangle\langle j| - \tfrac{1}{2}\{|j\rangle\langle j|, \rho_S\}\Big) \;+\; O(\tau^2),$$

with a **dephasing rate** $\gamma \sim g^2\tau$ (the exact prefactor depends on short-time details; the scaling in g, τ is robust).

Passing to the stroboscopic limit (n steps, total time $t = n\tau$, keeping γ fixed) yields the **Lindblad master equation on** S:

$$\dot{\rho}_S \;=\; -i[H_S, \rho_S] \;+\; \gamma \sum_j \Big(L_j \rho_S L_j - \tfrac{1}{2}\{L_j, \rho_S\}\Big), \qquad L_j = |j\rangle\langle j|.$$

That is **pure dephasing in the site basis** with rate γ, while **hopping** J remains intact. Importantly, we did **not** pinch S itself and we did **not** project away the hopping blocks; we only dephased Z so that S experiences classical phase noise.

> **Why pinching = dephasing here?**
> The pinching on Z removes off-diagonals of the joint state in the σ^x basis after each short interaction. This turns the coherent ancilla back-action into **classical, memoryless phase kicks** on the $|j\rangle$ occupation of S, which is exactly site-dephasing.

Step 4 — Consequence: Anderson localization disappears

For the disordered H_S, unitary evolution localizes. Under **any** $\gamma > 0$ site-dephasing, the interference responsible for localization is destroyed; the long-time, large-scale dynamics becomes **diffusive** (for moderate γ) with an effective diffusion constant

$$D \;\sim\; \frac{J^2}{\gamma} \quad (\text{intermediate-}\gamma\text{ regime}).$$

As $\gamma \to \infty$ you hit the Zeno freeze (transport suppressed again), so there is an **optimal finite** γ where diffusion is maximal. But crucially, **localization is gone** once $\gamma > 0$.

Step 5 — Why simpler "pinch S in site basis" fails

If you instead pinch S in the site basis, the Zeno Hamiltonian is
$H_Z = \sum_j |j\rangle\langle j|H_S|j\rangle\langle j| = \sum_j \epsilon_j |j\rangle\langle j|$
(the hopping is projected out). That **freezes** motion (no transport), i.e. you haven't "cured" localization—you've made it absolute.
Our construction avoids that by pinching **only the ancillas** that *mediate dephasing* while leaving system hopping untouched.

Figure 8.20: anderson

3) Zeno-specific upper bounds on $I(A{:}B)_{\rho_{AB}(t)}$

Let $m = \#\mathrm{supp}(\mathbf{p})$.

3.1 Label-entropy cap (time-invariant)

Introduce an explicit classical register C storing α:

$$\omega_{CAB}(t) := \sum_\alpha p_\alpha \; |\alpha\rangle\langle\alpha|_C \otimes \sigma_A^{(\alpha)}(t) \otimes \tau_B^{(\alpha)}(t).$$

Since $A \perp B \mid C$ (conditional independence),

$$I(A{:}B)_\omega \;=\; I(A{:}C)_\omega + I(B{:}C)_\omega - I(AB{:}C)_\omega \;\le\; \min\{I(A{:}C)_\omega,\; I(B{:}C)_\omega\}.$$

By data processing (discarding C cannot increase correlations),

$$I(A{:}B)_{\rho_{AB}(t)} \le I(A{:}B)_\omega \le \min\{I(A{:}C)_\omega,\; I(B{:}C)_\omega\} \le H(\mathbf{p}) \le \log m.$$

Thus

$$\boxed{I(A{:}B)_{\rho_{AB}(t)} \;\le\; H(\mathbf{p}) \;\le\; \log m, \quad \text{and } H(\mathbf{p}) \text{ is time-invariant.}} \tag{U1}$$

This is **nontrivial and Zeno-specific**: the cap depends only on the *initial* block weights and the chosen pinching.

3.2 Holevo-refined cap (often strictly tighter)

Recall the Holevo information of an ensemble $\{p_\alpha, \sigma_A^{(\alpha)}(t)\}$:

$$\chi_A(t) := S\Big(\sum_\alpha p_\alpha \sigma_A^{(\alpha)}(t)\Big) - \sum_\alpha p_\alpha S\big(\sigma_A^{(\alpha)}(t)\big).$$

A standard cq-identity gives $I(A{:}C)_\omega = \chi_A(t)$ and $I(B{:}C)_\omega = \chi_B(t)$. Hence

$$\boxed{I(A{:}B)_{\rho_{AB}(t)} \;\le\; \min\{\chi_A(t), \chi_B(t), H(\mathbf{p})\}.} \tag{U2}$$

This upper bound **reflects the time-dependence** (via the internal block dynamics), but never exceeds the **time-invariant** label cap $H(\mathbf{p})$.

4) Lower bounds on $I(A{:}B)_{\rho_{AB}(t)}$

The mixture representation (2.1) shows that $I(A{:}B)$ is entirely due to how much the *marginals* depend on α. We give two rigorous families of lower bounds.

4.1 State-intrinsic Pinsker lower bound

Use the identity

$$I(A{:}B)_\rho \;=\; D\big(\rho_{AB} \,\|\, \rho_A \otimes \rho_B\big),$$

and Pinsker's inequality $D(\xi\|\zeta) \ge \frac{1}{2\ln 2}\|\xi - \zeta\|_1^2$ to get

$$\boxed{I(A{:}B)_{\rho_{AB}(t)} \;\ge\; \frac{1}{2\ln 2}\,\big\|\rho_{AB}(t) - \rho_A(t)\otimes\rho_B(t)\big\|_1^2.} \tag{L1}$$

This is fully rigorous, basis-free, and requires no choices. It becomes *computable* once you can estimate that trace distance (e.g., by semidefinite programming bounds, or by evaluating it exactly for your finite sizes).

A useful identity (exact, no approximation) that sometimes helps to bound the norm from below/above is the **covariance form**

$$\rho_{AB}(t) - \rho_A(t)\otimes\rho_B(t) = \sum_\alpha p_\alpha \big(\sigma_A^{(\alpha)} - \rho_A\big) \otimes \big(\tau_B^{(\alpha)} - \rho_B\big) = \tfrac{1}{2}\sum_{\alpha,\beta} p_\alpha p_\beta \,\big(\sigma_A^{(\alpha)} - \sigma_A^{(\beta)}\big) \otimes \big(\tau_B^{(\alpha)} - \tau_B^{(\beta)}\big).$$

(First equality is a direct expansion; the second is the standard $\mathbb{E}[(X-\mathbb{E}X)\otimes(Y-\mathbb{E}Y)] =$

Figure 8.21: Caption

Sei $\hat{H}$ ein freier quadratischer Hamiltonian (z. B. auf einem Gitter oder im Kontinuum) für Fermionen oder Bosonen ohne nicht-triviale Wechselwirkungen, der durch eine geeignete Fourier-/Spektraltransformation diagonalisiert werden kann in Einteilchenmoden:

$$\hat{H} = \sum_k \varepsilon(k)\, \hat{\gamma}_k^\dagger \hat{\gamma}_k,$$

wobei k der Impuls (oder allgemeiner: Spektralparameter) ist und $\varepsilon(k)$ die Dispersionsrelation.

Dann gilt: Eine lokal eingeführte Störung oder Korrelation (z. B. $\langle \hat{c}_x(t)\hat{c}_0^\dagger(0)\rangle$) breitet sich nicht beliebig schnell aus, sondern zeigt eine *Front* bzw. einen *Lichtkegel*; die Grenze dieses Kegels ist durch die maximale Gruppengeschwindigkeit

$$v_{\max} = \max_k \|\nabla_k \varepsilon(k)\|$$

gegeben. Für $|x| > v_{\max} t$ ist der Beitrag der Wellenzahlmoden asymptotisch stark unterdrückt, während für $|x| < v_{\max} t$ stationäre-Phase-Beiträge dominieren. Das ergibt eine scharfe Ausbreitung mit Geschwindigkeit $v_{\max}$ (bei paarweise erzeugten Quasiteilchen kann effektiv $2v_{\max}$ auftreten, wenn zwei gegenläufige Moden auseinanderlaufen).

Figure 8.22: Caption

Gute Frage — der Punkt ist subtil: „Quadratisch“ bezieht sich nur darauf, dass der Hamiltonoperator in den Erzeugungs-/Vernichtungsoperatoren (oder Ort/Impuls) bis zur zweiten Potenz auftaucht. „Frei“ bedeutet aber zusätzlich, dass man das System durch eine lineare Transformation in unabhängige Moden diagonalisiert, d.h. es gibt *keine* effektiven Wechselwirkungen zwischen den Quasiteilchen. Man kann also einen Hamiltonian haben, der formal quadratisch aussieht, aber wegen einer nichtlinearen Nebenbedingung oder Projektion **nicht frei** ist.

Ein konkretes Beispiel:
Der enge-Bindungs-Hamiltonian mit *Gutzwiller-Projektion*, also ohne doppelt besetzte Gitterplätze:

$$\hat{H} = \sum_{\langle i,j\rangle,\sigma} t\, \mathcal{P}\, \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma}\, \mathcal{P},$$

wobei $\mathcal{P} = \prod_i (1 - \hat{n}_{i\uparrow}\hat{n}_{i\downarrow})$ der Projektor ist, der Doppelbesetzungen ausschließt. Formal ist $\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}$ quadratisch in Fermionoperatoren, aber durch das Einschieben von $\mathcal{P}$ wird das System *nicht frei*: die Projektion koppelt die Bewegung eines Teilchens an den Besetzungszustand der Umgebung, und es entstehen effektive Wechselwirkungen. Man kann diesen Hamiltonian nicht durch eine einfache unitäre Transformation in eine Summe unabhängiger Einteilchenmoden diagonalisiert bekommen.

...

Figure 8.23: what is a free quadratic hamiltonian and its speed

2. Herleitungsskizze (stationäre-Phase-Analyse)

Nehmen wir als Beispiel eine Einteilchen-Greensfunktion (oder Korrelationsfunktion) für ein freies Syst in eindimensionalem Gitter:

$$G(x,t) = \langle 0|\hat{c}_x(t)\hat{c}_0^\dagger(0)|0\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} dk\, e^{i(kx - \varepsilon(k)t)} f(k),$$

wobei $f(k)$ eine glatte Gewichtung ist, die vom Anfangszustand kommt (für eine lokalisierte Erzeugun ist sie typischerweise konstant oder langsam variierend).

Für große x, t mit festem Verhältnis $\frac{x}{t}$ analysiert man das Integral asymptotisch mittels **Methode der stationären Phase**. Die Phase ist $\Phi(k) = kx - \varepsilon(k)t$. Stationäre Punkte erfüllen

$$\Phi'(k) = x - t\varepsilon'(k) = 0 \quad \Longrightarrow \quad \frac{x}{t} = \varepsilon'(k) = v_g(k).$$

Das heißt: Nur diejenigen Impulse k^* tragen dominierend bei, deren Gruppengeschwindigkeit $v_g(k^*) = \varepsilon'(k^*)$ mit dem Verhältnis x/t übereinstimmt. Wenn es kein k mit $\frac{x}{t} = v_g(k)$ gibt, d.h. wenn $|x/t| > \max_k |\varepsilon'(k)| = v_{\max}$, dann gibt es keine stationäre Phase im Integrationsbereich, und das Integral fäl schnell ab (durch wiederholte Integration per Partes oder komplexe Analyse bekommt man eine starke Unterdrückung, typischerweise algebraisch oder exponentiell je nach Glattheit).

Dagegen: Für $|x| \le v_{\max} t$ existieren stationäre Punkte, und die Beiträge sind von Ordnung $t^{-1/2}$ (im eindimensionalen Fall) mit Oszillationen bestimmt durch $\Phi(k^*)$. Das ist der *ballistische Lichtkegel*: Korrelationen „erreichen“ Ort x zur Zeit t nur, wenn $|x|/t \le v_{\max}$.

Im mehrdimensionalen Fall wird $\varepsilon'(k)$ durch Gradienten ersetzt, und man fordert $\frac{x}{t} = \nabla_k \varepsilon(k)$ projekt auf die Richtung von x, was analog zum Lichtkegel führt mit

$$v_{\max} = \max_k \|\nabla_k \varepsilon(k)\|.$$

Schluss: Die „Ausbreitungsgeschwindigkeit“ ist also analytisch bestimmbar: sie ist exakt die maximale Gruppengeschwindigkeit, die aus der Dispersionsrelation $\varepsilon(k)$ folgt.

---

3. Voraussetzungen / warum das in *jedem* freien/quasi-freien Fall funktioniert

- **Diagonalisierbarkeit**: Ein freier quadratischer Hamiltonian lässt sich (z. B. durch Fourier- oder Bogoliubov-Transformation, je nach Paarbildun ↓ n unabhängige Moden bringen mit einer wohldefinierten Dispersionsrelation $\varepsilon(k)$

Figure 8.24: what is a free quadratic hamiltonian

Setup (kein Bridging $A \leftrightarrow B$)

Arbeitsraum $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_Z$ mit Zerlegung von Z durch paarweise orthogonale Projektoren $\{P_j\} \subset \mathcal{B}(\mathcal{H}_Z)$, $\sum_j P_j = \mathbf{1}_Z$.
"Kein Bridging" heißt: Jeder Summand von H hat Trägerschnitt nur in $A\otimes Z$ oder $B\otimes Z$ (plus lokale Terme). Das ist für den folgenden Varians-Vergleich nicht einmal nötig; entscheidend ist nur, dass gepinchт **auf** Z wird:

$$\mathcal{E}(X) \;:=\; \sum_j (\mathbf{1}_{AB}\otimes P_j)\, X\, (\mathbf{1}_{AB}\otimes P_j), \qquad H_{\text{Zeno}} := \mathcal{E}(H).$$

Zerlegung in Z-Blöcke

Schreibe

$$H = \sum_{j,k} H_{jk}, \qquad H_{jk} := (\mathbf{1}\otimes P_j)\, H\, (\mathbf{1}\otimes P_k),$$

und ebenso $\tilde H := H - \langle H\rangle_\psi \mathbf{1}$ mit $\langle X\rangle_\psi := \langle\psi|X|\psi\rangle$.
Dann

$$\Delta_\psi(H)^2 = \|\tilde H\psi\|^2 = \| \sum_{j,k} H_{jk}\psi\|^2 = \sum_j \Big\| \sum_k H_{jk}\psi\Big\|^2,$$

weil die Summanden mit verschiedenen **linken** Indizes j orthogonal sind (sie liegen in orthogonalen Z-Unterräumen $\operatorname{ran} P_j$).
Für das gepinchte $H_{\text{Zeno}} = \sum_j H_{jj}$ erhält man

$$\Delta_\psi(H_{\text{Zeno}})^2 = \Big\| \sum_j H_{jj}\psi\Big\|^2 = \sum_j \|H_{jj}\psi\|^2.$$

Damit folgt unmittelbar die **Identität**

$$\boxed{\Delta_\psi(H)^2 - \Delta_\psi(H_{\text{Zeno}})^2 = \sum_j \Big(\| \sum_{k\neq j} H_{jk}\psi\|^2 + 2\,\Re\langle H_{jj}\psi, \sum_{k\neq j} H_{jk}\psi\rangle\Big)} \qquad (\bigstar)$$

Der zweite Term verschwindet, weil

$$\langle H_{jj}\psi, H_{jk}\psi\rangle = \langle\psi|H_{jj}H_{jk}|\psi\rangle = \langle\psi|(\mathbf{1}\otimes P_j)\, H\, (\mathbf{1}\otimes P_j)\, (\mathbf{1}\otimes P_j)\, H\, (\mathbf{1}\otimes P_k)|\psi\rangle = 0 \quad (k\neq j).$$

Somit ist

$$\boxed{\Delta_\psi(H)^2 - \Delta_\psi(H_{\text{Zeno}})^2 = \sum_j \| \sum_{k\neq j} H_{jk}\psi\|^2 \;\geq\; 0} \qquad (\dagger)$$

und die behauptete Monotonie folgt. Gleichheit $\Leftrightarrow$, $\downarrow$ Off-Diagonal-Beiträge $H_{jk}\psi$ (für $k\neq j$) verschwinden, also $H\psi$ liegt schon in der blockdiagonalen Algebra.

Short answer: even **with bridging allowed**, nothing changes for non-selective Zeno via **pinching**.
For any orthogonal projections $\{P_j\}$ (local on Z or not), and any Hermitian H,

$$H_Z := \sum_j P_j H P_j \quad \Rightarrow \quad \Delta_\psi(H_Z) \le \Delta_\psi(H) \qquad (\text{all } \psi).$$

So you still cannot increase ΔE (hence cannot lower the Mandelstam–Tamm time) by pinching—bridging terms don't help.

Why this remains true (one-line proof)

Write $H = \sum_{j,k} H_{jk}$ with $H_{jk} := P_j H P_k$ and center $\tilde H := H - \langle H\rangle_\psi \mathbf{1}$. Then

$$\Delta_\psi(H)^2 = \|\tilde H\psi\|^2 = \sum_j \Big\| \sum_k \tilde H_{jk}\psi\Big\|^2 \ge \sum_j \|\tilde H_{jj}\psi\|^2 = \Delta_\psi\big(\sum_j P_j H P_j\big)^2,$$

because the vectors $\sum_k \tilde H_{jk}\psi$ (fixed left index j) lie in mutually orthogonal subspaces $\operatorname{ran} P_j$. This argument **does not** assume or use "no bridging"; it only uses orthogonality of the P_j.

Figure 8.25: Caption

Final statement (precise lower bound)

Proposition (Lower LR-like bound, finite OBC XX chain).
Fix $J > 0$. There exist universal constants $C > 0$ and T_0 such that for every chain length N and for every pair of bulk sites $1 \ll x < y \ll N$, if $t \geq T_0$ and

$$t \;\leq\; \frac{\min\{x-1,\, N-y\}}{2J}$$

(pre-reflection), then there exists a choice of local norm-one observables $A \in \mathrm{Alg}\{x\}$, $B \in \mathrm{Alg}\{y\}$ (e.g. $A = \gamma_x^{(1)}$, $B = B_t$ as above) such that

$$\big\|[A(t), B]\big\| \;\geq\; C\, t^{-1/3}.$$

Moreover, this lower bound is realized when the separation $\ell = y - x$ is tuned to the **Airy front**,

$$\ell \;=\; 2Jt + s_0 (2Jt)^{1/3} + O(1),$$

for a fixed s_0 at which $|\,\mathrm{Ai}(-s_0)| > 0$.

Comments / variants

- If you insist on **time-independent** B, choose $B = \gamma_y^{(2)}$ or $B = \gamma_y^{(1)}$. Then $\|[\gamma_x^{(1)}(t), B]\| = 2|\Im(e^{-i\varphi} U_{xy}(t))|$ for some fixed $\varphi \in \{0, \pi/2\}$. Since the phase of $U_{xy}(t)$ varies smoothly with t, one still gets a $t^{-1/3}$ subsequence (in t) with the same scaling. The **supremum** over local B gives the cleanest statement (8).
- Using number operators gives $\|[n_x(t), n_y]\| \asymp |U_{xy}(t)|^2$, hence a $t^{-2/3}$ lower bound; this is why one uses operators **linear** in the fermions (or Pauli X/Y) to see the $t^{-1/3}$ front.
- The argument above is entirely **model-explicit** (no abstract LR theory): the $t^{-1/3}$ originates from the cubic turning point at the maximal group velocity, i.e. a universal Airy front for 1D cosine bands.

If you want, I can also write the same derivation purely in the sine-sum form (1) via stationary phase around $k = \pi/2$ without passing through Bessel's closed form; it gives the same Airy law and the same $t^{-1/3}$ prefactor.

Setup (XX $\to$ free fermions)

Consider the length-N XX chain (I fix the usual sign convention; any overall sign of J does not affect the statements below):

$$H \;=\; -\frac{J}{2} \sum_{x=1}^{N-1} \left(\sigma_x^x \sigma_{x+1}^x + \sigma_x^y \sigma_{x+1}^y \right) \qquad (J > 0).$$

Under the Jordan–Wigner transform, H becomes a free nearest-neighbour hopping Hamiltonian

$$H \;=\; -J \sum_{x=1}^{N-1} \left(c_{x+1}^\dagger c_x + c_x^\dagger c_{x+1} \right),$$

with canonical anticommutation relations (CAR) $\{c_x, c_y^\dagger\} = \delta_{xy}$, $\{c_x, c_y\} = 0$.

Let h be the single-particle OBC hopping matrix $(h\psi)(x) = -J(\psi(x+1) + \psi(x-1))$ on $\{1, \dots, N\}$ with Dirichlet ends. Then in Heisenberg picture

$$c_x(t) \;=\; \sum_{y=1}^{N} U_{xy}(t)\, c_y, \qquad U(t) := e^{ith}.$$

The OBC eigenbasis is the **discrete sine transform**:

$$\phi_n(x) = \sqrt{\frac{2}{N+1}} \sin(k_n x), \quad k_n := \frac{\pi n}{N+1}, \quad n = 1, \dots, N,$$

with dispersion $\varepsilon(k) = -2J \cos k$. Hence

$$U_{xy}(t) = \sum_{n=1}^{N} \phi_n(x) \phi_n(y)\, e^{-ic(k_n)t} = \frac{2}{N+1} \sum_{n=1}^{N} \sin(k_n x) \sin(k_n y)\, e^{\,i\,2Jt \cos k_n}. \tag{1}$$

From kernels to commutators (choose local observables)

Define site-localized **Majorana** operators

$$\gamma_x^{(1)} := c_x + c_x^\dagger, \qquad \gamma_x^{(2)} := \frac{c_x - c_x^\dagger}{i}.$$

Both are self-adjoint and have operator norm $\|\gamma_x^{(a)}\| = 1$.

A short CAR computation gives, for any x, y,

$$\big[\gamma_x^{(1)}(t), \gamma_y^{(2)}\big] = 2i\, \Im U_{xy}(t), \qquad \big[\gamma_x^{(1)}(t), \cos\theta\, \gamma_y^{(2)} + \sin\theta\, \gamma_y^{(1)}\big] = 2i\, \Im\big(e^{-i\theta} U_{xy}(t)\big).$$

Optimizing over the phase $\theta = \arg U_{xy}(t)$ yields

$$\sup_{\substack{B \in \mathrm{Alg}\{y\},\, \|B\| = 1 \\ B \text{ linear in } c_y, c_y^\dagger}} \big\|[\gamma_x^{(1)}(t), B]\big\| \;=\; 2\,|U_{xy}(t)|. \tag{2}$$

Thus, an **existential lower bound** for commutators of strictly local, norm-1 observables follows from a lower bound on $|U_{xy}(t)|$.

(Remark.) If you prefer Pauli operators, one may use
$\sigma_y^x = \big(\prod_{z<y} (1 - 2c_z^\dagger c_z)\big)(c_y + c_y^\dagger)$,
$\sigma_y^y = \cdots$,
and obtain the same scaling (the JW string cancels in a two-point commutator and only contributes a

Figure 8.26: lower commutator bound xx chain

Great point — let me make the **propagator ⇒ commutator** step explicit and fully rigorous (for the finite OBC XX chain; afterwards I note the general quadratic case).

Step 1 — Heisenberg evolution of the site fermions

For the XX chain (after JW), the Heisenberg evolution is linear:

$$c_x(t) = \sum_z U_{xz}(t)\, c_z, \qquad c_x^\dagger(t) = \sum_z \overline{U_{xz}(t)}\, c_z^\dagger,$$

with the single-particle kernel $U(t) = e^{ith}$.

Introduce **Majoranas** at site x:

$$\gamma_x^{(1)} := c_x + c_x^\dagger, \qquad \gamma_x^{(2)} := \frac{c_x - c_x^\dagger}{i}.$$

They are self-adjoint and $\|\gamma_x^{(a)}\| = 1$.

Then

$$\gamma_x^{(1)}(t) = \sum_z \Big(U_{xz}(t)\, c_z + \overline{U_{xz}(t)}\, c_z^\dagger \Big).$$

Step 2 — Exact commutator identity

Using the CAR $\{c_z, c_y^\dagger\} = \delta_{zy}$, $\{c_z, c_y\} = 0$, one computes

$$\big[\gamma_x^{(1)}(t), \gamma_y^{(1)}\big] = 2i\, \Im U_{xy}(t), \quad \big[\gamma_x^{(1)}(t), \gamma_y^{(2)}\big] = 2i\, \Re U_{xy}(t).$$

Therefore, for any phase θ,

$$\Big[\gamma_x^{(1)}(t),\; B_\theta\Big] = 2i\, \Re\big(e^{-i\theta} U_{xy}(t)\big), \qquad B_\theta := \cos\theta\, \gamma_y^{(2)} + \sin\theta\, \gamma_y^{(1)}.$$

Choose $\theta = \arg U_{xy}(t)$ to align the phase; since $\|B_\theta\| = 1$,

Bibliography

[Con90] John B. Conway. *A Course in Functional Analysis*. 2nd ed. Springer, 1990.

[Bha97] Rajendra Bhatia. *Matrix Analysis*. Vol. 169. Springer, 1997.

[Bur+20] Daniel Burgarth et al. "Quantum Zeno Dynamics from General Quantum Operations". In: *Quantum* 4 (July 2020), p. 289.

[MW19] Tim Möbus and Michael M. Wolf. "Quantum Zeno effect generalized". In: *Journal of Mathematical Physics* 60.5 (May 2019).

[BZ18] Norbert Barankai and Zoltán Zimborás. *Generalized quantum Zeno dynamics and ergodic means*. 2018.

[Car10] Eric Carlen. "Trace inequalities and quantum entropy: An introductory course". In: 529 (Jan. 2010).

[Tom16] Marco Tomamichel. *Quantum Information Processing with Finite Resources*. Springer International Publishing, 2016.

[Chr+04] Matthias Christandl et al. "Perfect State Transfer in Quantum Spin Networks". In: *Phys. Rev. Lett.* 92 (18 May 2004), p. 187902.

[Zwi+11] Analia Zwick et al. "Robustness of spin-coupling distributions for perfect quantum state transfer". In: *Phys. Rev. A* 84 (2 Aug. 2011), p. 022311.

[CC06] Pasquale Calabrese and John Cardy. "Time Dependence of Correlation Functions Following a Quantum Quench". In: *Phys. Rev. Lett.* 96 (13 Apr. 2006), p. 136801.

[AC18] Vincenzo Alba and Pasquale Calabrese. "Entanglement dynamics after quantum quenches in generic integrable systems". In: *SciPost Phys.* 4 (2018), p. 017.

Appendix A

Codes

```
import qutip as q
import numpy as np

def projector(U, r):
    if isinstance(U, q.Qobj):
        U = U.full()
    D = np.zeros((4, 4))
    D[:r, :r] = np.eye(r)
    return q.Qobj(U @ D @ U.conj().T)

d = 16
sx = q.sigmax()
sy = q.sigmay()
sz = q.sigmaz()
s0 = q.qeye(2)
e1 = q.basis(2, 0)
e2 = q.basis(2, 1)



def PABZ(P):
    temp = q.tensor(s0, s0, P)
    temp = q.Qobj(temp.full())
    temp.dims = [[[2,2], [2,2]], [[2,2], [2,2]]]
    return temp

def state_ev(rho,H,t):
    U = (-1j * H * t).expm()
    return U*rho*U.dag()

def T(H,P):
    id = q.qeye(16)
    id.dims = [[[2,2], [2,2]], [[2,2], [2,2]]]
    return P*H*P + (id - P)*H*(id - P)

def zeno_state_ev(rho,H,P,t):
    return state_ev(T(rho,P), T(H,P), t)
```

Listing A.1: A few methods for calculating the Zeno dynamics

the systems are ordered 1423

```
import time
import math
from definitions import *

plt.rcParams.update({
    "text.usetex": True,
    "font.family": "serif",
    "text.latex.preamble": r"\usepackage{braket}"
})

start_time = time.time()

d = 16
```

```
rho_0 = q.tensor(e1, e1, e1, e1)
rho_0 = rho_0.proj()
rho_0.dims = [[[2,2], [2,2]], [[2,2], [2,2]]]


H = q.tensor(sx, s0, sx, s0) + q.tensor(sy, s0, sy, s0) + q.tensor(s0, sx, s0, sx)
     + q.tensor(s0, sy, s0, sy)
H.dims = [[[2,2], [2,2]], [[2,2], [2,2]]]
randomU = q.rand_unitary(4)
randomU.dims = [[2,2], [2,2]]
t_span = np.linspace(0, 15, 5000)

arr = []
for i in range(1,5):
    print(i)
    P = PABZ(projector(randomU, i))
    P.dims = [[[2, 2], [2, 2]], [[2, 2], [2, 2]]]
    temp = np.array([q.entropy.negativity(q.ptrace(zeno_state_ev(rho_0, H, P, t),
    [0,1]), 1) for t in t_span])
    temp[np.isclose(temp, 0, atol=1e-10)] = 0.0
    arr.append(temp)

arr = np.array(arr)
Negativ = np.array([q.entropy.negativity(q.ptrace(state_ev(rho_0, H, t), [0,1]),
    1) for t in t_span])
Negativ[np.isclose(Negativ, 0, atol=1e-10)] = 0.0

fig, axs = plt.subplots(2, 2, figsize=(10, 8))  # 2 Zeilen, 2 Spalten
for i in range(len(arr)):
    axs[math.floor(i/2), i % 2].plot(t_span, arr[i], label="Zeno")
    axs[math.floor(i / 2), i % 2].plot(t_span, Negativ, label="no Zeno")
    axs[math.floor(i/2), i % 2].legend()
    axs[math.floor(i/2), i % 2].set_title(r'Rank $r = {}$'.format(i+1))


plt.savefig('first_try.pdf',dpi=600)
plt.show()


plt.figure()
f = np.array([f(rho_0, H, t) for t in t_span])
f_zeno = np.array([f_zeno(rho_0, H, P, t) for t in t_span])
plt.plot(t_span, f)   # Linienplot
print('hallo2')
plt.plot(t_span, f_zeno)
print('hallo3')
plt.legend(['no Zeno', 'Zeno'])
plt.xlabel("t")           # Achsenbeschriftung
plt.ylabel("f")
plt.title(r'Overlap with $|\Phi^+ \rangle $')
plt.grid()
plt.show()
```

Listing A.2: Calculating and plotting the negativity in the XX model

```
from definitions import *
import time
import qutip as q
import numpy as np
import cma
import sys, os

# Disable
def blockPrint():
    sys.stdout = open(os.devnull, 'w')

# Restore
def enablePrint():
    sys.stdout = sys.__stdout__

start_time = time.time()
rho_0 = q.tensor(e1, e1, e1, e1)
rho_0 = rho_0.proj()
rho_0.dims = [[[2,2], [2,2]], [[2,2], [2,2]]]
```

```
t = 15

def qf(Z):
    Q, R = np.linalg.qr(Z)
    s = np.sign(np.diag(R))
    s[s == 0] = 1.0
    return Q @ np.diag(s)


def theta_to_P(theta, r=2):
    theta = np.asarray(theta, dtype=float)
    re = theta[0:4*r].reshape(4, r)
    im = theta[4*r:8*r].reshape(4, r)
    Z = re + 1j * im
    U = qf(Z)
    return U @ U.conj().T  # Projektor

def cost_from_projector(P):
    P = PABZ(P)
    return -float((q.entropy.negativity(q.ptrace(zeno_state_ev(rho_0, H, P, t),
    [0,1]), 1).real))  # Platzhalter

def objective(theta, r=2):
    return cost_from_projector(theta_to_P(theta, r=r))

# --- 4) CMA-ES Wrapper ---
def optimize_cma(r=2, sigma=0.5, restarts=5):
    dim = 8 * r  # reelle Parameter
    best = (np.inf, None)
    for _ in range(restarts):
        theta0 = np.random.randn(dim)
        es = cma.CMAEvolutionStrategy(
            theta0, sigma,
            {'bounds': [-5, 5], 'popsize': 24, 'tolfun': 1e-7}
        )
        res = es.optimize(lambda th: objective(th, r=r)).result
        if res.fbest < best[0]:
            best = (res.fbest, res.xbest)
    theta_star = best[1]
    P_star = theta_to_P(theta_star, r=r)
    return P_star, best[0], theta_star

t = 20
for i in range(1,4):
    print(i)
    blockPrint()
    P_opt, fval, theta_opt = optimize_cma(r=i)
    enablePrint()
    print(q.Qobj(P_opt))
    print(fval)
    print(q.entropy.negativity(q.ptrace(zeno_state_ev(rho_0, H, PABZ(P_opt), t),
    [0,1]), 1))
end_time = time.time()
duration = end_time - start_time
print(f"Duration: {duration:.2f} seconds")
```

Listing A.3: Finding the projection such that the Zeno state maximizes negativity for a fixed time